

Global Facilities – Design & Construction Standard - Electrical – Electrical Specification

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ECN Area: ENGINEERING

Owner(s) ENGINEERING

Target Audience: All Facilities

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[ECN History](#)[Submit a Change Request](#)

1.0 Purpose

- 1.1 This document serves as a System Standard document that provides specifications that shall be followed during the Design, Engineering and Construction of new Electrical System as part of Greenfield FAB, Assembly and Test, other manufacturing facilities Construction and Brownfield expansion projects..
- 1.2 This document is a Micron Specification which provides the general principles to be followed in the electrical system and equipment engineering design, construction and testing/commissioning. It applies to all Greenfield and Brownfield expansion projects. This document is a principle reference for brown field revamp and rejuvenation projects.
- 1.3 Design and engineering of the electrical installation shall satisfy all statutory requirements of the national and local authorities of the country in which the installation is to be located.
- 1.4 The electrical installation shall be suitable for the site conditions as specify by the owner.
- 1.5 The contents of this specification and of standards and publications referred herein shall be adhered to, except where amended by specific requirements provide by owner of related installation and is allowed under statutory requirements mentioned above. (clause 4.1)

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- 1.6 Any contradiction between the requirements of this specifications and IEC/IEEE/ANSI or national standards shall be subject to clarification and approval by the Owner.
- 1.7 Standardize equipment of similar nature or identical materials/components and of similar or identical construction should be from the same Manufacturer (e.g., power and convenience outlets, luminaries, junction boxes, switches.).
- 1.8 The Contractor shall obtain from the Manufacturer's test reports in accordance with the equipment specifications, e.g., for generators, motors, HV/MV and LV switchgear, UPS equipment (DUPS/SUPS), SMS, ATS, Capacitor banks, MV Starters, surge arresters, Condition Monitoring devices, HV/MV cables and transformers.

2.0 Scope

- 2.1 The document covers the new Base build System installations only and does NOT cover tool installation. Tool installation projects may use the relevant sections of this document as reference.
- 2.2 Site (s) Impacted:
This standard specifies requirements and provides recommendations for the design and construction of fixed electrical installation in Micron. It applies to all Micron Facilities employees, contractors, and vendors at all Micron sites for new Greenfield FAB Construction projects.

3.0 Definitions and Acronyms

Term/Acronym	Definition/Description
The Contractor	The party that carries out all or part of the engineering design, purchasing, installation, testing and commissioning of a project. The Owner may undertake all or part of the duties of the Contractor.
The Manufacturer/ Supplier	The party that manufactures or supplies equipment and services to perform the duties specified by the Contractor.
The Owner	The party that initiates the project and eventually pays for it. The Owner may also include an agent or consultant authorized to act for, and on behalf of, the Owner.
Shall	The term shall indicate a minimum technical requirement.
Should	The word should refer to a recommendation.

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Term/Acronym	Definition/Description
CoHE/LOTO	Control of Hazardous Energy/ies or sometimes called Lockout/Tagout. Preventing machinery/equipment from releasing hazardous energy while worker/s performing servicing activities
Distribution of electricity	The transfer of electricity to consumers within an area of consumption 601-01-10, IEC60050
Emergency lighting	Lighting provided when the supply to the normal lighting fails. (Non-US site)
Emergency egress lighting	Lighting provided when the supply to the normal lighting fails. (US sites)
Escape lighting	Part of Emergency Lighting provided to ensure that the escape route is illuminated in the event of a major incident.
Emergency service	A service, which, if it fails in operation or when called upon, will affect the continuity, quality or quantity of the product or could result in equipment damage or extended downtime, but would not harm people or the environment.
Flywheel	A revolving wheel of a machine used to increase machine's momentum for stability or reserve of available power.
Greenfield	The term Greenfield refers to buildings/manufacturing plants constructed on fields that were literally new by Companies
High Voltage (HV), 601-01-27, IEC 60050	The set of voltage levels in excess of low voltage. Although not covered by an IEC definition, it has become industry practice to use the term Medium Voltage (MV) for voltages from 1 kV up to 35 kV.
Installed capacity	The sum of the rated powers of equipment of the same kind (generators, transformers, converters, etc.) in an electrical installation.
Interruptible, emergency electricity supply	A source of electrical power which is backed up by a second (emergency) source of power, such as to provide a supply of electricity that may be interrupted for no more than 10 seconds.
Islanding	Disconnecting the external (public utility) grid supply, so that all or part of the plant electrical system will operate

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Term/Acronym	Definition/Description
	in an "island" with the main on-site generators as the source of supply.
Low Voltage (LV), 601-01-26, IEC 60050	A set of voltage levels used for the distribution of electricity and whose upper limit is 1000 V AC
Non-essential service	Non-essential service is any electrical service that is neither critical nor essential.
Point of Common Coupling (PCC)	The point of coupling at the public utility networks, to which the system under consideration is, or is to be, connected. Other systems (consumers) may also be connected to or near this point.
Safety Switch	A switch in the vicinity of a consumer used to disconnect the equipment for safeguarding purposes.
Site conditions	The external factors, e.g. altitude, air temperature, wind velocity, vibrations, earthquakes, black body temperature, relative humidity, etc., which may influence the operation of a machine or apparatus.
Stand-by capacity	The capacity provided for the purpose of replacing equipment which may be withdrawn from service under planned or unplanned circumstances. Similar to US term, as "Redundant capacity".
Above Grade	Not buried in ground and not embedded in concrete slab on grade
Below Grade	Buried in ground or embedded in concrete slab on grade
Concealed	Inside building above grade and located within walls, furred spaces, crawl spaces, attics, above suspended ceilings, in concrete, etc. in general, any items not visible or directly accessible.
Connect	Complete hookup of item with required services, including conduit, wires and other accessories.
Exposed	Either visible or subject to mechanical or weather damage, indoors or outdoors, including areas such as mechanical and storage rooms. In general, any item that is directly accessible without removing panels, walls, ceilings, or other parts of structure.
Furnish	Supply and deliver complete
Install	Place, secure and connect as required to make fully operational.
EMT	Electrical Metallic tubing

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Term/Acronym	Definition/Description
FMC	Flexible metal conduit
GFE	Global Facilities Engineering
IMC	Intermediate metal conduit
LFMC	Liquid-tight flexible metal conduit
RNC	Rigid non-metallic conduit
SMS	Sag Mitigation System
DUPS	Dynamic Un-interruption Power Supply
SUPS	Static Un-interruption Power Supply
RAM	Risk Assessment Matrix
GFE&C	Global Facilities Engineering and Construction
EHS	Environmental, Health, Safety.
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKM's
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team

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4.0 Roles and Responsibilities

Roles	Responsibilities
GFE Lead (Discipline Engineer)	• Develop and maintain System Standard document • Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard • Perform periodic document review and update System Standard
GFE&C Lead (Discipline Engineer)	• Review and provide technical feedback on System Standard • Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to GFE Lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard.
Site Facilities Teams	• Use System Standard as technical reference • Review and provide technical feedback

5.0 Electrical System Design

5.1 Introduction

In order to provide electrical service to the facilities and tools, to determine what type of system is available from utility company or existing facility owned and operated system by complex. Once it is confirmed, it is important to understand the characteristics of the system – not only voltage, capacity, and available fault current but also the operational, reliability, and relative cost characteristics inherent to the system by virtue of its configuration or arrangement. Knowing the characteristics associated with the system arrangement, the most appropriate service and distribution system for the application can be determined.

5.2 Design Philosophy

The design shall meet the specific design criteria, objective stated in the project definition phase, e.g. the basic of design (BOD) document or project specification.

The BOD or project specification document relating to the electrical system shall documented by a system design description, a key line diagram, basic layout drawings and functional/outline specifications.

Prior to purchasing electrical equipment, a preliminary Short Circuit Analysis shall be performed to determine the magnitude of prospective currents flowing throughout the power system at various time intervals after a fault occurs at each bus in the system. This information shall be used to determine momentary and interrupting duties based on the maximum available fault current, selection of protective devices, and to assist in setting those devices. Selection of protective devices should account for the possibility of harmonic current and voltages on the system equipment and that they be resistant to their effects on the device functionality.

System studies and protection reports, including software files shall be provided in support the design. The system studies consist of:

- Short circuit calculation
- Load Flow Study
- Protection coordination study including relays setting schedules and arc flash energy.
- Dynamic performance studies under motor starting (base load plus starting of largest motor) and/or loss of generation.

The design and selection of electrical equipment should take into consideration optimizing capital and operating cost. Total life cycle cost calculation should be included in the design for assets that have a design life of 20 years or longer.

5.3 Power Reliability Assessment

Fully redundant power sources shall supply the site, comprising of either multiple separate power sources from the utility or at least a single utility source and a separate cogeneration source. There shall be adequate transmission, distribution, or cogeneration capacity on each of these sources to carry the entire site load.

Double Radial Feeder configuration should be consideration for the distribution network unless the justification due to economic consideration provided, to be approved by Facilities VP level or higher.

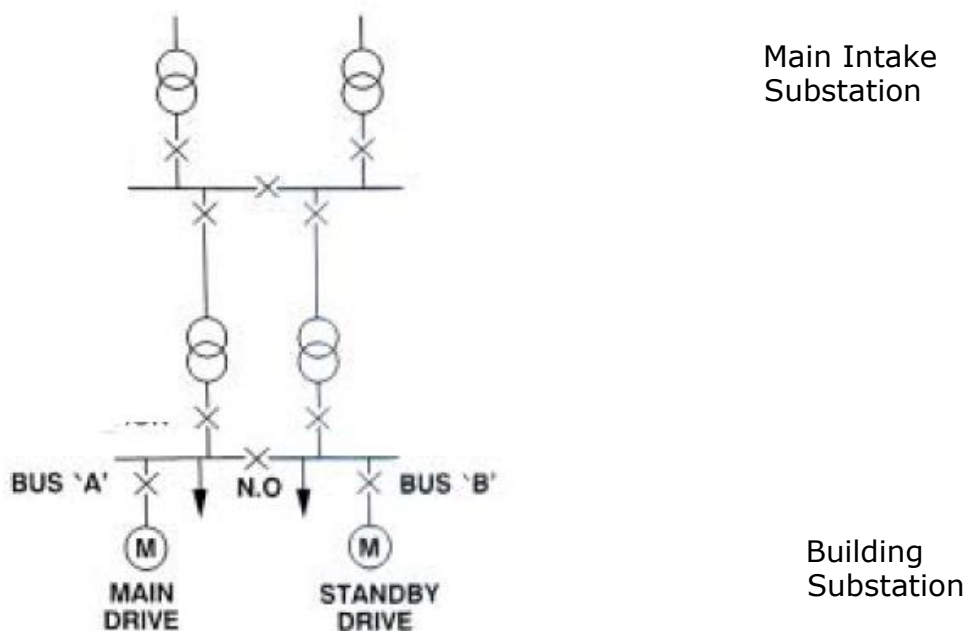
The reliability of a supply from a single feeder may be increased by providing automatic changeover facilities (ATS) to switch the load to an alternative feeder, however the method application to be fulfilled the follows:

Comprehensive testing during FAT and SAT. ATS to be used only at LV main switchboard level or MV motor control switchgears only.

If the load contains large induction motors, automatic changeover could entail re-closing onto partially de-energized motor. Because the motors would be slipping behind the supply frequency during the period of interruption, the supply voltage on reclosure could be up to 180o out of phase with the motor emf. Magnetization currents much larger than normal current could flow, with the possibility of mechanical damage.

Re-acceleration study to consider the automatic change over activity.

Typical Distribution Diagram



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- 5.3.1 Reliability of utility power sources shall be evaluated for the frequency of service interruptions, the average duration of the interruptions, sag and swell data, and any other power quality information available for the site to assist in determining the system design and configuration.
- 5.3.2 The anticipated initial and future load, load demand and diversity, and characteristics of the load shall be considered in sizing the system components.
- 5.3.3 Full-redundant power sources shall supply the site, comprising of either multiple separate power sources from the utility or at least a single utility source and a separate cogeneration source. There shall be adequate transmission, distribution, or cogeneration capacity on each of these sources to carry the entire site load.
- 5.3.4 Lightning and surge protection will be designed for the conditions at the particular site of installation and will be installed at critical points in the electrical system.
- 5.3.5 Main Power Intake Substations shall be constructed in breaker and a half or ring bus configuration Double bus configurations to be considered for less critical applications and where specific limitations
- 5.3.6 The system shall be configured so that loss of a source or major system component shall not inhibit the ability to supply power to the site.
- 5.3.7 The power source for the tripping circuit for circuit breakers shall be from a DC source. The DC source shall be in conformance with the DC supply in section 6.6
- 5.3.8 The protective devices in the system shall limit the extent and duration of the service interruption whenever equipment failure, human error, or adverse natural events occur on any portion of the system. In addition, they shall minimize damage to the system components involved in the failure
- 5.3.9 Equipment installed outdoors shall be equipped with thermostatically controlled space heaters or air conditioning, ventilation, and any other protective safeguards deemed necessary for the site location, altitude, and ambient conditions. Outdoor equipment to be rated at least IP55 or above.
- 5.3.10 Start-up and auxiliary power sources for cogeneration equipment shall be configured so the power source is available if the generator is off-line or not synchronized.

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- 5.3.11 The secondary power supply shall be designed or configured so that upon the loss of the utility source, the site electrical load shall be supplied by the secondary power source. Any automatic switching shall allow protective devices to clear faults before transfer and will not operator in the event of momentary voltage anomaly.
- 5.3.12 The redundant power supplies from the utility shall be routed on separate pathways to the site to improve reliability. Separation for routing shall be minimum of 10 FT.
- 5.3.13 If a cogeneration system is used in conjunction with the utility, it shall be capable of being operated in island mode without shedding FAB, critical production, or support services for those loads.
- 5.3.14 A risk analysis shall be performed to determine the amount of fuel storage required to operate the cogeneration system in island mode in an emergency condition.
- 5.3.15 A grounding/earthing system will be installed to provide an effective path for ground/earth-fault circuit, bonding of electrical equipment, and safety of personnel.
- 5.3.16 HV Power transformers serving the site shall be equipped with automatic load tap changers that operate in auto or master follower mode for the case of parallel operation
- 5.3.17 Electrical equipment and protective devices shall be selected, and setting determined based on a short circuit and coordination study, load flow study, and arc flash study.
- 5.3.18 A motor starting analysis shall be performed and the voltage drop on a medium voltage bus shall not exceed 3.5%.
- 5.3.19 System utilizing cogeneration shall have protective devices configured so the operation of the utility system will not adversely affect the generation equipment.
- 5.3.20 A sag mitigation system shall be incorporated into the design to protect FAB loads, the degree of protection the system provides shall be determined by evaluating power quality at the site.
- 5.3.21 Electrical loads shall be regularly monitored, and future load projections made to forecast system needs.
- 5.3.22 Circuit and system redundancy shall be maintained to allow maintenance to be performed without service disruption to the load.
- 5.3.23 Spare shall be allocated, and equipment sized to allow for future expansion.
- 5.3.24 Bypass scheme (Internal and External are required.) for Sag Mitigation system (SMS), dynamic and Static UPS and other

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power quality improvement equipment shall be provided to improve availability and maintainability. And the system should be alarmed and immediately switched to internal bypass without load impact when any control, communication error or during an internal failure Segregation of DVR and UPS/DUPS and EDG loads into A & B scheme

- 5.3.25 Segregation of DVR and UPS/DUPS loads into A & B scheme
- 5.3.26 Load test provision for UPS/ DUPS & Emergency power system.
- 5.3.27 Where appropriate, power factor shall be corrected to meet local or the supplying utility's requirements. If capacitors are utilized, consideration shall be given to potential problems in system self-excitation of induction machines, resonance, and waveform distortion in rectifying equipment.
- 5.3.28 A secondary selective protection scheme may be used in combination with the primary selective protection scheme to provide a higher degree of reliability.

5.4 Load category and consumption

Tools loads and Facility loads shall be categorized base on the service as either critical, essential and normal.

The critical manufacturing tools and facility load, which means there are high economic impact (RAM) and personnel safety/environmental impact.

Failure of any facility load and manufacturing tools, which consequence may result in severe safety, environmental and or economic impact shall be classified as Critical load. Classification of severity shall be based on organisational Risk Assessment Matrix (RAM).

The critical facility loads shall be supplied via SUPS and emergency power supply system.

The critical Tools load shall derive power from un-interruption source such as DUPS equivalent, and the rest of the FAB tools shall be protection by Sag mitigation system for transient disturbance.

5.4.1 General

Electrical loads shall be classified as performing a service, which is 'Critical', 'Essential', or 'Normal'. (emergency, optional standby, code required standby in USA)

5.4.2 Critical service

The requirements for power supply to 'Critical' services is un-interruption.

Examples of 'Critical' services are as follows:

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- Endangers employee safety or cause occupational illness or physical injury. Examples include: FAB lighting or other select lighting
- Would cause Environmental impact.
- Will cause significant scrapped wafers, significant rework of multiple wafers, could cause damage to the tool, or require more than 4 hours to restart and re-qualify. Single Wafer Tools not meeting these criteria shall not be placed on UPS systems.
- Will cause impact to Facilities systems, that in turn will directly impact Production. Examples include: VOC Controls, Process Cooling Water Pumps, PLC Cabinets, Pollution Abatement, Select Instrumentation and Control devices, Toxic Gas Monitoring Systems, Security System components, and IS Machine Room equipment.
- SCADA systems and all communication equipment to have UPS or DUPS coverage

5.4.3 Essential service (interruption within minute)

For essential services, the extent of partial or complete duplication of the energy source, of the lines of supply or of the equipment, or the introduction of automatic restarting or changeover facilities shall be determined by the owner.

Examples of essential services are as follows:

- Product transport by means of duplicated pump sets with account being taken of pump maintenance requirements;
- Power supply to process analysers by means of a duplicate supply system with changeover facility;
- Power supply to security lighting and plant area lighting.
- Dryers control of CDA service
- Reference to APPENDIX H: Facility Electrical Power Supply Matrix.

5.4.4 Normal service

Normal service is not backed up power supply.

Examples of normal services are power and lighting supplies to offices, warehouses, residential areas, etc.

5.4.4.1 Maintainability (A / B) Distribution Scheme

An electrical distribution scheme for the maintainability of the MV (6.6kV or 4.16KV) and LV (480V) electrical systems and mitigation of production curtailment in the event of an outage in the system.

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This design scheme requires half the equivalent mechanical and production equipment to be fed from one side of the main switchboard and half from the other side of the main switchboard.

The overall philosophy is to enable shutting down half of the production equipment and associated mechanical systems for maintenance, while the other half of similar equipment remains in operation. Recommend using this scheme where cost effective or necessary for reliability.

5.4.5 Load assessment and electricity consumption

- a. A schedule of the installed electrical loads, the maximum normal running plant load and the peak load, expressed in kilowatts and kilovars and based on the plant design capacity when operating under the site conditions specified, shall be prepared using Standard Form.
- b. The schedule of installed electrical loads shall be prepared and updated throughout the design phases of the project and will form the basis for provision of the necessary electricity supply and distribution system capacity.
- c. During the concept phase of a project development, a separate allowance shall be made to account for an increase in the electrical load owing to uncertainties in the process design.

This is particularly important where own generation is the main source of electricity. Typically, this allowance may be in the order of 10% to 25%, depending on how much of the process design is based on existing operating experience. The allowance is normally reduced at distinct project phases as the design develops.

- d. The formulae for determining the total electrical loads:
 - Maximum normal running plant load = $x(\%)E + y(\%)F$
 - Peak load = $x(\%)E + y(\%)F + z(\%)G$where
E = sum of all continuously operating loads
F = sum of all intermittent loads
G = sum of all stand-by loads
x, y and z are diversity factors
- e. Load growth allowance and values of the diversity factors (x, y, and z) shall be appropriate to the type of plant and determined by the Principal.

- f. Values of the diversity factors (x, y and z) shall take account of the individual drives or consumers which make up the continuous, intermittent and stand-by loads, respectively. For example, y(%)F or z(%)G cannot be less than the largest individual intermittent or stand-by drive or consumer.
- g. The various loads shall be classified as follows:
- E - "Continuous"
All loads that may continuously be required for normal operation, including lighting and workshops
NEC defines continuous as running for more than 3hours
 - F - "Intermittent"
Loads required for intermediate pumping, storage, loading, etc.
NEC defines continuous as running less than 3 hours
 - G - "Stand-by"/redundant
All loads required in emergencies only, such as fire-water pumps or those of normally not running electrically driven units in stand-by mode for normally running steam-driven ones, e.g. charge pumps, boiler feed pumps, etc.
 - Spare pumps etc., e.g. the "B" pump of an A-B combination, are not to be considered as "Stand-by" loads.
 - Even though not to be considered as stand-by loads, the largest motor of the spare duty/redundant shall be considered in the motor start studies in (3.1) with the duty motor running.
- h. The following default values may be used for initial load assessments, or if the diversity factors have not been finalised:
- x = 100%.
At rated plant throughput all driven equipment is assumed to be operating at its duty point. However, some diversity may be applied to non-process loads, e.g., offices and workshop power and lighting, typically 90%.
 - y = 30%.
 - z = 10%.
- i. A separate schedule shall be prepared for each switchboard, the total of all switchboard loads being summarised as required to arrive at the maximum normal running and peak loads for each substation and for the plant overall.
- j. All loads to be shed during an underfrequency condition shall be identified as such in the 'remark' column.
- k. All loads to be automatically restarted after a voltage dip shall be identified as such in the 'restarting' column.

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- l. Where a group of drives operate as a unit, it shall be considered as an individual consumer.
- m. The power consumption of electric motors shall be based on motors rated in accordance with the requirements of mechanical standard.

The percentage of total intermittently operating load that contributes to the maximum normal running load will depend on FAB operations.

5.5 Electrical Protection

System studies shall be performed using similar software program, to align/extend, to existing site. If there is currently no software program use on subject site, ETAP shall be used for all Micron sites.

System studies and protection reports, including software files, shall be provided in support of the design, for approval by GFE&C (Global Facilities Engineering and Construction Team) depending on the type, size and complexity of the installation, such studies may comprise the following:

- i. Load flow studies;
- ii. Short circuit studies complying with IEC 60909 Parts 0, 1, 2, 3, 4; or IEEE standard 3002.
- iii. System dynamic stability (transient stability) studies under three phase fault conditions;
- iv. Dynamic performance studies under motor starting and/or loss of generation conditions;
- v. Protection grading studies (selectivity), including relay setting schedules; an arc flash study should be carried out for low voltage switchboards and for high voltage switchboards only if the protection grading studies show that arcing faults cannot be cleared within the Internal Arc Classification time of the switchgear;
- vi. Arc flash study to be carried out with arc-flash energy calculation for all electrical equipment that needs live electrical work or exposed electrical work with appropriate Arc Flash labels and PPE, complying with NFPA 318 – Standard for the Protection of Semiconductor Fabrication Facilities and NFPA 70E – Standard for Electrical Safety in the Workplace.
- vii. Harmonic analysis studies;
- viii. Motor re-starting and re-acceleration;
- ix. Critical Supply recovery following total power failure, including emergency diesel generator(s), UPS, I&C, and FCMS.

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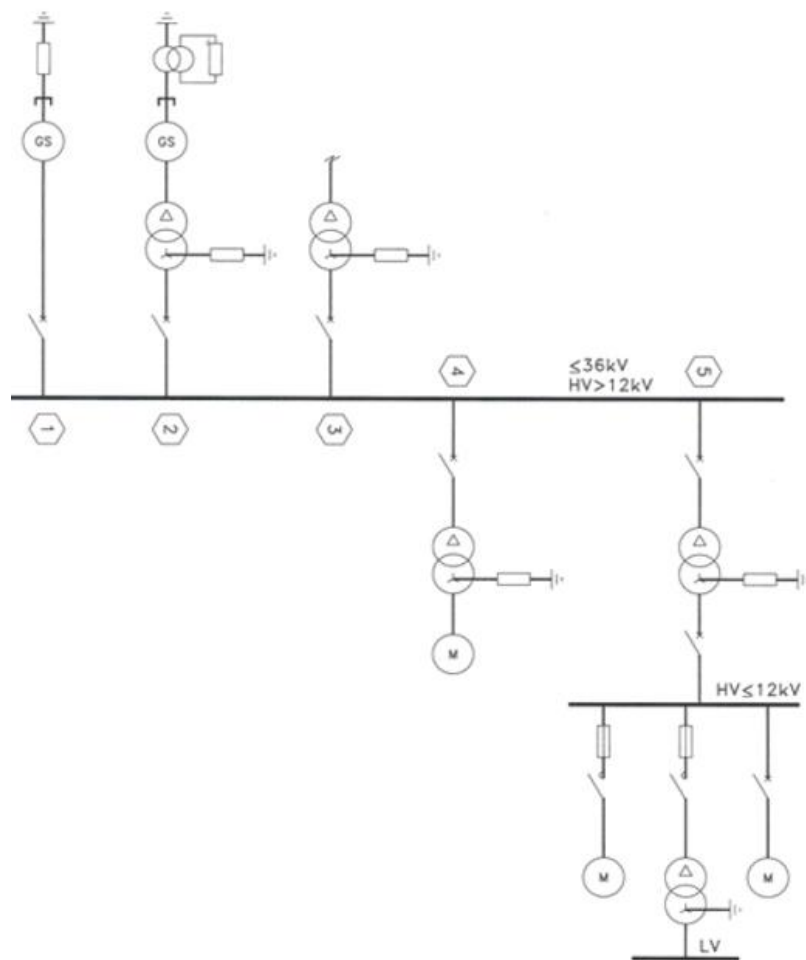
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5.6 Electrical Earthing/Grounding System

The purpose of an earthing/grounding system is to provide an acceptable path for earth/ground fault current to return to the system neutral. This has to be implemented in a method ensures 'safety to personnel', i.e., without giving rise to dangerous touch, step or transferred potentials.

The earthing/grounding system arrangement below is preferred:

System Earthing/Grounding Neutral Diagram



1. Direct Connected Generators
2. Transformer connected generators
3. Public utility incoming feeder
4. HV motor with unit transformer
5. Transformer feeder to HV switchboard with circuit breaker
outgoing circuits only or mixed outgoing circuit switching devices.

5.6.1 Earthing/grounding System Requirements:

Permanent earthing/grounding system for the electrical network and equipment for the installation shall have at least one common earthing/grounding grid connected to at least two earth/ground electrodes.

The resistance to earth/ground of this common earthing/grounding grid shall be as low as practicable and not exceed 1 Ohms.

Earth/ground wires shall be of standard copper conductor with green/yellow PVC sheathing. Conductor shall be sized to carry the rated fault current of the distributed switchgear for its rated short-time duration without deterioration.

For mechanical reasons main earth/ground wires shall be at least 100 mm² and branch earth/ground wires 25 mm², unless adequate mechanical protection is provided by other means, e.g., wire installations in conduit, or earth/ground conductors forming part of a cable, allowing a smaller size.

The connections between earth/ground electrode and conductors should be made so that the earth/ground resistance of each individual electrode can be readily inspected and tested, without disconnecting any part from the earthing/grounding system.

The earth/ground bar in each switchboard and sub-distribution switchboard shall be connected with earth/ground wires of size, capable handling system fault and suitable for 100% duty, to the common earthing/grounding grid. The minimum earth/ground cable size shall be 70 mm².

5.7 Code and Regulations

All work and materials shall comply with the latest rules, codes and regulations, including, but not limited to the following:

- a. Local Electrical Code and Regulations
- b. Occupational Safety and Health Act Standards (OSHA).
- c. NFPA 70E – Standard for Electrical Safety in the Workplace.

For European Standard:

- a. IEC_ International Electrotechnical Commission
- b. ATEX_ European Standard for Hazardous Equipment (Applicable for Intrinsic Safety Barriers installed in Solvent CDU's only)

For North America Standard:

- a. NFPA #70 - National Electric Code (NEC).
- b. NFPA #101 - Life Safety Code.
- c. IEEE_ Institute of Electrical and Electronic Engineers
- d. NEMA_ The National Electrical Manufacturers Association
- e. API RP505_ Recommended Practice for Classification of Locations for electrical Installations
- f. UL 1604 Electrical Equipment used in Class I and II, Hazardous Locations
- g. International Energy Conservation Code (IECC)
- h. NFPA 318 – Standard for the Protection of Semiconductor Fabrication Facilities
- i. NFPA 70E – Standard for Electrical Safety in the Workplace
- j. International Building Code (IBC)

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6.0 Electrical Standard and Specification Design for Equipment

6.1 General requirement

- i. All electrical equipment and materials shall comply with all local code and international standards as specified in detail project specification as well as refer to this specification.
- ii. In general, all electrical equipment and materials should be located indoor; The required ambient conditions shall be as follows:
 - In general, all electrical equipment and materials should be located indoor; The required ambient conditions shall be as follows:
 - Ambient temperature – 10 Deg Celsius (50oF) to 40 Deg Celsius (104oF) for electrical equipment design and required the electrical room temperature is 24+/- 2 Deg Celsius (71.6~78.8 oF) unless specified in the project specification.
 - Humidity 0% to 70% with average room humidity is 65% unless specifically mentioned in the project specification.
- iii. If equipment and materials need to be installed/stored outdoor, it shall be equipped with thermostatically controlled space heaters and/or air conditioning, ventilation, and any other protective safeguards deemed necessary for the site location, altitude, and ambient conditions.
- iv. Indoor/Outdoor area Installation: It requires electrical equipment and components shall be protected from any intrusion of liquids. Solutions include rain caps, appropriate NEMA/IP ratings, or other containment.
- v. Special condition area Installation (Corrosive, Hazardous, Explosion): It requires electrical equipment and components shall be protected from corrosive, hazardous, and explosion environment. Detail of class, division and group which indicate area of special condition shall be provided by General Contractor to ensure safety performance under special condition.
- vi. Switchgear and or other electrical equipment shall be installed so that there is sufficient space, as defined by manufacturer, to facilitate constructability, operation and maintenance requirements. Safe clearance distance between such equipment to walls and/or other equipment shall be adhered to below or in accordance to local requirements/codes. Any deviation shall be made in consultation and with approval from Owner, before work can proceed. Below the minimum clearance recommended, and if the manufacturer/s and or local regulation requires a greater distance, it shall be adhered

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Description	Distance (m)
Front of High Voltage Switchgear	2.0 (Minimum)
Rear of High Voltage Switchgear	0.8 (Minimum)
Between High and Low voltage Switchgear	2.0 (Minimum)
Sides of switchgears (HV/MV/LV)	1.0 (Minimum)
Front of Low Voltage Switchgear	1.5 (Minimum)
Rear of Low Voltage Switchgear	0.6 (Minimum)
Between two LV switchgear/switchboards	1.5 (Minimum)
Above, top clearance, switchgear	1.0 (Minimum)
Around Oil type transformers	1.0 (Minimum)
For dry type transformer	Use clearance from manufacturer's recommended design data.
Provide safe escape with minimum distance between two equipment with their doors opened	Shall be more than 700 mm

vii. In general, below AC voltage classification is used to categorize voltage type.

- High voltage: AC Voltage more than 35,000 volts.
- Medium voltage: AC Voltage between 1,000 up to 35,000 volts.
- Low voltage: AC Voltage lower than 1,000 volts.
- Extra low voltage: AC Voltage lower than 50 volts.

viii. Special condition area installation: I.S. Machine room.

- The I.S. room shall be designed to meet the "Uptime Institute's Tier III Concurrently Maintainable Site Infrastructure" protocol.
- At a minimum, the primary power source, generators, and UPS System will be N+1.
- At a minimum, there shall be two separate primary power sources to serve the site or building housing the I.S. Machine Room. The preference would be that the separate sources come from different generation facilities.
- The power transformers serving the I.S. room shall be fed from medium voltage switchgear that automatically switches from a preferred power source to an alternate power source in the event of a power anomaly or outage.
- The electrical system from the utility source to the I.S. equipment shall be completely redundant.

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- Two simultaneously active paths from the distribution transformers to the I.S. equipment shall be provided, one normal and one UPS, or both from a UPS source.
- Facilities equipment will have two separate sources from the electrical distribution system, so that system maintenance can be performed without affecting the I.S. room or equipment.
- Equipment providing redundant pathways to the I.S. equipment shall not be loaded beyond 40% of capacity.
- All other equipment shall not be loaded beyond 80% of capacity.
- Branch panels for I.S. equipment shall be equipped with CTs to measure the current on the individual circuits serving the I.S. equipment. The circuits will be monitored in the Control Room and will alarm if the current exceeds 40% of the breaker's capacity.
- Branch circuits for I.S. equipment shall not be loaded beyond 40% of the breaker's capacity.
- The electrical system will be designed based on an I.S. equipment load of 1615 watts per sq. m (150 watts per sq. ft.) minimum and validated with the I.S. group.
- Branch circuit and network raceways for the I.S. equipment shall be installed above the backside of the equipment racks, 600 mm (24 in.) above them.
- If premise transformers are installed in the I.S. equipment area, they shall be FM-approved and either dry-type, or filled with a non-combustible dielectric medium.
- Separate service outlets will be provided in the I.S. Machine Room for maintenance or construction work.
- UPS units shall be reviewed with the I.S. Machine Room Owner. At a minimum, the UPS DC source shall have adequate run time to allow the generator a minimum of three attempts to start without affecting the I.S. room or equipment.
- Generators shall have sufficient fuel storage to operate at full load for 8 hours minimum.
- UPS shall have a minimum of 20-minute battery string.
- Lightning surge protective devices shall be provided on the main electrical service serving the I.S. room
- A lightning risk analysis shall be performed to evaluate the potential hazards and determine if lightning protection is required for the I.S. room or the building housing it.
- The I.S. room temperature shall be maintained between 23 °C (73.4 °F); plus 2 °C, minus 2 °C.
- Humidity shall be between 35% and 65%

6.2 High Voltage (HV) Switchgear

6.2.1 Switchgear GIS/AIS Type

- a. The HV switchgear shall be Sulphur Hexafluoride (SF6) or non SF6 gas insulated switchgear (GIS) type and meet requirements in IEC 62271-203, IEEE C37.122, IEEE C37.123. If there is any local utility provider specific requirement for air insulated switchgear (AIS) or GIS type; it shall be communicated with Micron Global Facilities in advance.
- b. The HV switchgear can be installed Indoor or Outdoor. For indoor installation, proper overhead crane must be provided as permanent installation to support future operation.
- c. The HV switchgear must be maintenance-free design with highest level of safety, performance and reliability.
- d. The HV bus scheme to be breaker and a half Scheme. Double-Bus Single Breaker configuration can be considered if there are any site-specific limitations. When extending either end of the switchgear, it shall only be necessary to isolate the associated bus extension switch gas compartment and remove end cover and spacer for the associated busbars. No other gas compartment shall be affected.
- e. The primary components of the HV switchgear equipment such as busbar, circuit breaker, disconnecting switch, earthing/grounding switch, voltage transformer, current transformer shall be enclosed in metal enclosure.
- f. The HV switchgear encapsulated type shall be single-phase or phase segregated model unless otherwise stated.
- g. A typical HV switchgear technical data sheet is shown in the appendix.

6.2.2 Circuit breaker

- a. The circuit breaker design, construction, testing and application shall satisfy ANSI/IEEE C37.04 and/or IEC 62271-100 as minimum requirement unless otherwise stated.
- b. The circuit breaker shall be gas single pressure type or vacuum type suitable for high speed reclosing service.
- c. Circuit breaker short circuit rating (making, interrupting, and braking) shall satisfy maximum prospective short circuit at present and future.
- d. Circuit breaker short circuit capacity shall consider full DC component at present and future.

- e. Required interlocks (w/ disconnecting switch and earthing/grounding switch) shall be provided to prevent dangerous operations of circuit breaker.
- f. Circuit breaker shall be arranged if gas pressure is dropping; it could be safely open at the rated current and voltage duty.
- g. Circuit breaker shall be automatically lockout and cannot be closed when the gas pressure is below the nominal gas pressure.
- h. Details of gas leakage rate for breakers and bus bar chambers shall be submitted.
- i. Service life of gas for breakers and bus bar chambers shall be stated.
- j. Gas pressure status shall be clearly indicated.
- k. High voltage circuit breaker Technical Data Sheet is shown in appendix.
- l. Circuit breaker to be operated electrically.

6.2.3 Disconnecting Switch

- a. The disconnecting switch shall be triple-pole single throw electrically and manually operated type.
- b. Disconnecting switch short circuit rating (making, interrupting, and braking) shall satisfy maximum prospective short circuit at present and future.
- c. Disconnecting switch short circuit capacity shall consider full DC component at present and future.
- d. Disconnecting switch shall be operated electrically and manually.
- e. Disconnecting switch shall be fully interlocked with circuit breakers to ensure that they can only be operated when the associated main circuit is not energized.
- f. Disconnecting switch shall be fully interlocked with earthing/grounding switches to prevent from unauthorized operation.
- g. Service life of gas for disconnecting switch chambers shall be stated.
- h. Gas pressure status shall be clearly indicated.
- i. High voltage disconnecting switch Technical Data Sheet is shown in appendix.

6.2.4 Earthing/Grounding switch

- a. Earthing/grounding switch shall be located on both sides of each circuit breaker and on each end of the circuit as shown on the single line diagram.
- b. Earthing/grounding switch shall be electrically and manually operated. Earthing/grounding switch on line side shall be high speed making proof type with a making capability to equal circuit breaker closing capability.
- c. The operating mechanism shall be enclosed in waterproof and dustproof metal housing. A mechanical position indicator and space heater with humidity controller to maintain the internal relative humidity below 60% shall be provided in the housing.
- d. Earthing/grounding switch shall be fully interlocked with the associated circuit breakers and disconnecting switches to prevent from earthing/grounding the live parts.
- e. Service life of gas for earthing/grounding/grounding switch chambers shall be stated.
- f. Gas pressure status shall be clearly indicated.
- g. High voltage earthing/grounding switch Technical Data Sheet is shown in appendix.

6.2.5 Protection and Metering

- a. The protective relay shall be intelligent electronic device (IED) type with built-in advanced microprocessor unit and modular communication module. All the relay need to provide supervision circuit and to be connected to SCADA for trip/pickup/warning and relay failure.
- b. The protective relay shall be able to operate continuously at minimum 10 years without any obsolete spare parts.
- c. Metering shall have energy and power quality measurement capability and included but not limited to these following functions:
 - Power Quality Analysis
 - Harmonic distortion
 - Voltage sag and swell detection/transient
 - Disturbance direction detection
 - Waveform capture
 - Programmability (logic and math functions)
 - Compliance monitoring
 - Up to the 63rd harmonic

- Applications for:
 - Equipment monitoring and control
 - Co-generation and IPP monitoring
 - Tariff metering
 - Energy pulsing and totalization
 - Demand/Max Demand and power factor control
 - Measurements
 - Voltage/Current/Frequency
 - Apparent power total/per phase
 - Power factor total/per phase
 - Active power total/per phase
 - Reactive power total/per phase
 - Data recording
 - GPS synchronisation
 - Sequence of event recording
 - Time stamping
 - Event logs
 - Data logs
 - Trending/forecasting
 - Min/max of instantaneous values
 - Other features:
 - Electromagnetic compatibility (EMC)
 - Transmission rate of at least 100/1000 Mbits/s
 - Memory capacity of at least 5 MB
 - With common communication protocol link to Scada
 - Accuracy class – voltage & Current of 0.1% and energy of Class 0.2
- d. The protective relay shall be full or semi flush-mounted and draw-out case, fitted with operation indicators and necessary operation button (i.e. acknowledge, reset, etc.) on the front plate.
- e. Pilot wire relays via fibre optic connection to local power authority feeder connection and it should include below requirement but limited to:
- Pilot wire relay system shall be designed to provide complete phase and ground fault protection for two terminal transmission line (Line protection).
 - Each pilot wire relay system shall meet minimum requirement which required by local power authority and if not specified, the pilot wire relay system could be consist of pilot wire relay, insulation transformer, pilot wire monitoring relay, drainage reactor, neutralizing reactor, pilot wire relay test switch, mini-ammeter or indicator.

- f. Line protection function (87L/21S) shall be provided.
- g. Transformer differential protection function (87T) shall be provided.
- h. Busbar differential protection function (87B) shall be provided.
- i. Undervoltage relay (27) shall be provided.
- j. Overcurrent phase fault relay with Instantaneous elements (50/51) shall be provided.
- k. Earth/ground fault relay with Instantaneous elements (50N/51N) shall be provided.
- l. The bus tie breaker relay shall be equipped with instantaneous overcurrent relay with timing and selector switch to adjust instantaneous or time delay over current protection.
- m. All available protection function as mentioned above or additional item as per local regulation shall be able to isolate faulty part as designed and to avoid total power outage of the whole HV switchgear system.
- n. All protection devices and metering shall be provided communication protocol via IEC 61850 and linked through single-mode fiber optic network.
- o. Test sockets, plugs and leads shall be provided without disconnecting the permanent wiring, each protective relay to have a provision for testing and calibrating the relay by current injection using external power source.
- p. Protection relay settings and logic to be reviewed and approved by Micron Engineer for safe and reliable switching and operations.

6.2.6 Voltage Transformer (VT) and Current Transformer (CT)

a. VT

- The VT rating and location shall satisfy ANSI/IEEE C57.13 and IEC 60044-2 as minimum requirement unless otherwise stated. Rated output shall match the maximum load of the equipment connected and a future capacity of 25%. Secondary voltage of transformer shall preferably be 110V.
- VT shall be electromagnetic type mounted outside or inside metal enclosure with gas insulation. VT compartments shall be fully segregated from other primary compartments of the HV switchgear.

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- The VT rating (Frequency, ratio, ampere, accuracy, insulation level, temperature rise) shall meet system design requirement.
- All secondary leads of VT shall be connected to local control panel (LCP). All connections with in the LCP shall be solderless connectors on suitable terminal block and allow to permit connection to remote control panel (RCP).
- The VT rating of metering outfit (MOF) shall be approved and passed the test by local power authority.

b. CT

- The CT rating and location shall satisfy ANSI/IEEE C57.13 and IEC 60044-1 as minimum requirement unless otherwise stated.
- CT shall be bushing type mounted outside or inside metal enclosure with full insulation between the primary conductor and current transformer core and secondary parts shall be provided by the SF6 gas insulation.
- The CT rating (Frequency, Ratio, Ampere, Accuracy, Insulation Level, Temperature Rise) shall meet system design requirement. Dedicated CT shall be provided for Protection relaying and Metering.
- CT shall be multicore design and its cores shall be made of cold-rolled grain-oriented steel or nickel-iron alloy.
- Ratio changing of CT shall be possible by means of taps on the secondary windings (min. 2 taps).
- All secondary and tap leads of CT shall be connected to local control panel (LCP). All connections with in the LCP shall be solderless connectors on suitable terminal block and allow to permit connection to remote control panel (RCP) which RCP shall be in different control room.
- The CT rating of metering outfit (MOF) shall be approved and passed the test by local power authority.

6.2.7 Control power

- a. Provide disconnect breaker for separate DC power supply feeder on each HV switchgear section.
- b. HV switchgear control power shall be supplied via DC power supply which incorporates redundant chargers and redundant battery system with more than two strings. The charger and battery capacity shall have the capability of sustaining the worst operating conditions.

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- c. DC power supply needs to be monitored and shall be alarmed on the Supervisory Control and Data Acquisition (SCADA) system.

6.2.8 Other accessories

- a. Gas Leakage detection

An effective gas leakage detection system shall be provided for each compartment and connected to an alarm in Local control panel, Remote control panel, and Power SCADA/IMS.

- b. Cable Bushing

The cable bushing between HV switchgear feeder and primary of main transformer shall be supplied by HV switchgear vendor. The cable bushing should have separate oil and air insulated sections and serve as direct connection between the cable and main transformer.

The GIS connection comprises a housing which appropriately adapted to the power cable, connecting electrode and fitting for any misalignment and vibration. The cable bushing gas space shall be combined and monitored with the gas compartment of the GIS module to which it is connected.

- c. Cable Termination

For power cable type and size of outgoings which GIS are connected; shall be shown and matched in design single line diagram. Power company feeder as Incoming power cables and terminations will be provided by approved and licensed local power company Contractor. The power cable from HV switchgear from feeders to main transformer shall be supplied by HV switchgear vendor/contractor. The connector shield rings, cable housing, mounting plated and necessary accessories required for the installation of all cable termination shall be provided by HV switchgear vendor/contractor. All details shall meet power company standard.

- d. Lightning Arrester

The lightning arrester shall be station class, metal-oxide or zinc-oxide type and insulated with gas filled-housing. The rating of lightning arrested shall meet project specified requirement as shown in data sheet.

- e. Online Partial Discharge Monitoring

To detect early stage of partial discharge inside the HV switchgear Gas filled-compartment, online partial discharge monitoring system (Online PDMS) is required to be installed. UHF-based detection technology or equivalent or higher is recommended. Comprehensive partial discharge analysis shall be done automatically and provide clear alarm/warning signal

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to integrating power monitoring system (IMS). The diagnostic capability of Online PDMS shall compliant with IEEE Std. 1434 and IEC 60270.

f. Surge Protective Devices (SPD)

Contractor to select the right category of SPD for the correct protection against damage. ANSI/IEEE C62.41 or IEC 61312-1 gives recommendations on protection level required.

g. Blind cover

The blind cover for each end shall be furnished. The covers will be used for blinding compartment during maintenance works.

h. Gas Filling Device

One set of gas filling device including pressure reducing valve, manometer, adapter and filling hose shall be furnished for filling SF6 gas.

6.2.9 Installation

6.2.9.1 General

Where necessary, owner or the Contractor shall submit Manufacturers' test reports, equipment certificates and site test reports, etc., for approval by the local authorities.

Unless specified, the minimum ingress protection for equipment shall be IP54 for outdoor installed equipment, IP41 for equipment installed indoors and IP2x for exposed parts of the equipment that has to be accessible during normal operation.

Switchgear shall be installed in accordance with Manufacturer's installation instruction and supporting drawings.

Foundation drawings, together with information on mass and loading data, shall be provided by manufacturer of Switchgear, to provide correct civil design of the relevant buildings and foundations. This include switchgear foundations, any inserts to be cast in, shall be within manufacturer's specific tolerances.

Switchgear arrangement and layout drawings shall be obtained from the manufacturer so that Contractor can incorporate into the overall arrangement drawings and cabling design.

Switchgear modifications required during installation and/or commissioning shall be approved by the Owner prior to implementation. All modifications shall be

recorded and information incorporated in the "as-built" drawings.

Switchgear shall be installed so that there is sufficient space, as defined by manufacturer, to facilitate operation, maintenance and constructability requirement. Refer to section 6.1 (f).

Switchgear shall only be installed when the switchroom/substation civil and building works are completed, so as to minimize the ingress of dust/dirt and moisture during or after erection.

Substation floors shall be smooth and level to permit the handling of equipment on rollers, regardless of crane facilities provided.

Concrete floors shall have a surface protection layer to avoid formation/creation of dust.

HV Switchgear shall be located in a separate room from the controls, protection and metering room.

An equipment/switchgear access door should lead directly to the outside for each switchroom.

Specific requirement for switchrooms with gas insulated switchgear

6.2.9.2 Transport, Storage and Erection

Assemblies supplied in transport units shall have these units clearly marked to facilitate assembly at site.

Instruction for transport, storage and erection of the equipment shall be supplied by the Manufacturer as an integral part of the order.

Special tools and equipment required for erection, commissioning and maintenance shall form part of the order and shall be shipped together with the assembly.

6.3 Medium Voltage (MV) Switchgear

6.3.1 Switchgear GIS/AIS Type

- a. The preferable MV switchgear shall be Sulphur Hexafluoride (SF6) gas insulated or other gas insulated switchgear (GIS) type and meet requirements in IEC 62271-203. If there is any project specific requirement for air insulated switchgear (AIS) type and meet requirements in IEC 62271-200; it shall be communicated with Micron Global Facilities in advance.
- b. The MV switchgear should be rated for indoor environment.
- c. The MV switchgear must be maintenance-free design with highest level of safety, performance and reliability.
- d. The MV bus scheme shall be Single-Bus arrangement.
- e. The MV switchgear shall apply *FAX-...-FA1-Main A-Tie-DS-Main B-FB1-...-FBY* configuration.
- f. When extending either end of MV switchgear, it shall only be necessary to isolate the associated bus gas compartment and be able to remove end panel for the associated busbars. No other gas compartment and overall system operation shall be affected.
- g. The primary components of the MV switchgear equipment such as busbar, vacuum circuit breaker, disconnecting switch, earthing/grounding switch, voltage transformer, current transformer shall be modularized and enclosed in grounded metal enclosure. Each primary component shall be constructed into separate gas cubicles with a pressure relief device.
- h. The MV switchgear shall have arc resistant capability.
- i. The internal components of MV switchgear shall be hermetically sealed with segregated single-phase enclosure provided.
- j. MV switchgear technical data sheet is shown in the appendix.

6.3.2 Circuit breaker

- a. The circuit breaker design, construction, testing and application shall satisfy ANSI/IEEE C37.04 and IEC 62271-100 as minimum requirement unless otherwise stated.
- b. Circuit breaker shall be vacuum type (VCB) with highest switching duties, able to operate both electrically and mechanically as well as able to withstand any impacts during any harsh conditions.
- c. Circuit breaker short circuit rating (making, interrupting, and braking) shall satisfy maximum prospective short circuit at present and future.
- d. Circuit breaker short circuit capacity shall consider full DC component at present and future.
- e. Required interlocks (w/ disconnecting switch, earthing/grounding switch, MV door compartment) shall be provided to prevent dangerous operations of circuit breaker.
- f. Provide local manual close and trip function
- g. Provide operating mechanism to automatically recharge the closing spring
- h. Provide control power trip switch for disconnection of the control power circuit breaker.
- i. Provide local-remote selector switch
- j. Provide operation counter
- k. Provide emergency machine off push button
- l. Circuit breaker for transformer feeder shall be equipped with Inter-trip and close prevention interlocks w/ transformer door enclosure.
- m. Medium voltage circuit breaker technical data sheet is shown in appendix.

6.3.3 Disconnecting and Earthing/Grounding Switch

- a. Disconnecting/Circuit breaker and earthing/grounding switch shall be built-in three-position (Closed-Open-Earth/Ground) switch type. It shall have highest switching duties, able to operate both electrically and mechanically as well as able to withstand any impacts during any harsh conditions.
- b. Mechanically and electrically block access to three-position switch operation during circuit breaker is in the closed position to prevent dangerous operations.
- c. Provide dedicated operation handle for changing Closed-to-Open or Open-to-Closed
- d. Provide dedicated operation handle for changing Open-to-Earth/Ground or Earth/Ground-to-Open

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- e. All operations status shall be monitored and indicated on the MV door compartment and Power SCADA.
- f. Provide locking point of Earthing/Grounding switch
- g. Provide visual verification window to allow actual visualization switching changes.
- h. MV disconnecting and earthing/grounding switch technical data sheet is shown in appendix.

6.3.4 Protection and Metering

- a. The protective relay shall be multi-function type with built-in advanced microprocessor unit and modular communication module.
- b. The protective relay shall be able to operate continuously at minimum 10 years without any obsolete spare parts.
- c. For Incoming portion need to be equipped with power quality grade metering device. The metering shall be power quality measurement grade with capability of power quality analysis function. Multi-function relays shall be provided as per clause 6.2.5 (c)
- d. All protection devices and metering shall be provided communication protocol via IEC 61850 and linked through single-mode fiber optic network.
- e. Transformer differential protection function (87T) shall be provided if applicable.
- f. Line differential protection (87) shall be provided, if applicable.
- g. Undervoltage relay (27) shall be provided.
- h. Overcurrent phase fault relay with Instantaneous elements (50/51) shall be provided.
- i. Lockout relay (86) shall be provided.
- j. Synchronizing relay function (25) shall be provided.
- k. Provide IED that suitable for motor type load.
- l. Provide IED that suitable for transformers (Oil-Immersed) type load.
- m. Earth/Ground fault relay with Instantaneous elements (50N/51N) shall be provided.
- n. Provide CT shorting blocks for relay maintenance purpose.
- o. If required, inter-trip and interlocking scheme for various protection shall be provided.
- p. MV protection and metering devices technical data sheet is shown in appendix.

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- q. All protective devices shall be capable of handling Inrush currents under all service conditions, without abnormal ageing or deterioration.
- r. All protection settings and its logic to be reviewed and approved by Micron engineer for safe, reliable switching and operation with proper selectivity.

6.3.5 Voltage Transformer (VT) and Current Transformer (CT)

a. VT

- The VT rating and location shall satisfy ANSI/IEEE C57.13 and IEC 60044-2 as minimum requirement unless otherwise stated. Rated output shall match the maximum load of the equipment connected and a future capacity of 25%. Secondary voltage of transformer shall preferably be 110V.
- VT shall be inductive single-phase busbar type mounted outside or inside metal enclosure with gas insulation. VT's compartment shall be fully segregated from other primary compartments of the MV switchgear.
- The VT rating (Frequency, ratio, ampere, accuracy, insulation level, temperature rise) shall meet system design requirement. Voltage transformers for measuring purposes shall be of accuracy class 1 for third parties (Utility) supply, Class 3 for internal supply and Class 0.5 for use with generator AVRs.
- VT shall be equipped with three-position switch (Close-Open-Earth/Ground).
- VT primary leads shall be protected by current limiting fuse which located in same VT gas-filled enclosure. VT for use with generator AVRs shall not be protected by fuses but shall be connected within the protection zone of generator differential protection. VT secondary leads shall be protected by MCCB and connected to low voltage compartment. All connections with in the low voltage compartment shall be solderless connectors on suitable terminal block.

b. CT

- The CT rating and location shall satisfy ANSI/IEEE C57.13 and IEC 60044-1 as minimum requirement unless otherwise stated.
- CT shall be toroidal type mounted outside or inside metal enclosure with full insulation between the primary

conductor and current transformer core and secondary parts shall be provided by the SF6 gas insulation.

- The CT rating (Frequency, Ratio, Ampere, Accuracy, Insulation Level, Temperature Rise) shall meet system design requirement. Dedicated CT shall be provided for Protection relaying and Metering. CT for measuring purposes shall be accuracy class 3 except those for measuring the supply from third parties (Utility), which shall be of class 1.
- Ratio changing of CT shall be possible by means of taps on the secondary windings (Min. 2 Taps).
- All secondary and tap leads of CT shall be connected to integral CT lugs inside the low voltage compartment. All connections within low voltage compartment shall be solderless connectors on suitable terminal block and no slide-on connection acceptable. CT shorting blocks shall be provided for maintenance or troubleshooting purpose.
- Current transformers shall have an appropriate VA rating and a saturation factor that will ensure the proper function of protective devices for all short-circuit currents up to the rated value of the switchgear.
- Where required for certain type of differential protection, the CTs shall be specified in terms of knee-point voltage, excitation current and secondary wiring resistance so as to ensure stability under through fault conditions.

6.3.6 Control power

- a. Control power arrangement for closing, tripping and protection shall be either:
 - Combined DC supply unit/s with battery back-up for closing, tripping, protection and monitoring or
 - DC supply units with battery back-up for tripping and protection and AC supply for closing.
- b. Provide disconnect breaker at least two separate DC power supplies with battery back-up feeders on each MV switchgear section. DC power supply shall be monitored for failure. A maintenance/failure selector switch, make-before-break, and blocking diodes for each DC source shall be supplied at the bus section/coupler.
- c. MV switchgear control power shall be supplied via dedicated DC power supply which incorporates redundant chargers and redundant battery system with more than two strings. The charger and battery capacity shall have the capability of sustaining the worst operating conditions.

- d. Output voltage ripple of DC supply shall not cause interference with the other equipment of the assembly.
- e. Closing and tripping of switching devices shall not result in a voltage of less than the permissible voltage specified for the installed protective relays.
- f. Preferred tripping and protection voltage supply shall be 110V DC. Each supply unit shall be capable of closing or charging at least two switching devices simultaneously and all others in succession per assembly. In addition, battery of each supply unit shall be capable of supply the entire assembly control circuit load which include electronic relays supply for a minimum period of 8 hours and then able to trip all connected switching devices in succession.
- g. Closing/switching voltages for circuit breakers and disconnect/earthing (grounding) switch shall be 230V AC or 110V DC in combination with tripping and protection supply units.
- h. Two separate 230V AC source should be provide to switchgear. Selector switch (break before make) at the bus coupler should allow change-over to alternate feed, if one supply is unavailable or taken out for maintenance.
- i. Switchgear manufacturer shall provide high endurance (mechanical and electrical) and di-electric strength relays (include sockets), for final operational elements. This is to prevent abnormal operation due to faulty relays or sockets.

6.3.7 Automatic and Manual transfer operation

- a. MV switchgear shall be equipped control logic that allow both automatic transfer system (ATS) and manual transfer system (MTS).
- b. Provide Selector switch to choose between ATS or MTS at Tie Cubicle.
- c. Conditions taken into consideration when design ATS and MTS for MV Switchgear.
 - If MV switchgear bus coupler (Tie) is always in closed position (NC), then ATS function is not required. MTS is still required.
 - If MV switchgear bus coupler (Tie) is always in open position (NO) and LV switchgear bus coupler is equipped with ATS function, then MV switchgear only require MTS.
 - If MV switchgear bus coupler (Tie) is always in open position (NO) and LV switchgear bus coupler is NOT equipped with ATS function, then MV switchgear require both function ATS and MTS.

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- d. MV switchgear shall be able to perform Auto-Transfer and Auto-Restore during testing and troubleshooting when ATS function is available.
- e. Automatic transfer system operation shall meet all criterion prior to initiation.
 - Both Incoming circuit breakers (NC), tie circuit breaker (NO), and disconnecting switch (NC) are in normal position, in-service, healthy, and disconnected from earthing/grounding switch.
 - Power outage occurs at one side of Incoming line. This must be confirmed by at least two detections of undervoltage relays (27) at different point of measurement (Incoming line voltage and Bus voltage). Another Incoming line must be healthy, normal voltage, and free from bus fault or any fault.
 - After confirming Item 2 above, dead side Incoming line shall be opened. Associated dead bus must be health and free from any bus fault or any fault.
 - After item 1~3 are met then load transfer can be initiated by closing tie circuit breaker.
- f. Automatic restore operation shall NOT be allowed. Only Manual Restoration is available.
- g. Manual Restoration/Transfer Operation
 - System is normal with no protective function activated.
 - Outage supply line is healthy and ready to take load with line undervoltage relay (27) normalized.

6.3.8 Other accessories (Online Partial Discharge Monitoring, Surge Protection Devices, etc)

- a. Gas Leakage detection
 - An effective gas leakage detection system includes gauges, shall be provided for each compartment and connected to an alarm in Local control panel, Remote control panel, and Power SCADA/IMS.
- b. Cable Bushing
 - The cable bushing between HV switchgear feeder and primary of main transformer shall be supplied by HV switchgear vendor. The cable bushing should have separate oil and air insulated sections and serve as direct connection between the cable and main transformer.
 - The GIS connection comprises a housing which appropriately adapted to the power cable, connecting electrode and fitting for any misalignment and vibration.

The cable bushing gas space shall be combined and monitored with the gas compartment of the GIS module to which it is connected.

c. Cable Termination

- For power cable type and size of outgoings which GIS are connected; shall be shown and matched in design single line diagram. Power company feeder as Incoming power cables and terminations will be provided by approved and licensed local power company Contractor. The power cable from HV switchgear from feeders to main transformer shall be supplied by HV switchgear vendor/contractor. The connector shield rings, cable housing, mounting plated and necessary accessories required for the installation of all cable termination shall be provided by HV switchgear vendor/contractor. All details shall meet power company standard.
- Cable glands/docking plate and mounting for single core cables shall be non-magnetic material and insulated from framework and switchgear mounting plate.

d. Lightning Arrester

- The lightning arrester shall be station class, metal-oxide or zinc-oxide type and insulated with SF6 gas filled-housing. The rating of lightning arrested shall meet project specified requirement as shown in data sheet.

e. Online Partial Discharge Monitoring

- To detect early stage of partial discharge inside the MV switchgear gas filled-compartment, online partial discharge monitoring system (Online PDMS) is required to be installed. UHF-based detection technology or equivalent or higher is recommended. Comprehensive partial discharge analysis shall be done automatically and provide clear alarm/warning signal to integrating power monitoring system (IMS). The diagnostic capability of Online PDMS shall compliant with IEEE Std. 1434 and IEC 60270.

f. Surge Protective Devices

- Surge Protective Devices shall prevent electrical equipment from over-voltage transient events by blocking or divert surge current to ground instead of damaging the electrical equipment.
- Surge Protective Devices shall be connected to line side of equipment.
- Selection of Surge Protective Devices to ensure equipment subjected to momentary over-voltage and duration is

within the equipment withstand. The remaining energy is release to ground.

g. Blind cover

- The blind cover for each end shall be furnished. The covers will be used for blinding compartment during maintenance works.

h. SF6 Gas Filling Device

- One set of gas filling device including pressure reducing valve, manometer, adapter and filling hose shall be furnished for filling SF6 gas.

6.3.9 Installation

6.3.9.1 Transport, Storage and Erection

Assemblies supplied in transport units shall have these units clearly marked to facilitate assembly at site

Instruction for transport, storage and erection of the equipment shall be supplied by the Manufacturer as an integral part of the order.

Special tools and equipment required for erection, commissioning and maintenance shall form part of the order and shall be shipped together with the assembly.

6.3.9.2 General

Where necessary, owner or the Contractor shall submit Manufacturers' test reports, equipment certificates and site test reports, etc., for approval by the local authorities.

Unless specified, the minimum ingress protection for equipment shall be IP54 for outdoor installed equipment, IP41 for equipment installed indoors and IP2x for exposed parts of the equipment that has to be accessible during normal operation.

Foundation drawings, together with information on mass and loading data, shall be provided by manufacturer of Switchgear, to provide correct civil design of the relevant buildings and foundations. This include switchgear foundations, any inserts to be cast in, shall be within manufacturer's specific tolerances.

Switchgear arrangement and layout drawings shall be obtained from the manufacturer so that Contractor can incorporate into the overall arrangement drawings and cabling design.

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Switchgear shall be installed so that there is sufficient space, as defined by manufacturer, to facilitate operation, maintenance and constructability requirement. Refer to clause 6.1 (f).

Switchgear shall only be installed when the switch room/substation civil and building works are completed, so as to minimize the ingress of dust/dirt and moisture during or after erection.

Substation floors shall be smooth and level to permit the handling of equipment on rollers, regardless of crane facilities provided.

Concrete floors shall have a surface protection layer to avoid formation/creation of dust.

MV Switchgear shall be located in a separate room from the controls, protection and metering room.

An equipment/switchgear access door should lead directly to the outside for each switch room.

6.3.9.3 Site Installation

Switchgear shall be installed in accordance with Manufacturer's installation instruction and supporting drawings.

GIS switchgear shall be supervised and installed by the manufacturer competent person and/or manufacturer's trained/qualified persons. AIS switchgear shall at least be supervised by an experience qualified person/s familiar with the switchgear or by the manufacturer.

Switchgear modifications required during installation and/or commissioning shall be approved by the Owner prior to implementation. All modifications shall be recorded and information incorporated in the "as-built" drawings.

A site test certificate shall be issued by the manufacturer to confirm satisfactory completion of the recommended site test.

6.4 DUPS -Dynamic Uninterrupted Power Supply System (DUPS)

6.4.1 General requirement

Dynamic uninterruptible power supply devices (DUPS) combine the functionality of a battery-powered or flywheel-powered UPS and a diesel generator. When mains electricity supply is within specification, an electrical generator with a mass functions as motor to store kinetic energy in an electro-mechanical flywheel. In combination with a reactor or choke coil, the electrical generator also works as active filter for all sorts of power quality problems, like harmonics, RFI, and frequency variations. When mains electricity supply fails, stored energy in the flywheel is released to drive the electrical generator, which continues to supply power without interruption. At the same time (diesel engine should start before the flywheel stored energy is depleted which include the run-up time of diesel generator to establish voltage and switch-over process), the diesel engine takes over from the flywheel to drive the electrical generator to make the electricity required. The electro-magnetic flywheel can continue to support the diesel generator in order to keep a stable output frequency.

The main advantages of DUPS equipment compared to battery-powered UPS combined with a diesel-generator are the higher overall system energy efficiency, smaller footprint, use of fewer components, longer technical lifetime (no use of power electronics) and the fact it does not result in chemical waste (no use of batteries).

The main disadvantages of DUPS equipment are a more frequent maintenance regimen due to the number of moving parts. DUPS are also typically installed in external buildings due to noise concerns from the generators.

A DUPS can provide a ride-through time of 15 seconds above. A flywheel UPS can be installed ahead of typical UPS battery systems to reduce the effects of lightning & switching transients and to increase battery life.

- a. The maximum design demand load allowance shall be 80% of the kilowatt rating of the DUPS if power factor is < 1.0.
- b. The DUPS unit rating shall reflect its 100% service duty.
- c. The available fault current at the point of service will be reviewed and the DUPS system and its components shall have suitable withstand rating.
- d. The DUPS system and its components will meet all seismic requirements for the location where it is being installed.
- e. The level of noise produced by the DUPS, operating at rated voltage and frequency, shall be below limitations which required by local regulation.

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- f. If conditions warrant, the unit will be ordered with topicalization and fungus protection.
- g. The manufacturer shall install electrical components in the DUPS system in conformance with either NFPA 79 or IEC 60204.
- h. The DUPS system shall be self-contained and not be dependent on outside utilities for system operation.
- i. The DUPS system shall be designed for continuous operation and shall be capable of supplying startup transient, surges, and unbalance associated with the loads it serves.
- j. Electrical equipment and components shall be protected from intrusion of liquids. Solutions include rain caps, appropriate IP ratings, or other containment.
- k. A coordination study shall be performed to allow for selective coordination of the system protective devices.

6.4.2 Motor-Generator Diesel Engine

- a. The motor/generator shall be three-phase type, brushless, self-excited, self-regulating, self-cooling, and synchronous machine which can deliver the specified output continuously without exceeding the temperature and other limitations specified.
- b. The generator of DUPS system shall be started by either an electric starter by batteries which batteries shall be lead-acid or another form of stored energy starting system which shall be approved by Global facility in advance.
- c. DUPS system components shall meet performance and design criteria in generator specifications.
- d. Insulation level of generator shall be F class with average winding temperature rise at 80 deg C.
- e. Individual unit shall be equipped with an electronic voltage regulator. The generator and voltage regulator components shall be protected against voltage transients induced by switching or lightning surges.
- f. Paralleled units shall have automatic control logic for active and reactive load sharing, voltage adjustments, circuit breaker control, and sequencing. Automatic Voltage Regulation shall be provided in association with the governor and synchronizing system to allow for automatic paralleling and load sharing between multiple DUPS systems.
- g. If not operated in parallel, A and B distribution shall be used on the DUPS system.
- h. The DUPS systems shall utilize either a closed transition transfer switch or the Life Safety loads be protected by

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adequate SUPS to allow complete generator testing as outlined in NFPA 110: Standard for Emergency and Standby Power Systems.

- i. Design and installations shall conform to Factory Mutual (FM) requirements for Emergency and standby system as well as energy storage system.
- j. Engine must sustain overload condition for at least 60 minutes for 110% capacity.
- k. Generator shall be equipped with re-greasing double sealed ball/roller bearing capable of operating for at least 200,000 hours continuously.
- l. The room or enclosure for all the components of the DUPS system shall be cooled, as necessary, to prevent overheating under conditions of the highest anticipated ambient temperature at the installed elevation, when being operated under full load.
- m. The room or enclosure for all the components of the DUPS system shall be heated, as necessary, maintain the required temperature to allow for cold start and load acceptance by the system.
- n. The heating and cooling systems shall be sized to ensure that while operating during either normal or emergency conditions, the environment in the space or room for the DUPS system will be maintained within the DUPS manufacturer's ambient temperature specifications.
- o. Indoor installations shall have provisions to provide fresh air that meets the manufacturer's requirements for prime mover combustion.
- p. The fuel system shall be designed with a fuel strainer and filter to provide a clean fuel supply to the prime mover.
- q. Fuel tanks shall be sized large enough that the DUPS system can operate at full load for 8 hours minimum. Fuel shall be consumed within the storage life, provisions made to replace stale fuel with clean fuel, or the fuel filtered and treated to maintain the required quality. Fuel stored in cold climates shall be either blended or stored at temperatures at least 9.5 degrees C (15 degrees F) above the pour point of the fuel. Appropriate containment shall be provided for the fuel system.
- r. Adequate space shall be provided around the cooling system radiator on the prime mover for air circulation.
- s. The prime mover shall be located, or the exhaust diverted to avoid the exhaust fumes from the unit being introduced into a building's air handling system while the generator system is running.

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- t. Appropriate containment shall be supplied for the prime mover lubrication system and cooling system.
- u. For battery storage rooms with Wet Cell, lead-acid batteries, temperature shall be maintained at 25 degrees C (77 degrees F) to limit the production of hydrogen gas and ventilated to prevent an explosive accumulation of hydrogen gas.
- v. All auxiliary devices and mechanical equipment that maintains the conditions of DUPS system shall be served from AC UPS or DC Battery system.

6.4.3 Clutch

- a. Clutch should be maintenance free and dry type.
- b. The clutch shall provide seamless transfer from flywheel mode to generator supply mode and vice versa.
- c. The clutch shall be adequately flexible to transmit torque smoothly and to take up any angular or axial misalignment which may occur due to generator running clearances.

6.4.4 Flywheel/Energy Kinetic Unit

- a. The stored energy system in the DUPS system should be maintenance-free flywheel type. Batteries energy storage type and mixed battery-flywheel type shall be approved by global facility team in advance.
- b. The energy kinetic unit shall be manufactured by DUPS system manufacturer (1st). 2nd manufacturer of energy kinetic unit shall be approved by Global Facility team in advance.
- c. Kinetic energy stored shall be capable of momentary delivering the full load via the alternator until the diesel engine runs up (nominal speed and rated voltage established) to handle the load.
- d. Once the diesel engine accelerated to supply the load, it will at the same time charge the kinetic Unit (stored energy). Manufacture to perform test to demonstrate the discharge and recharge of the Kinetic unit meets the requirements/specifications.

6.4.5 Switchgear, step-up transformers (if applicable), and bypass switches

- a. DUPS switchgear shall be gas-insulated switchgear type for MV application and air-insulated switchgear type for LV application.
- b. DUPS switchgear shall consist of redundant incoming lines, outgoing lines, static bypass lines, and manual bypass lines. Additional up-stream, down-stream or tie-breakers can be added, if required.
- c. The DUPS system will switch to bypass automatically and without interruption in an event of an internal failure in the system.

6.4.6 Control panel

- a. Control panel for DUPS shall have simple to operate, controls and protective functions both on analogue or digital Human Machine Interface (HMI). A complete overview of the condition/status of the machine shall be provided, including fault events, transfer status, alarms, etc., Remote access and interface with Scada shall also be provided.
- b. Controller shall include but not limited to the following:
 - Keypad driven menu shall view system measurements and interact with the system.
 - Generator voltage and current reading and control
 - Active and reactive power measurements and control
 - Diesel engine speed and controls
 - Induction coupling control
 - Number of Diesel Engine starts
 - Real time information of system and list of historical events/status that changes state with time stamped.
 - Fuel tank level and control.
- c. These are the following alarms/status and trip functions:
 - Generator over frequency
 - Generator underfrequency
 - Generator overvoltage
 - Generator undervoltage
 - Start failure
 - Unintended Stop
 - Mains rotating field mismatch
 - Maintenance run hours
 - Generator breaker close failure

- Generator breaker open failure
- Main breaker close/open failure
- All circuit breakers status

6.4.7 Control power

All supply sources for supply security shall be taken from the DUPS output to the critical load. Supply can also be taken from nearest secure DC power supply, provide location is in close proximity.

6.4.8 Installation

- a. Power module should preferable delivered pre-mounted on robust base frame. For specific site constrain/condition, include lifting and installation, split frames can be supplied.
- b. Mounting of engine exhaust, fuel and cooling system shall be similar to installation works for emergency/stand-by diesel generator.
- c. UPS shall be installed with sufficient clearance for ventilation and access for maintenance.
- d. Power cables especially single core cables should be segregated and or shielded from controls and signaling cables to avoid electromagnetic interferences (EMC).
- e. UPS shall be as close to the critical load as possible to reduce the risk of damage.
- f. Motor-Generator (M-G) and control cabinet assemblies shall be installed on a level surface and secured to the floor or frame as recommended by manufacturer.
- g. Access to all components in cabinets installed shall ensure ease of maintenance and replacement. This include side and rear access, if components cannot be replaced from the front. Refer to manufacturer recommendation for minimum clearance required.
- h. Earthing/Grounding and bonding of dynamic rotary UPS shall not be effective for protection but also shielding against EMC and static charge.
- i. Battery installed shall be in well ventilated area/room and temperature controlled so that capacity derating or lifespan should not be compromised. Also refer to static UPS battery (6.5.3).
- j. Installation checks shall be performed prior to powering up the system for testing. Field inspection include components nameplate, inspection of insulators for crack/damage, components for damage/cleanliness, alignment, anchorage and grounding, fuses and circuit breakers, mechanical operation

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and interlock, insulation liquid/gas level/pressure, lubrication type and level, verification of safety equipment in place, etc.,

- k. Start-up and testing/commissioning shall be performed to ensure components and system functionality. This include site calibrations and remote alarm/controls check.

6.5 SUPS – Static UPS

6.5.1 Specific requirement

- a. On-line double conversion UPS shall be provided.
- b. The maximum design demand load allowance shall be 80%.
- c. The system size shall be based on the estimated total steady state load, load power factor, inrush current requirements of the load, and enough run time in the event of an outage to protect the Critical load.
- d. This specification is intended for 3 phase UPS units, 30 kVA and larger. UPS system characteristics and selection criteria for Critical load shown as below:
 - Selection criteria for a load being placed on UPS is safety, production, and economics.
 - Process tools and associated equipment must meet the following criteria to qualify for connection to UPS power. Where a loss of power:
 - Endangers employee safety or cause occupational illness or physical injury. Examples include: FAB lighting or other select emergency lighting.
 - Would cause environmental impact.
 - Will cause significant scrapped wafers, significant rework of multiple wafers, could cause damage to the tool, or require more than 4 hours to restart and re-qualify. Examples include: Diffusion Furnaces , Wet Hoods (controller only), Implanters, Photo clusters (Track and Scanner, excluding EUV scanners), SEM Tools and RDA Tools. Single Wafer Tools not meeting these criteria shall not be placed on UPS systems.
 - Will cause impact to Facilities systems, that in turn will directly impact Production. Examples include: VOC Controls, Process Cooling Water Pumps, PLC Cabinets, Pollution Abatement, Select Instrumentation and Control devices, Toxic Gas Monitoring Systems, Security System components, and IS Machine Room equipment.
- e. Equipment shall be capable of operating in an area where the temperature is 0 degrees C to 40 degrees C (32 degrees F to

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104 degrees F), with Relative Humidity 0% to 95% (non-condensing) without being derated.

- f. The heating and cooling systems shall be sized to ensure that while operating during either normal or emergency conditions, the environment in the space or room for the UPS system will be maintained within the UPS manufacturer's ambient temperature specifications.
- g. Electrical equipment and components shall be protected from intrusion of liquids. Solutions include rain caps, appropriate IP ratings, or other containment.
- h. The UPS System shall be capable of operating from either Utility power or power supplied from a standby source.
- i. Units shall be self-cooling or use self-contained forced air-cooling fans.
- j. The available fault current at the point of service will be reviewed and the UPS system and its components shall have suitable withstand rating.
- k. The UPS System and its components will meet all seismic requirements for the location it is being installed at.
- l. Output transformers (where used) shall be electrostatically shielded.
- m. If conditions warrant, the unit will be ordered with tropicalization and fungus protection.
- n. Safety equipment, including, but not limited to interlocks for by-pass switches and shielding or guarding of live components, shall be provided.

6.5.2 Rectifier & Inverter, Static by-pass, and Maintenance by-pass

- a. Power source to Rectifier/Inverter shall be independent line and backed by generator.
- b. Power source to Static by-pass and Maintenance by-pass shall be independent line. Not required backed by generator.
- c. The SUPS system will switch to bypass automatically and without interruption in an event of an internal failure in the system.
- d. For UPS and other power quality improvement devices especially those use the DSP(Digital Signal Product), check with OEM on the possibility of malfunction due to internal data corruption and any software validation code to detect data corruption and switch the equipment to safe mode without load interruption.
- e. Pulse-width modulated, insulated-gate bipolar transistor (IGBT) for Inverter and twelve pulse SCR for Rectifier.

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- f. Output voltage shall be +/- 1% steady state, frequency regulation +/- .1%, with < 5% input THD at full load, voltage distortion of 4% maximum for non-linear loads, and 2% maximum for linear loads at the output.
- g. Transient voltage response shall be < 5% from nominal voltage for 100% load step, full load re-transfers, and full load drop on battery.
- h. Power Factor to be 0.95 lagging minimum with full load and nominal input voltage.
- i. The minimum overload rating of the unit will be 125% of rating for ten minutes and 150% of rating for 10 seconds.
- j. The UPS system shall operate continuously within the specified voltage and frequency limits, from no-load to full-load conditions without thermal degradation.

6.5.3 Battery system

- a. The battery shall be UL recognized.
- b. Highly recommended Lithium-Ion Battery (LIB, or Battery hereafter) to be used and refer to "Global Facilities - Electrical BKM-Lithium -Ion Battery for SUPS Back-up".
- c. Primary and secondary connection of battery system shall be equipped with DC breaker type MCCB/MCB to allow protection and isolation during normal and maintenance operation.
- d. ~~Valve regulated lead acid (VRLA) or wet cells to be used switchgear control applications.~~
- e. Sealed battery system shall be maintenance-free, 10 years minimum life-time, and can be replaced easily without any interruption to UPS operation.
- f. Liquid electrolyte/flooded/wet-cell lead acid battery type is less preferable for UPS application and to be provided and approved by Global facility team in advance. Wet cells are recommended for switchgear controls and engine starting applications.
- g. Flooded cell battery system shall be maintenance-free except for electrolyte part, 20 years minimum life-time, and can be easily refilled without any interruption to UPS operation.
- h. Rooms that house flooded cell batteries shall be exhausted or sufficient air changes shall be made to reduce hydrogen gas accumulations.
- i. Ventilation equipment serving rooms that house flooded cell batteries shall be powered from a source backed by generator.
- j. Battery system/rack shall be segregated and isolated for each UPS system to avoid catastrophic problem.

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- k. A minimum of two battery string with individual disconnecting means shall be installed to reduce the risk of a single string failure.
- l. Battery racks or cabinets shall be installed, anchored, and seismically braced per the manufacturer's installation directions and recommendations and the contract documents.
- m. Battery racks shall be constructed to afford it protection from corrosion; any coatings or materials used shall be flame resistant.
- n. Containment systems shall be provided for battery strings, IEEE Recommended Practice for Stationary Battery Electrolyte Spill Containment and Management (Std. 1578) shall be used as a reference for selection and performance criteria for new battery systems.
- o. Rooms housing storage batteries shall be constructed with using the appropriate fire rating.
- p. Rooms housing storage batteries as well as UPS/DC system shall be monitored with either heat or smoke detectors monitored by the building fire alarm system or protected by an approved fire suppression system.
- q. Adequate clearances shall be maintained around all equipment and personnel protected from Live Parts. Adequate paths of Egress shall be provided and maintained in rooms with storage batteries.
- r. Battery jars shall be constructed with high impact-resistant, self-extinguishing material.
- s. Valve Regulated Lead Acid battery (VRLA) systems shall be provided with a listed device or other monitoring method to preclude, detect, and control thermal runaway. IEEE Guide for Selection and Use of Battery Monitoring Equipment in Stationary Applications (Std. 1491) shall be used as a reference for selection and performance criteria for new battery systems.
- t. Provide online battery monitoring system on the battery banks to monitor each individual battery cell condition (DC Voltage, internal resistance, temperature, etc). The online battery monitoring system shall monitor battery parameters and generate alarms that are linked to SCADA system.
- u. Battery breaker status shall be monitored by either UPS or by SCADA system.
- v. UPS to have full monitoring capability and monitored through the SCADA system for the early detection of internal issues.

6.5.4 Switchboard

- a. Supplies for critical services shall, if required by the load, be uninterrupted on failure of one energy source.
- b. Two totally different AC supply sources shall separately supply to each SUPS. Supply sources should preferable from different section of the upstream switchboard. This is to allow one supply (UPS/bypass) to be taken out for maintenance.
- c. Means of isolation, protection and clear identifications shall be provided for all power supply to the SUPS at both ends.
- e. Downstream switchboard of SUPS (critical load switchboard) shall be capable of continuous operation on full load and short circuit withstand above the SUPS outgoing protection. The largest protective device (fuse/MCCB) for this switchboard, when activated due to a fault, shall not affect the SUPS including the transfer to bypass or shutdown.

6.5.5 Control panel

- a. UPS shall incorporate all the necessary equipment to enable operation, protection and control of the UPS accordance with this specification. To safeguard the unit and its components from the consequences of internal and external short circuits, overvoltage and any main or control malfunctions.
- b. Operating, status and diagnostic indicators shall be provided on a human-machine interface, such as liquid crystal displays (LCD) or equivalent, with keypad driven menu and light-emitting diodes (LEDs)
 - Keypad will be used to view system measurements and interact with the system.
 - Alarms and indication panel displays possible faults and alarms (with respective LED)
 - Unit will have self-diagnostics capabilities, event logging with capability of communication to Scada. Preferable communication protocol to meet IEC 61850 or similar with installed Scada.
- c. Each LED/alarm circuit shall have in-built test facilities, with the exception of LED associated with PCBs. Failure of an LED/LCD shall not cause UPS maloperation or affect the correct functioning of the remote common alarm signal.

6.5.5.1 Control Panel indications/status:

At least these following indications and measuring instruments, shall be provided on the front outside panel of the unit to verify the operating status of the UPS. Indications shall be superimposed on a mimic diagram of the UPS to identify the relevant components of circuit. In addition these indications should be available including to remote signaling in Scada.

- AC input supply available;
- Alternative/Bypass supply available;
- Rectifier ON
- Inverter ON
- Load on inverter
- Load on bypass
- Inverter & bypass synchronized
- Battery on boost/high rate charge

6.5.5.2 Measurement:

All measuring instruments shall be digital.

This following information shall be provided on the front panel LCD display and remote signaling to Scada:

- DC circuit voltage;
- Battery charge and discharge current;
- UPS output voltage per phase;
- UPS output current per phase;
- UPS output frequency;
- Battery Autonomy Time remaining

6.5.5.3 Protection and Alarms

All alarms with trip functions shall be hand reset. Protection functions and alarm status shall be visually verifiable by appropriate indicators with first failure feature.

These are the following required signals:

Signals	Actions
Rectifier Failure	alarm and trip rectifier
DC overvoltage	alarm and trip rectifier
DC undervoltage	alarm and trip rectifier
Inverter failure	alarm and trip rectifier

Signals	Actions
Inverter output voltage deviation	alarm and trip rectifier
Cubicle/inverter stack overtemperature	alarm and trip rectifier
Cubicle fan failure	fan failure alarm & pre-alarm
AC input phase undervoltage	alarm only
Charge failure/battery discharging	alarm only
Battery disconnected	alarm only
Inverter overcurrent	alarm only
Inverter/bypass not synchronized	alarm only

Above alarms shall be connected to a terminal block so as to provide one normally open and one normally closed, potential-free, contacts for remote cabling to a common alarm. In addition, the remote signalling via interface to Scada should be available.

6.5.6 Control power

Control power for UPS shall be taken from mainly from secured DC bus and should be design by UPS vendor to ensure safe and reliable operation of the UPS unit.

External power supply (DC preferable) for alarms shall be wired to UPS's relays (alarm) potential free contact/s.

6.5.7 Installation

- UPS units will be installed on 100 mm (4 in.) housekeeping pads.
- The UPS System and its components shall be in areas accessible to only trained, qualified personnel.
- Locations of UPS units will be reviewed, and appropriate action taken if there is vibration-sensitive equipment located nearby.
- Move-in paths for UPS units shall be maintained to allow removal of the unit without complete disassembly.
- Maintain enough clearances to allow for maintenance and ventilation to the UPS unit and its components.

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- f) All systems foreign to the UPS system shall be kept from entering or passing through dedicated rooms or spaces or over the equipment. Mechanical, control, and fire-suppression systems shall serve the UPS room or space only.
- g) Battery terminals & connectors shall be protected against inadvertent contact by anyone.
- h) The main connections between the rectifier/inverter units, battery system, static bypass, maintenance bypass and outgoing switchboard shall be provided by PVC or XLPE insulated cable of adequate mechanical strength and current carrying capacity and terminating in two ring type terminations on the battery, unless other method of mechanical protection is used.

6.6 Uninterrupted DC Power Supply (DC UPS)

6.6.1 Specific requirement

- a) DC battery bank/power supply shall specifically design to be used to provide power supply to protection, control, and monitoring for substation critical electrical equipment (HV/MV/LV).
- b) Highest performance and reliability of DC battery system/power supply must be provided.
- c) Each major component of DC battery system/power supply (Incoming supply, charger, DC bus, battery system) has its redundancy, flexibility and fault tolerant design which allowing future load transferring, troubleshooting, maintenance, and expansion.
- d) The type of DC source shall be based on the duration of protection required, reliability, cost analysis, and space constraints.
- e) IEEE Guide for Batteries for Uninterruptable Power Supply Systems (Std. 1184) shall be used as a reference for selection and performance criteria for new battery systems.
- f) Overcurrent protective devices will be provided to protect the DC source from overloads or short circuits.
- g) The available DC short circuit fault current at the point of service will be reviewed and the DC battery system and its components shall have suitable withstand rating.
- h) DC components, associated raceways or wireways, and racks or supports shall be grounded and bonded to provide an effective path for ground-fault circuits, bonding of electrical equipment, and safety of personnel.

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- i) A minimum of two battery string with individual disconnecting means shall be installed to reduce the risk of a single string failure.
- j) Battery load test and recharging provision to be provided for the strings or entire bank.
- k) 100V or higher rated battery string shall be used for switchgear protection control power to reduce the risk of under voltage.
- l) Mechanical systems shall provide heating and cooling to maintain an ambient temperature of 25 degrees C (77 degrees F), +/- 5 degrees C (+/- 9 degrees F) for areas where storage batteries are located.
- m) Ventilation shall be provided to rooms or cabinets housing batteries to sufficiently diffuse and ventilate the batteries to prevent the accumulation of combustible vapor.
- n) If adequate ventilation cannot be provided, a gas detection system designed to activate when flammable gas levels exceed 25 percent of the lower flammable limit (LFL) or where the level of toxic or highly toxic gas exceeds 1/2 the Imminent Danger to Life and Health (IDLH) levels. Alarms shall sound locally to evacuate the area and at the Main Control Room via the Facilities Monitoring Control System (FMCS).
- o) Provide emergency or standby power for space conditioning in areas where temperature or humidity are critical in maintaining operation of electrical equipment serving life safety equipment.
- p) Electrical equipment and components shall be protected from intrusion of liquids. Solutions include rain caps, appropriate IP ratings, or other containment.

6.6.2 Rectifier/Charger system

- a) Power source to rectifier/charger shall be independent line and backed by generator.
- b) Pulse-width modulated, six pulse SCR/ IGBTs, Rectifier with isolation transformer.
- c) Output DC voltage tolerance shall be maintained at +/- 2% for changes in rated line frequency of +/- 5%, in rated voltage of +/- 10%, and in load from 0-100% of rated. The charger shall maintain +/-0.5% or better regulation for combined variations of line, load, and frequency.
- d) The minimum overload rating of the unit will be 125% of rating for ten minutes and 150% of rating for 10 seconds.

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- e) The DC charger system shall operate continuously within the specified voltage and frequency limits as mentioned in c), from no-load to full-load conditions without thermal degradation.
- f) The DC charger shall be a "battery eliminator" type with additional filtering to allow the charger to be operated into connected loads without the battery. The charger in the eliminator mode shall have an output ripple less than 30mV RMS.
- g) Unit will have self-diagnostics capabilities, event logging, and local LCD or LED display.

6.6.3 Battery system

See section 6.5.3 Battery system

6.6.4 Switchboard

For DC UPS supplies security and maintainability, AC supply sources shall be feed for different sections of the LV switchboard. Means of isolation and protections for AC supply sources shall be provided at both ends connecting to the DC UPSs.

DC earth/ground fault monitoring shall be provided on DC output busbar or circuit. There should only be one voltage-divider type of earth/ground leakage monitor on each DC UPS.

Selector switches with blocking diodes shall be provided for each outgoing DC circuits at the equipment end (e.g. switchgear control panel).

DC distribution board with means of isolation, protection and earth/ground leakage detectors shall be provided if DC UPS is supplied to many outgoing loads.

Protections for DC load should be sized within the capability of any line components of the UPS, such that in any event of a short circuit on the load, the respective protection should activate, isolate and no damage or deterioration should happen to any of the UPS components.

6.6.5 Control panel

DC UPS shall incorporate all the necessary equipment to enable operation, protection and control of the UPS accordance with this specification. To safeguard the unit and its components from the consequences of internal and external short circuits, over-voltages and any main or control malfunctions.

Operation, status and diagnostic indicators shall be provided on a human-machine interface, such as liquid crystal displays (LCD) or equivalent, with keypad driven menu and light-emitting diodes (LEDs)

- Keypad will be used to view system measurements and interact with the system.
- Alarms and indication panel displays possible faults and alarms (with respective LED)
- Unit will have self-diagnostics capabilities, event logging with capability of communication to Scada. Preferable communication protocol to meet IEC 61850 or similar with installed Scada.

Each LED/alarm circuit shall have in-built test facilities, with the exception of LED associated with PCBs. Failure of an LED/LCD shall not cause UPS maloperation or affect the correct functioning of the remote common alarm signal.

a) Control Panel indications/status

At least these following indications and measuring instruments, shall be provided on the front outside panel of the unit to verify the operating status of the UPS. Indications shall be superimposed on a mimic diagram of the UPS to identify the relevant components of circuit. In addition these indications should be available, including to remote signalling in Scada.

- AC input supply available;
- Rectifier ON
- Load ON
- Battery on boost/high rate charge

b) Measurement

All measuring instruments shall be digital.

This following information shall be provided on the front panel LCD display and remote signaling to Scada:

- AC supply voltage;
- Battery charge and discharge current;
- UPS output voltage;
- Battery Autonomy Time remaining

c) Protection and Alarms

All alarms with trip functions shall be hand reset. Protection functions and alarm status shall be visually verifiable by appropriate indicators with first failure feature.

These are the following required signals:

Signals	Actions
Rectifier Failure	alarm and trip rectifier
Rectifier output voltage deviation	alarm and trip rectifier
Cubicle fan failure	fan failure alarm & pre-alarm
AC input undervoltage/overvoltage	alarm only
Battery discharging	alarm only
Battery disconnected	alarm only
Rectifier overcurrent	alarm only
DC earth/ground fault monitoring (both +ve & -ve)	alarm only

Above alarms shall be connected to a terminal block so as to provide one normally open and one normally closed, potential-free, contacts for remote cabling to a common alarm. In addition, the remote signaling via interface to Scada should be available.

6.6.6 Control power

All control power for the DC UPS unit shall preferable taken from the DC output with battery backup to prevent loss of power. Any alternative source of control power shall be agreed by the Owner and manufacturer shall prove that supply is secured for all operational modes, include power supply outage.

6.6.7 Installation

- To prevent personal injury or equipment damage, take extreme care when handling and transporting UPS cabinet and equipment.
- Unpack, lift and transport UPS according to manufacturer instruction.
- Ensure all covers, doors are in place and internal cover/s secured before attempting to lift or transport the system.
- Check all tools and equipment required the installation of the UPS are available, as per the manufacturer's instruction manual.

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- e. Verify nameplate/type labelled and system ordered, include check on input/output voltages and rating.
- f. Confirm cabinet/UPS and accessories (battery/racks) dimensions as design layout/footprint. Ensure compliance of minimum free access, maintenance and ventilation free space meets the local regulation, specification and manufacturer requirement.
- g. Connect the UPS and check voltages and phase rotation (for 3 phase unit) of mains input voltage.
- h. Check earthing/grounding system and earth/ground connection before powering up the UPS.
- i. Connect all the battery cables, output/load cables, alarm cables, etc., before powering up UPS and commissioning it in accordance with drawings/instruction manual provided.

6.7 Transformers

6.7.1 General requirement

- a) The heating and cooling systems shall be sized to ensure that while operating during normal conditions, the environment in the space or room for the transformer will be maintained within the transformer manufacturer's recommended ambient temperature specifications.
- b) Adequate space shall be provided around the transformer to allow for air circulation and natural ventilation.
- c) Transformers shall be protected against potential condensation, leaks, spills, or other liquids.
- d) Fire suppression systems shall meet all applicable local codes and standards.
- e) Equipment shall be braced to meet seismic requirements and regulations.
- f) Transformers serving non-sinusoidal loads shall be designed with adequate K factor ratings.
- g) The enclosure and fittings shall provide adequate protection from the environment where it is installed.
- h) Transformer design shall meet seismic requirements for the locale where it is being installed.
- i) Transformers shall be suitable for switching by all types of switching devices, i.e., vacuum or gas breakers without detrimental effect on the windings.
- j) The manufacturer shall be responsible to ensure the unit has been fully derated to function in the site conditions.
- k) Structure loading shall be reviewed before selection and placement of transformers.

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6.7.2 Gas or Liquid-filled type HV main transformer

6.7.2.1 Specific requirement

- i. kVA ratings of transformers specified in data sheets are continuous and based on not exceeding the temperature rise limits.
- ii. kVA rating is based on rated secondary voltage, being maintained by impressing on the primary tap indicated a voltage higher and its rated voltage to compensate the effect of regulation of the transformer.
- iii. The transformer shall deliver rated kVA continuously at 105% voltage and operate continuously at 10% above rated voltage at no-load, without exceeding the specified temperature rises.
- iv. When force-cooled ratings are specified, all current carrying parts including bushings shall be sized for highest rating.
- v. Oil-immersed power transformer shall use FR3 fluid type or Less Flammable Dielectric Liquid by FM Global (FM) and Underwriters Laboratories® (UL) for use in complying with the National Electric Code® (NEC) and insurance requirements. for its Tank and OLTC Tank.
- vi. On-line DGA monitoring system with full gas spectrum is required.
- vii. Average temperature rise by resistance test for the oil-immersed power transformer shall not exceed 65 deg C based on an average ambient of 30 deg C for any 24-hour period.
- viii. The transformer impedance is measured at self-cooled rating and shall be specified in data sheet.
- ix. Core Iron loss at fundamental frequency 50/60Hz shall be preferably less than 1.25 watts/kg.
- x. The level of noise produced by the transformer, operating at rated voltage and frequency, shall be below limitations which required by NEMA TR-1.
- xi. Transformer windings and core shall be braced to be able withstand without damage thereunto the electromagnetic stresses produced by the maximum short circuit current of 2 Second.
- xii. Insulation voltage class and related BIL-basic impulse insulation level for each transformer shall be specified in data sheet.

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- xiii. The core shall be built up of cold rolled, oriented grain non-aging silicon steel laminations having a high permeability and low hysteresis loss.
- xiv. The core shall be rigidly clamped and braced to resist distortion caused by short circuit stresses or transportation handling, and to prevent the shifting of core laminations with the resulting deterioration or damage to winding insulation and laminations.
- xv. Winding conductors shall be Cooper material. It shall be free from scale, burrs and splinters, properly rounded corners and uniformly insulated.
- xvi. Turns shall be wound, and coils shall be formed and braced so that finished windings are rigid, capable of withstanding short circuit stresses without deflection injurious to the insulation, and free from the hazards of sagging or slipping of turns.

6.7.2.2 Ratings

- i. The application/duty of the transformer will be mentioned in the requisition. Example generator transformer, motor unit transformer, etc.,
 - ii. Power transformer use for supply to group of motors shall be capable of withstanding in frequent restarting loads of up to 1.8 times the rated transformer current.
 - iii. IEC60076-1 or IEEE C57-12 or equivalent standard rated transformer should be preferred.
 - iv. Percentage Impedance voltage based on rated power should be as follows, unless an overriding design requirement.
- Impedance voltages for transformers rating above 25,000kVA shall be specified in the requisition, depending on system design and/or system studies

Rated Power (kVA)	Impedance Voltage (%)
Up to 630	4.00
631 to 1,250	5.00
1,251 to 3,150	6.25

Rated Power (kVA)	Impedance Voltage (%)
3,151 to 6,300	7.15
6,301 to 12,500	8.35
12,501 to 25,000	10.00

v. Transformer and System Voltage

- o Rated voltage on the high voltage side is generally equal to the system nominal voltage in which the high voltage winding is connected.
- o The correlation between the rated voltage (at no load) of the low voltage winding and the corresponding system voltage is as below:

System nominal Voltage (V)	Rated Secondary Voltage (No Load)
200	208
208	214
380	400
400	420
415	433
440	460
480	504
3300	3450
6600	6900
11000	11500
33000	34500
66000	72500

vi. Nameplate Details:

Rating plate shall be stainless steel or others that is corrosion resistance metal. It should be permanently fixed to a readily accessible and non-removable part of the transformer, such as cooling fins.

Nameplate shall have the following information:

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- o Name of Manufacturer
- o Serial number
- o Order No./Project number
- o Rated Capacity: kVA/MVA Rating
- o Primary voltage/s at various transformer tapping/connections
- o Secondary voltage
- o Full load currents (Primary & Secondary)
- o Rated Frequency
- o Number of Phases
- o Vector group (connection/displacement)/Diagram
- o Connection Diagram
- o Impedance rating/Percentage Impedance
- o Cooling of transformer (e.g. ONAN/ONAF)
- o Protection category/type (e.g. NEMA 1 Enclosure/IPxx)
- o Quantity of insulating gas/liquid (litres/gallons of oil) and type
- o Insulation/temperature class of winding insulation.
- o Manufacturer tested date
- o Total weight (kg)

6.7.2.3 On-load Tap Changer (OLTC)

- i. Main transformer shall be equipped with integrated on-load tap changer placed in the primary windings. Each tap of OLTC shall be continuously operated under +/- 2.5% variation of Primary incoming voltage.
- ii. Overall tapping range +/- 1.25% (+/- 17 steps)
- iii. Maximum step voltage 2.5%
- iv. Tolerance over tapping range +/- 0.5% of rated voltage
- v. The tap changer shall be installed in separate oil compartment in the transformer casing
- vi. The OLTC shall have local and remote control with local manual operating devices. When the manual operating device is in use, the remote control system shall not influence the tap changer.

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vii. The tap changer shall be capable of automatic or manual control. It shall be mounted on the transformer casing and at least consist of: electrical supply, electrical motor, drive shaft for tap changer, mechanical counter w/ reset button, auxiliary relays/contractor/limit switches, space heater w/ thermostat.

6.7.2.4 Other main transformer parts and accessories

- i. Bushings
- ii. Terminal Chamber
- iii. Tank
- iv. Oil preservation system
- v. Insulation Oil
- vi. Forced cooling device
- vii. Pressure relieve device
- viii. Bucholz relay or Sudden pressure relay
- ix. Gas detector relay
- x. Gas sampling device
- xi. Common terminal box
- xii. Liquid level gauge
- xiii. Oil thermometer
- xiv. Pressure-vacuum gauge and bleeder
- xv. Globe type combination drain and upper/lower filter valve with sampling device
- xvi. Lifting eyes for cover
- xvii. Facilities for lifting core and coil assembly from tank
- xviii. Lugs for lifting, jacking facilities, moving facilities.
- xix. Tank grounding pads
- xx. Nameplates
- xxi. On-line oil purifier and filter equipment
- xxii. On-line Partial discharge measurement sensor and system (if required)
- xxiii. On-line dissolved gas analysis monitoring.

6.7.2.5 Control panel (OLTC)

- i. Tap changer remote control panel shall be located indoor. Tap changer local panel outdoor use with enclosure protection IP55.
- ii. A remote-control panel for remote and automatic operation and indication of tap changer shall be supplied by qualified maker that proposed by Transformer manufacturer and approved by local authority company.
- iii. Control panel shall be equipped with important apparatus such as: remote-local change over switch, push button for tap position 'raise/lower', manual tap changing mechanism, mechanical tap position indicator, AVR relay, indication lamp, PTT, CTT, etc.
- iv. AVR-Automatic Voltage Regulating relay shall be intelligent electronic type with high adjustable capability for set point range, sensitivity, pick-up time, under voltage blocking and line drop compensator.

6.7.2.6 Control power

- i. Control power shall be supplied via DC power supply which specified in datasheet and details in section 3.6

6.7.3 Liquid-filled/Dry-type cast resin type MV distribution transformer

6.7.3.1 Specific requirement

- i. kVA ratings of transformers specified in data sheets are continuous and based on not exceeding the temperature rise limits.
- ii. kVA rating is based on rated secondary voltage, being maintained by impressing on the primary tap indicated a voltage higher and its rated voltage to compensate the effect of regulation of the transformer.
- iii. The transformer shall deliver rated kVA continuously at 105% voltage and operate continuously at 10% above rated voltage at no-load, without exceeding the specified temperature rises.
- iv. When forced-air cooled ratings are specified, all current carrying parts including bushings shall be sized for highest rating.

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- v. Oil-immersed distribution transformer shall use less flammable FR3 fluid.
- vi. Average temperature rise by resistance test for the oil-immersed distribution transformer shall not exceed 65oC based on an average ambient of 30oC for any 24-hour period.
- vii. Average temperature rise by resistance test for the dry-type distribution transformer shall not exceed 80oC based on an average ambient of 30oC for any 24-hour period.
- viii. The transformer impedance is measured at self-cooled rating and shall be specified in data sheet.
- ix. The level of noise produced by the transformer, operating at rated voltage and frequency, shall be below limitations which required by NEMA TR-1.
- x. Transformer windings and core shall be braced to be able withstand without damage thereunto the electromagnetic stresses produced by the maximum short circuit current of 2 Second.
- xi. Insulation voltage class and related BIL-basic impulse insulation level for each transformer shall be specified in data sheet.
- xii. The core shall be built up of cold rolled, oriented grain non-aging silicon steel laminations having a high permeability and low hysteresis loss.
- xiii. The core shall be rigidly clamped and braced to resist distortion caused by short circuit stresses or transportation handling, and to prevent the shifting of core laminations with the resulting deterioration or damage to winding insulation and laminations.
- xiv. Winding conductors shall be Cooper material. It shall be free from scale, burrs and splinters, properly rounded corners and uniformly insulated.
- xv. Turns shall be wound, and coils shall be formed and braced so that finished windings are rigid, capable of withstanding short circuit stresses without deflection injurious to the insulation, and free from the hazards of sagging or slipping of turns.

6.7.3.2 Ratings

(Refer to 6.7.2.2)

6.7.3.3 Accessories

- i. Provide fan cooling equipment (fans, power/control wiring, Local selector switch, controller, fuse, indicating light, relay)
- ii. Three-phase winding temperature indicator with contacts to control fans and alarm contacts
- iii. Provide Class B Stranded Type SIS power/control conductors
- iv. Copper ground pads
- v. Power monitoring connection
- vi. Nameplates
- vii. Grounding resistor as required by designer and specified in data sheet
- viii. Bushing Current Transformer type as required by designer
- ix. For liquid filled distribution transformer, shall provide additional item as minimum.
 - o Pressure-vacuum gauge with bleeder
 - o Liquid-level gauge with alarm contacts
 - o Dial-type temperature gauge with alarm contacts
 - o Drain valve with sample port
 - o Fill plug
 - o Pressure relief device
 - o Sudden pressure relay
 - o Winding Temperature relay

6.7.3.4 Control power

- a) Self-powered from secondary bushing single phase

6.7.4 Dry-type cast resin type LV distribution transformer

6.7.4.1 Specific requirement

- i. Transformers shall be continuously rated for maximum output in all operating modes (ONAN).
- ii. The maximum design demand load allowance for redundant units shall be 40% and 80% for non-redundant units.
- iii. Transformers used for redundancy or maintenance activities shall be suitable for parallel operation.
- iv. The transformer core shall be constructed of high-grade, non-aging, silicon steel laminations with high magnetic permeability and low hysteresis and eddy current losses.
- v. The transformer shall be constructed of individually cast primary and secondary coils, and coaxially mounted.
- vi. The high voltage winding shall be vacuum cast in epoxy with the low voltage windings wound separately and be either pressure injected, vacuum cast, or hermetically sealed with foil or sheet conductors.
- vii. Insulation shall be Class F and shall incorporate intrinsic fire extinguishing properties.
- viii. Random wound coils are not acceptable for use in medium voltage windings.
- ix. Windings shall have minimal internal splices or joints.
- x. Transformers serving non-sinusoidal loads shall be designed with adequate K factor ratings, i.e. K-4 type for facility load and K-13 type for process load are required base on harmonic study to verify the suitable type of transformer.
- xii. Transformers door (HV side and LV side) should not be equipped with interlock function with its upstream and downstream protection device (Circuit breaker), it includes transformers door opening/closing shall not interrupt normal operation. If required by local code or regulation, the transformers doors opening can provide with an alarming function (sent via SCADA or locally sounded) to inform the site the transformers door has been opened. It is further recommended any trip interlock function associated with transformers door access to be disabled, unless required by Local Code or Regulation.

6.7.4.2 Ratings

- i. The application/duty of the transformer will be mentioned in the requisition. Example motor unit transformer, transformer for variable speed drive etc.,
- ii. Power transformer use for supply to group of motors shall be capable of withstanding in frequent restarting loads of up to 1.8 times the rated transformer current.
- iii. IEC60076-1 or IEEE C57-12 or equivalent standard rated transformer should be preferred. In order to limiting the LV switchgear capacity to 5000A Facility or Production transformers maximum rating should be as follows:
MV/208V – 2MVA,
MV/480V – 4MVA.
- iv. Percentage Impedance voltage based on rated power should be as follows, unless an overriding design requirement.

Rated Power (kVA)	Impedance Voltage (%)
Up to 630	4.00
631 to 1,250	5.00
1,251 to 3,150	6.25
3,151 to 6,300	7.15
6,301 to 12,500	8.35
12,501 to 25,000	10.00

Impedance voltages for transformers rating above 25,000kVA shall be specified in the requisition, depending on system design and/or system studies

- v. Transformer and System Voltage
 - o Rated voltage on the high voltage side is generally equal to the system nominal voltage in which the high voltage winding is connected.
 - o The correlation between the rated voltage (at no load) of the low voltage winding and the corresponding system voltage is as below:

System nominal Voltage (V)	Rated Secondary Voltage (No Load)
200	208
208	214
380	400
400	420
415	433
440	460
480	504

vi. Nameplate Details:

Rating plate shall be stainless steel or others that is corrosion resistance metal. It should be permanently fixed to a readily accessible and non-removable part of the transformer, such as cooling fins.

Nameplate shall have the following information:

- o Name of Manufacturer
- o Serial number
- o Order No./Project number
- o Rated Capacity: kVA/MVA Rating
- o Primary voltage/s at various transformer tapping/connections
- o Secondary voltage
- o Full load currents (Primary & Secondary)
- o Rated Frequency
- o Number of Phases
- o Vector group (connection/displacement)/Diagram
- o K-Factor rating
- o Connection Diagram
- o Impedance rating/Percentage Impedance
- o Cooling of transformer (e.g. ONAN/ONAF)
- o Protection category/type (e.g NEMA 1 Enclosure/IPxx)
- o Quantity of insulating gas/liquid (litres/gallons of oil) and type (Not applicable for Dry type cast Resin Transformer)

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- o Insulation/temperature class of winding insulation.
- o Manufacturer tested date
- o Total weight (kg)

6.7.4.3 Accessories

- i. Provide fan cooling equipment (fans, power/control wiring, Local selector switch, controller, fuse, indicating light, relay) if needed.
- ii. Three-phase winding temperature indicator with contacts to control fans and alarm contacts
 - o Temperature sensing elements shall be imbedded in the windings, with means of replacement in the event of failure.
 - o In the event of an increase in transformer operating temperature, a means of alarming (sent via SCADA or locally sounded) shall occur. It is recommended the temperature alarm set point be 20 degrees C less than the overtemperature trip set point or the maximum nameplate temperature rating on units without a trip function.
 - o In the event of a temperature sensor failure, it would be preferable to fail into a safe mode (the transformer remaining in operation), with an alarming function (sent via SCADA or locally sounded) to inform the site the sensor has failed. It is further recommended any trip interlock function associated with failure of a temperature sensor be disabled, unless required by Local Code or Regulation.
- iii. Provide Class B Stranded Type SIS power/control conductors
- v. Copper ground pads
- vi. Power monitoring connection
- vii. Nameplates (Refer to 6.7.4.2)

6.7.4.4 Control power

- i. Self-powered from secondary bushing single phase
- ii. Three phase power supply, from LV distribution board, to power the cooling fans and temperature controls for the transformer. Cooling fans should be monitored, for any failure, when activated by high loading or high temperature. Any fan failure that is activated shall be monitored locally and remotely via Scada

6.7.5 Installation

6.7.5.1 Safety:

- i. Only qualified/competent technician should work on electrical equipment.
- ii. Risk assessment should be conducted to determine the hazards and mitigations to minimize the RISK to ALARP.
- iii. Proper personal protective equipment (PPE) should be worn when performing the installation work.
- iv. Any person with pacemaker, metal implant, or wearing jewelry should not be close (3 metres/10 feet) to energized/operational transformer.
- v. Applied Electrical Safety Rules/procedures, such as safe isolation and earthing/grounding, lock-out-tag-out, permit-to-work, etc., for all work on Electrical equipment.
- vi. Take all necessary safety precautions in handling and transporting of equipment for installation such as lifting, siting, transportation route, techniques, barricade, etc., Refer to local safety requirements, owner, manufacturer and contractor safety procedures.
- vii. Ensure sound earthing/grounding of electrical equipment and metallic structure/s.

6.7.5.2 Site installation:

- i. Ventilated dry-type transformers should not be located in polluted environments and or areas with possible dripping water.
- ii. Dry-type transformers installed in high humidity location should be specified to the manufacturer and precautions taken to manage it when transformer is de-energised over a long period of time.
- iii. Location should have sufficient space, clearance to walls and no obstruction to the transformer air circulation for its cooling.
- iv. High voltage and low voltage cables connections to the transformer should be clearly marked/identifiable.
- v. Cables should be properly clamped to avoid stress to the insulators/connectors/termination.
- vi. Recommended torque value shall be applied to all connections and bolting.
- vii. Installation of transformers and other precautions shall be in accordance to Manufacturer's instruction manual on installation and the QA/QC shall be approved by the Manufacturer's authority before energized.

6.8 Voltage Sag Mitigation System (SMS)

6.8.1 General requirement

- a) Voltage Sag Mitigation system shall have the capacity to supply the total load demand and an additional 20% of spare capacity.
- b) The SMS rating shall reflect its 100% service duty.
- c) The available fault current at the point of service will be reviewed and the SMS system and its components shall have suitable withstand rating.
- d) The SMS system and its components will meet all seismic requirements for the location where it is being installed.
- e) The level of noise produced by the SMS, operating at rated voltage and frequency, shall be below any limitations which required by local regulations.
- f) If conditions warrant, the unit will be ordered with topicalization and fungus protection.

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- g) The manufacturer shall install electrical components in the SMS system in conformance with either NFPA 79 or IEC 60204.
- h) The SMS system shall be self-contained and not be dependent on outside utilities for system operation, it shall not impact the operation due to utility source power outage.
- i) The SMS system shall be designed for continuous operation for the duration designed and shall be capable of supplying start-up transient/inrush, surges, and unbalance associated with the loads it serves and when switched into service.
- j) Electrical equipment and components shall be protected from intrusion of liquids/chemicals and environment (i.e. indoor/outdoor; hazardous; dust, etc.,). Solutions include rain caps, appropriate IP or NEMA ratings, or other containment.
- k) A coordination study shall be performed to allow for selective coordination of the system protective devices.
- l) The system shall be modular and ready for system expandability.
- m) During transient power disturbances on the distribution network, the sag mitigation system automatically corrects the voltage sag, swell or unexpected load changes, by maintaining the voltage sensitive process loads within an acceptable tolerance.
- n) The system has been designed to mitigate voltage sags down to a magnitude 40% of nominal for duration of approximately 1~2 minutes depends on the power quality event historical data to determine the magnitude and duration.
- o) The function of sag mitigation system is to temporarily and dynamically keeps the voltage at an acceptable level during intense voltage sags (both balanced and unbalanced droops) and swells seen on the mitigation system input terminals.

6.8.2 Performance requirement

- a) SMS system shall provide not only voltage protection against momentary disturbance but also correction for other power quality events at the point of common coupling (PCC).
- b) SMS system shall consist of a line reactor, power factor correction, voltage injection transformer, voltage source converter, low-pass filter, power protection, controller and operator interface module.
- c) SMS system ratings and performance criteria apply to the normal continuous range of AC voltage for operating conditions as specified in the datasheet, and at specified faults levels conditions for ambient temperature.

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- d) SMS manufacturer shall determine peak dynamic and continuous rated capacitive/inductive reactive power output of the SMS System based on the system analysis results.
- e) Under steady state conditions the SMS System is capable of continuously supplying enough normal power supply (NPS) current to maintain the NPS voltage and NPS current magnitudes at the MV bus to within the limits in the datasheet. The DVR System's overload capability can be used to address short duration swells in NPS magnitude provided the overload is available for the necessary duration.
- f) Under steady state conditions, with the voltage between 1.0 - 1.03 per unit (adjustable range by HMI), the SMS System output is zero or near zero so that the entire reactive capability is available for the dynamic conditions. SMS system controller controls additional shunt capacitor and/or a reactor bank(s) at the PCC under steady state conditions to maintain requirements defined above. The SMS System mitigates step voltage effects of switching the shunt devices controlled by the SMS System.
- g) The droop slope of the characteristic shall be continuously adjustable between 0.5% and 10% over the entire output range.
- h) The SMS System shall continue to generate reactive power and support the transmission system during a temporary under-voltage down to the value 0.2 per unit, per item 20 in the datasheet. The SMS System may be blocked if the voltage falls below 0.2 pu.
- i) The SMS System shall continue to absorb reactive power during a temporary overvoltage up to the value 1.25 pu and may be tripped for overvoltage of greater than 1.25 pu, per the datasheet.
- j) The SMS System shall resume automatic operation within 30 cycles once the voltage returns within the defined operating band as defined in the V-I characteristic.

6.8.3 Components ratings

6.8.3.1 Inverter design

- i. Each phase of the Voltage Source Converter (VSC) shall withstand maximum overvoltage and overcurrent stress due to system faults and switching.
- ii. SMS System shall be controllable up to the maximum voltage listed in the datasheet. The VSC

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shall be capable of blocking if the voltage exceeds the value at which SMS System may be blocked.

- iii. SMS System shall have sufficient redundant modules to meet the guaranteed reliability, availability, performance and the harmonic performance criteria under all possible operation modes with failed modules.
- iv. The inverters shall be 3-phase type.
- v. The inverters shall be IGBT semiconductor based PWM modules, switched at a minimum frequency of 2 kHz.
- vii. Inverters shall have independent phase control as an option for transient events.
- viii. Inverters module shall be fully integrated air-cooled, four-quadrant power converters and dc bus.
- ix. Each Inverter module shall employ capacitors, suitable sensing, driving, and isolating circuitry, and a microprocessor or DSP based controller.
- x. The inverter module shall be capable of producing capacitive and inductive VARs with the added ability to operate at 3 times their continuous output current for up to two seconds.
- xi. The design of the inverter structure shall permit easy access for visual inspection and maintenance. Each single-phase module (or equivalent) shall be parallel connected, including all necessary heat sinks, snubber circuits and gate drivers.
- xii. Inverters shall be arranged in parallel fashion. The Parallel inverters shall be provided such that the loss of one inverter shall not reduce the capability of the remaining inverters for operation.
- xiii. Reactors shall be supplied as required for filtering.
- xiv. Inverters shall be fused for protection requirements.
- xv. There shall be a monitoring system that monitors the status of the modules. The monitoring system shall be able to detect module failures and the detected failures shall be displayed on the HMI monitor, recorded and printed out in the control room, including the location of the failed module, time of failure and, if to the extent possible, cause of failure. There shall be an alarm when all the redundant modules in any phase have failed.

6.8.3.2 Cooling system

- i. Cooling system maintains full SMS System rated capacity at maximum ambient temperature and maximum heat losses.
- ii. Cooling for semiconductors by natural convection or forced air. Forced air requires full redundancy and removable filters at air inlets.
- iii. Air-cooled system monitoring alarms:
 - o Blower transfer
 - o High exhaust air temperature or high heat sink temperature
 - o High differential pressure across the filter
 - o Low air flow
- iv. Minimum Tripping States
 - o Excessive exhaust air temperature or high heat sink temperature
 - o Loss of air flow

6.8.3.3 Reactors

- i. The reactors shall be of single phase, air-core design, self-cooled and suitable for outdoor installation (ambient conditions in datasheet shall be considered).
- ii. Reactors shall be capable of withstanding short circuit forces based on maximum design fault levels.
- iii. The reactor voltage rating shall be as required by the SMS System design. The bidder shall determine the BIL rating as required by the SMS System design and the insulation requirements.
- iv. Audible noise levels from the reactors shall be coordinated to meet the requirements for the SMS System installation and the local regulation.
- v. Reactor to be installed with proper clearance and with proper insulation / bearers to prevent interference from, nearby structure, birds, animals, and other creatures.
- vi. Fencing, supports and other metal structure grounding to make sure the circulation currents are

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minimized without compromising the induced voltage level safety limit.

6.8.3.4 Capacitors

- i. The capacitor units shall be constructed with materials resulting in minimum losses and maximum reliability. Each unit shall be free of PCB.
- ii. Reactors for limiting of inrush currents (if required) shall be connected in series with the capacitor banks if required.
- iii. The capacitor voltage rating shall be as required by the power system characteristics and overall SMS System design. The BIL rating shall be determined by the bidder as required by the SMS System design.
- iv. The individual capacitor units can be internally or externally fused or fuseless. Supplier shall specify with their bid.
- v. Audible noise levels from the capacitor banks shall be coordinated to meet the requirements for the SMS System installation and the applicable local regulation.
- vii. Capacitor dielectric losses shall not exceed 0.15 W/kVar.
- viii. The capacitor units shall use internal discharge resistors that reduce the residual voltage to below 50V in not more than 5 minutes.

6.8.3.5 Transformers and Switchgears

- i. Provide Transformers as per specification 6.7
- ii. Provide Switchgear s as per specification 6.3
- iii. Reactive Current Rating: 100% full load amps.
- iv. Designed for harmonic currents voltage levels associated with the SMS System under normal operating conditions without loss of life.

6.8.3.6 Bypass line

- i. A bypass line will be switched into service when the load current demands exceed an allowable limit, to protect the SMS. Events such as short circuit fault

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at load side, high inrush current, over current and SMS failure.

- ii. This bypass thyristor switch is connected in anti-parallel in between the inverter and series injection transformer. Its function to take SMS offline and online from distribution system.

6.8.4 Accessories

- a) Accessories interface with the SMS module/s should be clearly define and marked by the manufacturer in the instruction manual/drawings. Accessories include heating and ventilation system requirements, switchboard external power requirements, other auxiliary power and signal/alarms, etc.,
- b) Cable accessories and termination diagram shall be obtained from manufacturer with interfaces/responsibility clearly marked/defined.
- c) Interface for communication and parameters to Scada. Loop check and full functional test to be performed with the vendor on site.
- d) Force air cooling intake, if needed, shall have sufficient safety space/clearance and away from other hazardous source. (such as pollution, chemical, heat, etc.,)
- e) Grounding/earthing and bonding of all the modules shall be effective.

6.8.5 Control panel

- a) ON/OFF sequencing of the DVR System
- b) Emergency OFF control
- c) Control mode selection
- d) Voltage control through three phase voltage regulation
- e) Filter switching strategy
- f) Coordinate Capacitor Banks switching control;
- g) Coordinate Reactor Banks switching control;
- h) HV overvoltage
- i) HV undervoltage
- j) LV bus undervoltage
- k) DVR System current supervision
- l) Trip Conditions
- m) Alarms

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6.8.6 Control power

- 6.8.6.1 Line supply use to detect the presence of voltage (magnitude and phase/s) sags in the system. Power quality to signal inputs to the controller/control unit to initiate voltage sag correction.
- 6.8.6.2 High current line detection, detects the current when a fault occurred on the downstream line.
- 6.8.6.3 Other control power and detections are taken from internal of the SMS unit.

6.8.7 Installation

6.8.7.1 Safety:

Only qualified/competent workers should work on Electrical equipment. They should work closely with the manufacturer to safely install, test and commission the SMS modules/system to the electrical network.

Refer to Clause 6.7.5.1 on installation electrical safety and shall comply to local codes/regulations that's applicable.

6.8.7.2 Equipment Installation:

- i. Installation work to be performed by the SMS manufacturer or supervisor by the manufacturer's trained qualified person. Installation and precautions shall be in accordance to the manufacturer's instruction manual.
- ii. The manufacturer shall install electrical components in the SMS system in conformance with either NFPA 79 or IEC 60204. (Electrical Safety standard for machineries/equipment)
- iii. Prepare Sag Mitigation System Modules (SMS) to be set on site for installation. Check concrete reinforcement pad with mounting studs has been installed and flooring levelled.
- iv. Proper lifting devices/equipment shall be used to safely lifted and place the SMS onto the installation pad.
- v. Location should have sufficient space, clearance to walls and no obstruction to the access and equipment cooling.

- vii. Inspect SMS modules/components for damage and cleanliness.
- viii. High voltage and low voltage cables connections to the SMS should be clearly marked/identifiable.
- ix. Cables should be properly protected and clamped to avoid stress to the insulators/connectors/termination.
- x. Recommended torque value shall be applied to all connections and bolting.
- xi. All special tools, equipment, materials and spares from the vendor are in place and properly stored.

6.9 Emergency Generator

6.9.1 General requirement

- a) If specified as standby duty, generator system shall be provided to supply power to emergency load that included as life safety system, legally required standby, and others critical load which required by other process and mechanical system.
- b) The maximum design demand load allowance shall be 80% of the kilowatt rating of the generator if power factor.
- c) The generator unit rating shall reflect of its 100% service duty.
- d) The generator set shall be able to start automatically upon receiving a normally-open, close to start dry contact signal from an automatic transfer switch.
- e) The available fault current at the point of service will be reviewed and the generator system and its components shall have suitable withstand rating.
- f) The generator system and its components will meet all seismic requirements for the location where it is being installed.
- g) The manufacturer shall install electrical components in the generator system in conformance with either NFPA 79 or IEC 60204.
- h) The generator system shall be self-contained and not be dependent on outside utilities for system operation.
- i) The generator system shall be designed for continuous operation and shall be capable of supplying start-up transient, surges, and unbalance associated with the loads it serves.
- j) Electrical equipment and components shall be protected from intrusion of liquids. Solutions include rain caps, appropriate IP ratings, or other containment.

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- k) A coordination study shall be performed to allow for selective coordination of the system protective devices.
- l) The room or enclosure for the generator system shall be cooled, as necessary, to prevent overheating under conditions of the highest anticipated ambient temperature, at the installed elevation, when fully loaded.
- m) The room or enclosure for the generator system shall be heated, as necessary, to maintain the required temperature to allow for cold start and load acceptance by the system.
- n) Indoor installations shall have provisions to provide fresh air to the prime mover for combustion.
- o) Adequate space shall be provided around the cooling system radiator on the prime mover for air circulation.
- p) The generator system shall be sited to avoid the exhaust from the unit being introduced into a building's air handling system while the generator system is running.
- q) Appropriate containment shall be supplied for the prime mover lubrication system and cooling system.
- r) All auxiliary devices and mechanical equipment that maintains the conditions of Emergency Generator system shall be served from AC UPS or DC Battery system.

6.9.2 Engine

- a) The generator system shall be started by either an electric starter or a stored energy starting system.
- b) The prime mover if required shall be appropriately de-rated for altitudes, ambient temperature, fuel energy content, accessory losses, and site conditions as recommended by the manufacturer.
- c) The generator system shall be supplied with all appropriate listings, certifications, and test reports.
- d) The generator shall be a three-phase, 50Hz or 60Hz, single pre-lubricated sealed bearing type, with brushless exciter, built with Class F insulation on the stator and rotor, and both shall be further protected with 100% epoxy impregnation and an overcoat of resilient insulation material to reduce possible fungus and/or abrasion deterioration. The generator shall incorporate reactive droop compensation.
- e) A generator-mounted volt-per-Hz type temperature compensated solid state regulator shall be provided to match the characteristics of the generator and prime mover. Voltage regulation shall be +/- 0.5% from no load to full rated load. Readily accessible voltage drop, voltage level, and voltage gain

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controls shall be provided. Voltage level adjustment shall be a minimum of +/- 5%.

- f) The solid-state regulator module shall be shock mounted and epoxy encapsulated for protection against vibration and atmospheric deterioration. Voltage regulation shall maintain precise control of the generator with harmonic loads of up to 20%. Provide overexcitation protection.
- g) Generator systems with a reciprocating engine using a radiator for cooling the engine shall use a solution of 50% water and 50% ethylene glycol coolant, with a blower-type fan, sized to maintain safe engine temperature when the generator is fully loaded at ambient temperature and at the specified altitude. Coolant additives shall be appropriate and approved for the region and conditions where it is being utilized. The coolant selection and its additives shall be recommended by OEM instead of coolant supplier.
- h) Generator systems with a reciprocating engine used as the prime mover shall utilize a thermostatically-controlled lubrication oil heater or a separate thermostatically-controlled jacket heater if ambient conditions at the installation site warrant their use.
- i) A generator-mounted, vibration isolated, dead front control panel shall be provided. All switches, lamps, and meters shall be oil-tight and dust-tight, with the enclosure door gasketed. The panel shall contain at, a minimum, a voltmeter and ammeter with 2% accuracy, a frequency meter, starting controls, fault indicators, oil pressure and engine temperature gauges, a run time, and provisions to monitor the status of these items.
- j) Voltage regulation shall be +/- 0.5% for any constant load between no load and rated load.
- k) Random voltage variation with any steady load from no load to full load shall not exceed +/- 0.25%.
- l) Frequency regulation shall be isochronous from steady state no load to steady state full load.
- m) Random frequency variation with any steady load from no load to full load shall not exceed +/- 0.25%.
- n) The alternator shall produce a clean AC voltage waveform, with no more than 5% total harmonic distortion at full linear, when measured line to neutral, and not more than 3% in any single harmonic.
- o) The time to automatically start, accelerate to speed and voltage, synchronize to the system bus, and pick-up the load on a normal power failure shall not exceed local code requirements if life safety systems are serviced from the generator system.

6.9.3 Exhaust, silencer and fittings

- a) Exhaust system shall include a flexible section, together with all necessary clips, cleats and supports.

The exhaust system shall also include a spark arrester and an efficient exhaust silencer. The exhaust ducting shall be arranged to carry all the diesel exhaust gases out of the building. If installed inside, to determine the height in order to avoid exhaust intake by the building.

- b) The ceramic exhaust cleaning device shall be provided.
- c) Critical-type silencer with muffler companion flanges made of carbon steel with an internal heat blanket and flexible stainless-steel exhaust fitting.
- d) Fittings, exhaust pipe, and adapter shall be selected which suitable for specified orientation, sized in accordance with engine OEM. It shall be mounted without supported/attached by engine.
- e) Size of exhaust pipe is determined so back pressure not exceeding engine specifications.

6.9.4 Fuel tank and fittings

- a) The fuel system shall be designed with a fuel strainer and filter to provide a clean fuel supply to the prime mover.
- b) Fuel tanks shall be sized large enough that the generator system can operate at full load for 8 hours (12 hours for extreme weather locations) minimum. Fuel shall be consumed within the storage life or provisions made to replace stale fuel with clean fuel, or the fuel shall be filtered and chemically treated to maintain the required quality. Appropriate containment shall be provided for the fuel system. Extended operations to support 48hours to be planned by reducing the load sequentially during extreme weather conditions and extended outage forecast.
- c) Fuel level in tank shall have means of site monitoring (either by level gauge or sight glass) and level transmitter wired to Master Control panel and Scada.
- d) Fuel pump motor and associated fuel piping design shall consider harsh operation condition.
- e) Dual wall contained fuel tank shall be provided and located nearby generator room.
- f) Provide leak detector in dual wall tank to detect failure of the primary tank for shutdown pump motor.
- g) Provide wire-reinforced flexible fuel lines with threaded connections and swivel fittings for each fuel connection on

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engine. Coordinate fuel supply and return line sizes from remote main tank to meet engine OEM min. installation requirements and max. tank pumping capacity.

- h) Provide diesel engine OEM standard fuel system, incl. replaceable spin-on-type fuel filter conveniently located for servicing.

6.9.5 Starting system, battery and charger

- a) The starting system shall use electrical motor which fed from maintenance free batteries.
- b) The batteries and charger shall be provided.
- c) The battery charger shall be suitable for continuous connection to the battery with the battery connected to the engine ready for instantaneous use.
- d) The batteries shall be floating charged under normal condition. When the batteries discharged, the charger shall automatically switch to equalizing charge mode. This switching can also be activated manually.
- e) Battery charger common alarm shall be connected to Scada system for monitoring. Alarms include supply (AC) failure, charger failure (DC voltage low), high DC voltage, etc.,
- f) Batteries for the electric starter system shall be nickel-cadmium or lead-acid.

6.9.6 Automatic voltage regulator (AVR), Exciter and Governor

- a) AVR shall be solid-state type generator mounted regulator with SCR control to match engine and generator characteristic.
- b) Voltage regulation shall be +/- 0.5% for any constant load between no load and rated load. Random voltage variation with any steady load from no load to full load shall not exceed +/- 0.25%.
- c) AVR shall recover the voltage for the momentary voltage variation within 0.5 seconds to stable operation.
- d) Exciter shall be brushless type using a 3-phase rotating rectifier circuit.
- e) Exciter shall be capable of supplying the field excitation current requirements for the generator.
- f) Governor shall be isochronous type for parallel operation. Engine speed is maintained at +/- 0.5% at any constant load, including no load.
- g) Governor shall adjust droop from isochronous to 3%.
- h) Governor system shall provide actuator, control module, dedicated magnetic pickup, idle/run switch for warmup after maintenance, three-phase isochronous load-sharing module,

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and auto synchronizer. Those shall be integrated into engine generator control panels by OEM.

6.9.7 Switchgear, step-up transformers (if applicable)

- a) Emergency system switchgear shall be gas-insulated switchgear type for MV application and air-insulated switchgear type for LV application.
- b) Emergency system switchgear shall consist of redundant incoming lines and outgoing feeders. Additional up-stream, down-stream or tie-breakers can be added, if required.
- c) Emergency system switchgear shall be independent system and mainly serve emergency load only. Normal load system shall NOT be supplied from emergency system switchgear although load shedding controller is provided.

6.9.8 Control panel

6.9.8.1 Master engine generator control panel (MGCP)

- i. For normal power loss, MGCP shall automatically start, parallel and supply emergency generator power to emergency loads. When normal power returns, initiate automatic power transfer through automatic transfer switches, then shutdown engine generators and place back on stand-by.
- ii. During engine generator fail to start or failure during running, MGCP shall be able to automatically shutdown an engine and initiate load shedding scheme.
- iii. Provide selector switch for Manual and Automatic operation mode.
- iv. During manual operation:
 - o When MGCP is set at Manual Mode. Entire emergency generator system will be started and stopped from each individual LGCP.
 - o Built-in synchronizing panel shall be provided and used during manual mode.
- v. During automatic operation:
 - o When MGCP and LGCP are set at Automatic Mode. Upon receipt of generator start signal from Automatic Transfer System after 2 seconds from under voltage detection confirmed. Entire emergency generator system will be started and synchronized less than 1 minutes of start signal. During

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- synchronization process, no-load condition shall be established.
 - Outgoing feeders will be staggering closing with 2 seconds interval time after all emergency generator system synchronized.
 - When normal power returns, after 10 seconds time delay, automatically transfer all critical loads back to normal power via automatic transfer switch and the process shall be staggering with 2 seconds.
 - After critical load transfer is completed. Generator circuit breakers can be opened, and generator unit is ready for shutdown and cooldown.
 - Should a generator fail to Sync. during automatic mode. MGCP automatically shutdown engine generator and lock out system after time delay. MGCP shall allow individual unit generator to set at manual mode and manually sync. to bus.
- vi. Master control PLC shall be designed to provide 3 different operation mode: Auto, Manual, and Testing.
- vii. Automatic load shedding
- viii. MGCP shall be furnished with listed below functions and capabilities to monitor and control the emergency system operation.
1. Automatic engine-starting/stopping and alarm failures
 2. Load-sharing type electronic governor to control engine speed
 3. Automatic synchronizer
 4. Mode selector: lockout/reset-off/cooldown-auto-run/test
 5. Voltage-adjust function
 6. Speed-adjust function
 7. Generator and Feeder circuit breaker control
 8. Load shedding function and bypass
 9. Synchronization check relay
 10. Voltage regulator controller
 11. HMI Display touchscreen:
 - Incoming Line: AC Voltage (L-L/L-N), Current, Frequency, Energy, Power Factor.

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- Generator: Generator Voltage (L-L/L-N), Current, Frequency, Energy, Power Factor.
- Generator Mechanical: Bearing Temp., Stator Winding Temp., Voltage Regulation, Excitation Voltage/Current.
- Visual indication of phase angle difference between generator bus and utility voltages during synchronization.
- Frequency meter control
- Alarm indication
- Communication interface with FMCS

6.9.8.2 Local engine generator control panel (LGCP)

Each individual generator unit shall be furnished with local digital engine-generator control panel enclosure with engine/generator control function and indicators as following:

- i. Lockout/reset function
- ii. Off/cooldown function
- iii. Manual/Auto mode
- iv. Run/Test mode
- v. Frequency meter
- vi. AC output voltmeter
- vii. AC output ammeter
- viii. Output voltage adjustment
- ix. HMI Display:
 - Engine: Lubrication oil pressure and temperature, CW temperature, speed, crank, fuel oil pressure, set points, low/high jacket temperature, battery status, engine control position, generator breaker, tachometer, running time meter.
 - Incoming Line: AC Voltage (L-L/L-N), Current, Frequency, Energy, Power Factor.
 - Generator: Generator Voltage (L-L/L-N), Current, Frequency, Energy, Power Factor.
 - Generator Mechanical: Bearing Temp., Stator Winding Temp., Voltage Regulation, Excitation Voltage/Current.
- x. Engine selector switch and EMO
- xi. Output contacts for alarm signal: Running, Auto, Load Shedding, Alarm, Charger, Air Intake/Exhaust louver control

6.9.9 Control power

- a) Secured power supplies, at least two sources either from AC UPS or DC UPS, shall be provided for generator switchgear and control gears. Supply shall be monitored and alarms send to Scada system.
- b) Generator output voltage for synchronising, speed regulations and transfer operation (undervoltage relay).
- c) Switchgear VTs for transfer switching, synchronising, undervoltage relay, metering, etc.,
- d) External or internal switchboard power supplies for spring charging, heaters and others.

6.9.10 Load Bank

- a) Provide a three-phase load bank, sized a minimum of 100% of the generator's rating, and be connected to the system with an isolation breaker. Load bank will be manually exercised only, with control to allow 200 KW load steps. If an permanently mounted load bank is not practical, it will be permissible to use either portable load banks or actual load to perform testing.

6.9.11 Installation

- a) Prepare the Diesel Generator (DG) set on site for installation. Check concrete reinforcement pad with generator mounting studs has been installed and flooring levelled.
- b) Emergency generator installed site and type shall comply with local code/regulation, if there is a requirement.
- c) Install the Emergency generator set and controls as instructed by the manufacturer in their installation manual.
- d) Proper lifting device that will allow the diesel generator set to be safely lifted and placed on the installation pad.
- e) Diesel generator set shall be placed on insulating/anti-vibration pad, if site requires to reduce noisy and or vibration.
- f) Install master engine generator control panel (MGCP) and connect all cables according to wiring diagrams from manufacturer and interface wiring to switchboard, remote alarms/status, power supply, etc.,
- g) Ensure all connections are secured and effective, such as to fuel line, making electrical connections, grounding, accessories, etc.,
- h) Inspection DG set for damage and cleanliness, which include diesel engine, filters, radiators, starter, charger, governor, AVR, generator, controls, etc., prior to testing and commissioning.

- i) All special tools, equipment and materials from the vendor are in place.

6.10 Electric Motors

6.10.1 Design and Selection of Electric Motors

- 6.10.1.1 Rotating Electrical machines shall comply with IEC 60034-1 or NEMA MG1. Owner shall be consulted if there are deviates or amendments to the standard.
- 6.10.1.2 Electric drives shall be controlled by fused contactors except for motor rating exceeds the maximum rating of a fused contactor, a circuit breaker shall be used.
- 6.10.1.3 The minimum/maximum power ratings of electric motors in relation to system voltage should be as follows:

Switchboard Nominal Voltages	Maximum LV motor rating	Minimum HV motor rating
480V/400V	225kW/185kW	
690V	315kW	
6.6kV		160kW

- 6.10.1.4 The auxiliary drives to the main unit shall be connected to the same section of the switchboard as the main unit in order to obtain optimum availability of the total system.
- 6.10.1.5 For low speed applications and installations that requires power factor compensation, synchronous motors of less than 3 MW may be economically justifiable. Contractor to demonstrate the cost benefit (capital and running cost) in comparison with a robust induction motor.
- 6.10.1.6 Synchronous motor driving pulsating torques machine during run-up may need to be addressed by the driven equipment Manufacturer.
- 6.10.1.7 For very large high speed motors, the use of magnetic bearings may be considered. The specification of these shall be agreed with machine and driven equipment manufacturers.
- 6.10.1.8 Electric drive shall be provided with a remote control unit (RCU) mounted local to the motor and or emergency stop.

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- 6.10.1.9 For LV motors with automatic control provided, then the auto on/off control signals shall be given via an IMCS or interposing relays.
- 6.10.1.10 For HV motors, automatic control shall be provided via protection devices (Intelligent Electronic Device – IED) or interposing relays.
- 6.10.1.11 Motors that are mounted in different level, and controlled from the other level for operational convenience, shall have safety switches located near the motors.
- 6.10.1.12 Where motors are fitted with anti-condensation heaters, a heater safety switch shall be installed adjacent to the motor's RCU.
- 6.10.1.13 Cage induction motors shall be switched direct on line or variable speed drive, if impractical or uneconomic, current-restricting start-up equipment shall be used.
- 6.10.1.14 Motors with auxiliary equipment, water cooling or special detection facilities, e.g., cooling air/water temperature detectors, water leakage detectors, shall be connected to an alarm annunciation system with first failure alarm.
- 6.10.1.15 Application of variable speed drive (VSDS) should be recommended where it demonstrate the benefits, such as design (gear box), process/operation, maintenance and energy efficiency.
- 6.10.1.16 The machine and all individual items forming part of it, shall have an expected lifetime of at least 20 years and shall be suitable for at least 4 years of uninterrupted operation under the condition specified.
- 6.10.1.17 The degree of protection (to ANSI/IEC 60529) shall be at least IP54 for machine and auxiliaries and IP55 for terminal box and bearing housing. Other special application and site condition may require higher degree of protection, such as near offshore, submerged drives.
- 6.10.1.18 Vertical motors with a downward drive-end shaft shall be provided with a canopy over any upward facing air inlets to prevent entry of water or any other contaminant.
- 6.10.1.19 All insulation materials shall be of Class F, as defined in IEC 60034-18 and rating for temperature rise for all parts of the machine windings shall be based on Class B.

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- 6.10.1.20 Operating conditions if not specified shall be as default conditions stated in standard. (IEC 60034-1 equivalent).
- 6.10.1.21 Maximum ambient air temperature, if not stated, shall be 40oC taken to be the "Mean Annual Extreme" for a warm and damp climate.
- 6.10.1.22 Water coolant temperature at the inlet to the heat exchanger shall not exceed 30oC and the minimum cooling water temperature will be 5oC.
- 6.10.1.23 Periods of running may alternate with idle (standstill) periods of maximum six months. At the end of such an idle period the machine shall be suitable for another running period without requiring additional inspection.
- 6.10.1.24 Machines shall be suitable for direct-on-line starting or re-acceleration with any voltage between 80% and 100% of the rated voltage applied at the machine terminals.

6.10.2 Additional requirements for Electric Machines

- a) Motors shall be at least super efficiency IE3 category, in which most countries adopt. Owner shall be informed and approve this deviation.
- b) Machines shall be rigid with first rotor bending critical speed not lower than 125% of the synchronous speed. Driven equipment vendor is to perform a torsional vibration analysis of the complete train (motor plus driven equipment) and motor manufacturer shall provide data required for this analysis.
- c) Every electric motor shall be provided with a rating plate or plates. Rating plate shall be of stainless steel and shall be securely fixed to a non-removable part of the frame.
- d) For machines to be used in hazardous areas the type of protection shall be provided/mark, in accordance with IEC 60079, for Machine, its terminal box and Auxiliary devices.
- e) Motors with mass greater than 25 kg shall have eye bolts or lugs identifiable for hoisting used.
- f) Machines weighing over 600 kg shall have jacking bolts or lifting facilities, to facilitate alignment of the machine with the driven equipment.
- g) Anti-condensation heaters shall be fully insulated design. Heaters shall be arranged to provide uniform heating of the stator and maintain a temperature of the winding at about 5k above ambient temperature.

6.10.3 MV Motors

- a) Steep fronted transient waves can be expected at the terminals of MV motors due to the result of switching activities.
- b) Both rolling element and sleeve bearings, temperature rise as measured by embedded temperature detectors (ETD), fitted in the bearing housing, shall not exceed 55 K at the end of the heat run.
- c) Forced lubricated sleeve bearings with lubricating oil temperature rise with the minimum allowable oil flow shall not exceed 28 K at the end of the heat run. However, the oil temperature and sump temperature of oil ring lubricated bearings shall not exceed 82 °C (API 617, 6th Edition 1995 and API 610, 8th Edition 1995).
- d) For machines supplied via a dedicated unit transformer, different requirements may prevail. In these situations the design of the machine-transformer combination shall be optimised with respect to the maximum allowable voltage drop at the busbar and the start/restart performance of the machine.
- e) At any voltage between 80% and 100% rated voltage, machines (including their unit transformers if applicable) shall be capable of:
 - three successive starts with the machine initially at maximum ambient temperature
 - two successive starts with the machine initially at full load operating temperature.
- f) Between successive starts the machine may be assumed to decelerate under operating conditions.
- g) After a cooling period of 30 minutes at standstill, another starting sequence of at least two successive starts shall be possible.
- h) Number of starts for machines (including unit transformer if applicable for HV machines) shall be suitable for one of the duties:
 - Normal: maximum 1 000 starts per year
 - Heavy: maximum 3 000 starts per year (e.g. drain pumps)
- i) The starting current shall not exceed 6.5 times the rated current of the machine, unless there is a more stringent limitation in the requisition.

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- j) The starting performance of cage induction motors at rated voltage shall not be less than the value below:

Lock Rotor Torque (LRT) – TL

Pull-out Torque – Tp

Breakdown Torque – Tb

*Note – all values are based on per unit of rated torque.

2 poles Machine	LRT (TL)	Tp	Tb
Rating up to 250kW	0.8	0.8	1.8
250 - 500kW	0.7	0.6	1.8
500 - 1600kW	0.6	0.5	1.8
4 poles Machine			
Rating up to 250kW	0.9	0.7	1.8
250 - 500kW	0.7	0.6	1.8
500 - 1600kW	0.6	0.5	1.8
6 Poles machine			
Rating up to 250kW	0.9	0.7	1.8
250 - 500kW	0.7	0.6	1.8
500 - 1600kW	0.6	0.5	1.8

- k) Torque speed characteristic of the motor shall be adequate for starting the driven load under most severe conditions (e.g pump discharge open), with rated frequency and applied rated voltage of 80% (motor terminals). Accelerating torque shall not be less than 10% of full load torque. This applies for driven equipment that exhibits quadratic torque-speed characteristic.
- l) Situation of starting driven equipment, other than above mentioned, that may not be possible to meet this requirement then owner shall be informed.
- m) Bracing of motors end windings shall be able to withstand three-phase short circuit at the terminals. Their shaft and iron core systems shall withstand two-phase short circuits. Manufacturer shall state the maximum transient air-gap torque for the above and include reconnection after power interruption.
- n) Manufacturer shall state the running-up time and the maximum allowable running up time, with 80% and 100% applied rated voltage at motor terminals.
- o) Machines with flexible, over-critical rotor bearing systems shall have the first critical speed below 80% of the motor machine synchronous speed and the second critical speed shall be greater than 125% of synchronous speed.

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- p) At whatever critical speed the bearing vibration of the machine shall not exceed twice the values specified. At two times frequency vibration the overall vibration shall not exceed 1.4 mm/sec (rms) and axial vibration severity of the motor frame, including terminal boxes, shall not be more than 4.5mm/sec (rms)
- q) The sound pressure level of the loaded motor shall not exceed 82 dB(A) in the work area, measured in accordance with ISO 1680-2.
- r) Embedded temperature detectors in the stator windings shall be provide for machines in excess of 1000 kW. At least six detectors shall be installed, two for each stator phase and be positions likely to show the highest temperature.
- s) Temperature detectors shall be PT-100 Platinum resistance elements to IEC 60751 and be wired to a separate terminal box mounted on the machine frame.
- t) Differential protection shall be provided on machines with rated outputs in excess of 3 500 kW
- u) When indicated on the requisition, RTDs shall be provided to measure bearing temperature.
- v) Whenever the shaft voltage measured across the ends of the shaft exceeds 250 mV (rms) for rolling element bearings or 400 mV (rms) for sleeve bearings, bearing insulation shall be provided.

6.10.4 LV Motors

- a) Machines shall be suitable for operation on a supply voltage and frequency variations as specified in IEC 60038 or ANSI C84-1.
- b) Number of starts for machines (including unit transformer if applicable for HV machines) shall be suitable for one of the duties:

Normal:	maximum 1 000 starts per year
Heavy:	maximum 3 000 starts per year (e.g. conveyors, drain pumps)
Extra heavy:	maximum 20 000 starts per year (e.g. cranes, hoists – LV only).
- c) The locked rotor apparent power shall comply with IEC 60034-12. However, for machines with a rated output in excess of 55 kW the starting current shall not exceed 7.0 times the rated current of the machine. Higher efficiency motors may have higher starting current, owner should be informed and consideration have to be taken in sizing of all the components (cable volt drop, contactor, protection, etc.,) in the string.

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- d) The starting performance shall comply with IEC 60034-12 or ANSI/NEMA MG1. Driven equipment which exhibits a quadratic torque/speed characteristic Design N shall apply. For equipment with other characteristics Design H shall apply.
- e) The sound pressure level of the loaded machine shall not exceed 77 dB(A) in the work area, measured in accordance with ISO 1680-2

6.10.5 Installation

6.10.5.1 General

- i. Motors stored outdoor and over a period of time before commissioning shall be provided with initial and thereafter periodic preservation procedures according to Manufacturer's instructions. As a minimum requirement, preservation shall include proper sealing of motor shaft, cable entries, turning of shafts and for those with anti-condensation heaters it should be energised.
- ii. Location of motor remote control unit (RCU) and or emergency switch should be on the opposite side of the terminal box and not be more than 1.5 meters away. Site ammeter, if installed, should be visible to operator when performing other activities, e.g. operating valve.
- iii. There shall not be any obstruction to the air flow of the motor or the motor heat exchanger.
- iv. Shim plates used should not be of non-magnetic material. Dowel pins may be installed to facilitate reinstallation after maintenance or repair work.

6.11 Low Voltage Switchgear and Control Gear

6.11.1 General requirement

- a) Switchgear assemblies shall be completely enclosed, self-supporting and suitable for floor or wall mounting, in multi-cubicle or multi-box type structures. The floor shall not be considered as being part of the enclosure.
- b) The assembly rated shall be suitable to form part of a system with coordinated insulation values in accordance with IEC 60664-1, overvoltage category IV, and shall be type tested accordingly.
- c) Partitions or barriers shall be provided in the assemblies to obtain Form 4 or 3B separation of compartments. Form 3B is only acceptable if all terminals have a degree of protection of at least IP 30. IEC 61439-2 defines forms of internal separation.

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- d) The degree of protection as per IEC 60529 for any assembly shall be at least:
- For indoor use (in enclosed buildings) with rain canopy IP 31
 - For outdoor use (protected by e.g. rain canopy) IP 55
- e) The requirements specified apply to type-tested assemblies (TTA). For assemblies used as sub-distribution systems, with incoming feeders protected by short-circuit current limiting devices of a nominal current of maximum 1600A at 480V.
- f) Non-metallic enclosed switchgear and controlgear may be used for LV sub-distribution switchboards, when protected by short circuit current limiting devices having a maximum nominal current of 400 Amperes.
- g) Insulated plate shall be provided for cable entry to the panel
- h) Where generators are directly connected to an LV switchgear, the ratio of peak to rms short circuit current is likely to be higher than normal allowance, as per IEC standard. The peak short circuit requirement shall be clearly stated in the project specification.
- i) Unless otherwise specified in the requisition, the assembly shall be arranged for a TN-S power system, i.e. with separate neutral and protective busbars throughout the system.
- j) The main busbars shall be rated for the nominal current over the entire length of the assembly and shall not be rated lower than the incoming and sectionalizer units. All the bus arrangement to be fully insulated where the air insulation gap is less than 50mm.
- k) A diversity factor of 1.0 shall be applied for temperature rise test or calculations for each switchboard panel containing only one functional unit. If a panel contains more than one two functional units, the rated diversity factor applied shall be not less than 0.9 when considering the circuit loads as known at the time of final inspection at the Manufacturer's works. Spare circuits and skeleton compartments shall be considered as being loaded to 90% of their rated current.
- l) All breakers and cables shall be loaded up to a maximum of 80% of its rated capacity.
- m) Additional protection relay shall be provided for incomer feeders rather than ACB/MCCB built-in protection.
- n) Switchgear nameplates including circuit labels, instruction and warning plates and caution notes shall be made of durable, corrosion resistant material.
- o) Nameplate information shall be mounted on the front of the assembly, e.g. on the sectionaliser or incoming feeder and include the following:

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- Name of Manufacturer,
 - Purchase order number,
 - Year of manufacture,
 - Type designation,
 - Rated voltage,
 - Rated frequency,
 - Rated current of busbar system,
 - Rated peak and short time withstand current and withstand time,
 - Degree of protection.
 - Other information to standards shall applied, include the relevant documents of the Manufacturer.
- p) Warning plate(s) or caution notice(s) shall be installed in location to identifying the danger point(s).
- q) LED (Light-Emitting Diode) lamps shall be provided for indication/trip/warning purposes.
- r) Each circuit identification tag/number of the assembly shall be provided with circuit labels installed on the front of the fixed and the removable part of a withdrawal or removable unit.
- s) Assemblies shall be provided with a durable one-line synoptic diagram, indicating the busbar circuit and the specific function of the different compartments.
- t) The Manufacturer shall supply at least the following drawings/documents in quantities specify in the requisition:
- Single-line diagram,
 - Schematic diagrams of all different types of circuits,
 - Documentation as specified in below for IMCS (Intelligent Motor Control Centre) only
 - General arrangement layout showing motor control unit/feeder control unit dimensions, central control unit dimensions, engineering workstation dimensions, mounting details of the units, connections, etc.
 - Cabling and wiring diagrams showing power input, output, terminals of motor/feeder control unit, central control unit, engineering workstation interface connections and include internal wiring cable schedule.
 - Setpoint setting tables for all units as set in the factory, hardcopy to be provided.
 - Configuration software for central/motor/feeder control unit.
 - Programming (logic) diagrams
 - Certificate of conformity for communication with FMCS/SCADA systems.

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- Arrangement drawing showing main circuits, main dimensions, panel layout, floor plan and shipping sections,
 - Minimum clearances during operation and maintenance around the assemblies for ventilation and safety,
 - Total mass of the assembly and of the individual shipping sections,
 - Transport, installation, commissioning, operation and maintenance instructions, limited and specific to the assembly and its components,
 - A list of recommended spare parts/components with model number,
 - Catalogs of protection relays, Power monitors, VFDs and other major components.
 - List of applicable type test certificates,
 - Test report of the final routine testing.
- u) Some of the above documents may be combined, provided all information indicated above is included.
- v) All documents shall show the relevant order number, item and Manufacturer's references, and shall be distributed as specified in the purchasing documents.
- w) Assemblies supplied in transport units shall have these units clearly marked to facilitate assembly at site.
- x) Erection instructions and any special tools required for erection and maintenance shall form an integral part of the order and shall be packed with the equipment.
- y) The language used shall be as specified in the requisition. Danger and warning labels shall also be provided.
- z) All structural work shall be adequately protected against corrosion. Frame and partitions may be of galvanised steel without a further paint coating. Those parts/covers requiring painting shall be properly pre-treated before the final coat(s) of oil resistant finishing paint is (are) applied.
- za) Terminals shall be suitable for copper conductors.
- zb) Adequate terminals and cable termination space shall be provided for the main cabling due to the substantial cable derating. Phase separation barriers shall be provided at terminal block.
- zc) All switchgear components requiring maintenance shall be easily accessible. All components in wall or back-to-back mounted switchgear shall be accessible from the front only.
- zd) Exposed parts within the assemblies which have to be accessible during normal operation, maintenance or equipping of spare compartments, shall either not be live in the opened position or shall be protected to a degree of protection of at least IP 20.

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- ze) Terminals of equipment installed on a compartment door and which can be live when the door is opened, shall be shrouded to a degree of protection of at least IP 20.
- zf) Compartment doors or covers shall be interlocked to prevent opening when the isolating switch/Circuit Breaker is in the 'ON' position. The isolating device/Circuit Breaker shall only be operable when the door is closed and when the withdrawable unit is in the inserted position.
- zg) All breakers in compartments shall be lockable in the OFF position to enable Lockout / tagout or CoHE.
- zh) Compartment doors and metal frame shall be electrically bonded to earth/ground point.
- zi) In the event of an internal fault in a functional unit, the damage shall be confined to that unit, so that the busbar system and all other functional units remain fit for further service. Likewise, an internal fault in a dropper system shall be confined to the associated panel, so that the busbar system and the other panels of the assembly remain fit for further service.
- zj) Assemblies shall comply with IEC/TR 61641 with regard to internal arc testing.
- zk) Busbar system (phase, neutral and earth/ground) shall be of hard or medium hard drawn, high conductivity copper. Busbar should preferably be located on top of the switchboard assembly.
- zl) Maintenance breaker or breaker slot to be provided for routine testing and maintenance of feeder breaker were shut down is not practical.
- zm) Switchboard construction to support reduce the live and exposed electrical work requirements and its related arcflash exposure risk.
- zn) Breaker lifting provisions to be provided where it is necessary.

6.11.2 LV Distribution Board (Fixed type)

- a) All distribution switchboards shall comply with IEC 61439-4.
- b) Main LV distribution switchboards supplied by one or more generating sets shall have an incoming switch for each generator. The outgoing circuits shall be protected by fuses or ACB/MCCBs having a maximum rating/setting of 2500 A.
- c) Each sub-distribution switchboard shall be provided with one incoming switch.
- d) Each outgoing feeder shall be provided with a switching device with short circuit and residual current protection. The maximum outgoing feeder rating shall be 125A.

- e) The sensitivity of the residual current circuit breakers shall be as follows:
 - 30 mA for circuits below 125 A;
 - 300 mA for circuits of 125 A.
- f) Fuses shall be accessible only if the fuse bases are protected to at least IP 2X, or if interlocking exists so that the fuse base is electrically isolated first.
- g) Switchboard with redundant power supply which allows for paralleling, shall be rated for maximum short circuit capacity for both sources.
- h) Plain cable feeders for LV sub-distribution shall be controlled and protected by fuse-switch combinations or by MCCBs incorporating short circuit and earth/ground fault protective devices.
- i) ACB/MCCBs and fuses shall coordinate with outgoing circuit protective devices on the receiving-end sub-distribution switchboard.
- j) The rating of any fuse-protected LV circuit, which derives a supply from a distribution switchboard, should not exceed 50% of the rating of a fuse-protected LV circuit, which energises the switchboard. This is to ensure that the circuit protective devices can be fast operating and selective in isolating short circuits.

6.11.3 LV Switchgear (Withdrawal type)

When a withdrawable unit has been removed from the assembly, the live parts inside the fixed compartment shall be protected against touch, with a degree of protection of at least IP 20. Where shutters are fitted to comply with above requirement, they shall be mechanically operated by the movement of the withdrawable unit and not be dependent on gravity

6.11.4 Circuit Breaker, Protective relay and other accessories

- 6.11.4.1 Circuit breakers in outgoing circuits shall comply with the requirements on operating accessibility, padlocking and phase and neutral switching.
- 6.11.4.2 Circuit breakers shall comply with and be type tested to IEC 947-2 and shall have utilisation category A, except for circuit breakers feeding sub-distribution switchboards which shall have utilisation category B.
- 6.11.4.3 Circuit breakers shall have a rated short-circuit making and breaking capacity in conformity with the prospective short-circuit current as specified in the requisition. Protective functions, such as short circuit,

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- overload and earth/ground-fault, should be part of built-in features of the circuit breakers.
- 6.11.4.4 Circuit breakers in a change-over supply duty should be withdrawable but shall at least have isolating facilities for maintenance and testing purposes.
- 6.11.4.5 Depending on the circuit/system it protects, protective relays shall be intelligent electronic device (IED) type with built-in communication module based on IEC 61850.
- 6.11.4.6 Protective relays selected shall be able to operate continuously with internal supervision and parts should not be obsolete in at least the first 10 years of service. Manufacturer selected should ensure backward compatibility.
- 6.11.4.7 The following list of protective relays may be applicable:
- i. Undervoltage relay (27)
 - ii. Overcurrent with instantaneous element (50/51), earth/ground-fault (50N/51N)
 - iii. Synchronizing relay (25)
 - iv. IED for motor protection (overload protection, earth/ground fault, stalled, etc.,)
 - v. Motor re-acceleration/restarting features (IED)
 - vi. Transformer differential (87T)
 - vii. Restricted/stand-by earth/ground fault
- 6.11.4.8 Motor earth/ground fault protection for 30kW and above, if it is not incorporated in IED shall be of core balance type. Relay shall be manually resettable after opening compartment. The sensitivity of the relays shall not exceed 3% of the rated nominal motor current or 6A, whichever is greater.
- 6.11.4.9 Switches/Isolators:
- Switches/Isolators shall be easily accessible and operable from the front panel of the LV compartments without having to open the door.
- Switches/Isolators shall be manual operating air-break with quick snap action in making and breaking the circuits.
- Switches/Isolators shall comply with and be type tested to IEC 60947-3. AC 22 utilization category shall be used when switching mixed resistive and inductive loads. Utilization category AC 23 shall be for highly inductive loads.
- Switches/Isolators shall be of the 4-pole type for 3-phases plus neutral system and 2-pole for single phase circuits. They shall be padlockable in the OFF position.

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Switches/Isolators shall be rated for short-circuit making capacity with prospective short-circuit current equal to required busbar system.

Switches/Isolators for incoming and bus coupler/section, if circuit breaker is not used, on a multi-section switchboard may have a short circuit rating based on actual distribution of the short circuit current. For example a coupled two-section board, about 50% of the maximum short circuit current will flow through each incomer.

Neutral pole of a switch/isolator in balanced 3 phase plus neutral system shall have current equal to other poles up to 63 amperes, inclusive. For devices above 63 A, neutral pole shall not be less than half of the current of other poles and a minimum rating of 63 A.

6.11.4.10 Metering:

- i. Each supply incoming unit shall be provided with one voltmeter connected to the busbar side between L1 and L3 phase. One ammeter measuring the L2 phase.

Metering supply to volt and watt-hour meters (if applicable) shall be protected with High breaking capacity (HBC-type) fuses of fully insulated pattern fixed directly onto the busbar system. A second set of fuses or miniature circuit breaker (MCB) with discrimination to the HRC fuses, shall be mounted locally near the instrument/s.

6.11.5 CT and VT

- a) Current transformers shall be in accordance with IEC 60185.
- b) Current transformers for measuring purposes shall be of accuracy class 3 (accuracy class 1 for measuring supplies from or to third parties) with an instrument security factor such that the instruments are not endangered at maximum fault current.
- c) The secondary current rating shall be 5 A or 1 A for ammeters and watt-hour meters mounted on the switchboard and 1 A for remote ammeters, either directly or via intermediate current transformers.
- d) Preferred primary ratings of instrument current transformers for remote ammeters are as follows:
10A, 20A, 30A, 50A, 75A, 100A, 150A, 200A, 300A, 400A, 500^a

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- e) Protective current transformers shall be of accuracy class 5P except in cases where class 10P is specified as adequate by the relay Manufacturer.
- f) The short-circuit rating shall be at least 1 second if applied for short-circuit protection.
- g) The wiring of the secondary circuits shall have a cross section of at least 2.5 mm².
- h) Secondary circuits of all current transformers shall be earthed/grounded.
- i) Voltage Transformer (VT), if required, shall be in accordance with IEC 60186 and shall be of class 3 (Class 1 for measuring supplies from or to third parties).
- j) Secondary side of the VT of metering transformers shall be 110V.
- k) The secondary circuits of the voltage transformers shall be earthed/grounded.

6.11.6 Control Power

- a) Control supply for circuit breaker tripping and protective functions/relays (including IED) shall be taken from the substation DC UPS (refer DC UPS).
- b) Two separate DC power supplies with battery back-up shall be wired to a selector switch at the LV Switchgear. Selector switch shall be make-before-break and each DC supply source shall be connected via a blocking diode.
- c) Loss of DC supply at the LV switchgear shall be monitored and alarm to FMCS (& SCADA) system.

6.11.7 Automatic and Manual transfer operation

- a) LV switchgear shall be equipped control logic that allow both automatic transfer system (ATS) and manual transfer system (MTS).
- b) Provide Selector switch to choose between ATS or MTS at Tie Cubicle.
- c) Conditions taken into consideration when design ATS and MTS for LV Switchgear.
 - If LV switchgear bus coupler (Tie) is always in closed position (NC), then ATS function is not required. MTS is still required.
 - If LV switchgear bus coupler (Tie) is always in open position (NO), then LV switchgear require both function ATS and MTS.

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- d) LV switchgear shall be able to perform Auto-Transfer and Auto-Restore during testing and troubleshooting when ATS function is available.
- e) Automatic transfer system operation shall meet all criterion prior to initiation.
 - Both Incoming circuit breakers (NC), tie circuit breaker (NO), are in normal position, in-service, healthy.
 - Permanent Power outage occurs at one side of Incoming line. This must be confirmed by at least two detections of undervoltage relays (27) at different point of measurement (Incoming line voltage and Bus voltage). Another Incoming line must be healthy, normal voltage, and free from bus fault or any fault.
 - After confirming item above, dead side Incoming line shall be isolated and opened. Associated dead bus must be health and free from any bus fault or any fault.
- f) After items are met then load transfer can be initiated by closing tie circuit breaker.
- g) Manual Restoration Operation should be recommended during normal power recovery.
- h) Manual Restoration/Transfer Operation
 - System is normal with no protective function activated.
 - Outage supply line is healthy and ready to take load with line undervoltage relay (27) normalized.

6.11.8 Installation

6.11.8.1 General Requirements

- i. Switchgear shall be installed in accordance with the installation instructions and supporting drawing provided by the Manufacturer. This include switchboard foundation and any inserts to be cast and within specific tolerances.
- ii. Switchboard shall be installed, as defined by Manufacturer, with sufficient space to facilitate maintenance.
- iii. Foundation drawings, together with information on mass and loading data, shall be provided to permit correct civil design of building and foundations.
- iv. If switchboards and/or controlgear are from different Manufacturer, one composite foundation drawing shall be supplied.
- v. Equipment arrangement and layout drawing shall be provided by Manufacturer or supplier for it to

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- be incorporated into the overall arrangement drawings and cabling design.
- vi. Substation floors shall be smooth and level to permit the handling of equipment on rollers. Concrete floors shall have a surface protection layer to avoid formation of dust.
 - vii. Switchboard shall only be installed after the completion of switchroom civil and building works, this is to minimise the ingress of dust and dirt during or after the erection.
 - viii. Any modifications required during installation shall be approved by Owner prior to implementation. All modifications shall be recorded and information incorporated into the "as-built" drawings.
 - ix. Switchboards shall be installed in location that minimise the risk of vibration at vicinity. If unavoidable, anti-vibration mountings may be used.
 - x. Switchboard installed shall avoid any pipes work in vicinity. If unavoidable, pipes should not have joints at that area.
 - xi. All cables shall be enter from the bottom or side of the switchboard, for cases where the installation is outside of the electrical room, unless the switchboard is installed in electrical or control room with properly insulated and covered, to avoid potential risk caused by water, liquid...etc.

6.12 Low Voltage Motor Control Centre (LV MCC)

6.12.1 General

- a) LV MCC shall be design and built to comply to IEC 61439 or UL845 or NEMA ICS 18 standard include arc resistance construction and arcflash containment.
- b) Only natural cooling shall be applied for LV MCC
- c) Refer to LV Switchgear and controlgear general requirement (6.11.1) shall applied.
- d) A space of at least 0.2 metre shall be available from the floor to any measuring apparatus, switch, indication light, push button or any other components. Maximum height of the above components shall not be more than 2 metres above the floor level. Also refer to NFPA 17 or UL standard 845 for compliances.
- e) Withdrawal units which is electrical and mechanical identical shall be interchangeable.

6.12.2 Switches/Isolators:

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- a) Switches/Isolators shall be easily accessible and operable from the front panel of the LV MCC compartments without having to open the compartment cover.
- b) Switches/Isolators shall be manual operating air-break with quick snap action in making and breaking the circuits.
- c) Switches/Isolators shall comply with and be type tested to IEC 60947-3. AC23 utilization category shall be used for motor switching loads.
- d) Switches/Isolators shall be of the 4-pole type for 3-phases plus neutral system and 2-pole for single phase circuits. They shall be padlockable in the OFF position.
- e) Switches/Isolators shall be rated for short-circuit making capacity with prospective short-circuit current equal to required busbar system.
- f) Switches/Isolators for incoming and bus coupler/section, if circuit breaker is not used, on a multi-section switchboard may have a short circuit rating based on actual distribution of the short circuit current. For example a coupled two-section board, about 50% of the maximum short circuit current will flow through each incomer.
- g) Neutral pole of a switch/isolator in balanced 3 phase plus neutral system shall have current equal to other poles up to 63 amperes, inclusive. For devices above 63 A, neutral pole shall not be less than half of the current of other poles and a minimum rating of 63 A.

6.12.3 Contactors:

- a. Contactors shall be of the holding coil type with operating voltage from phase and neutral.
- b. Motor starters and contactors shall conform to the requirements of intermittent and uninterrupted duty class 12 (12 operation per hour) at rated current. They shall have mechanical withstand of at least one million no load operation. Comply to IEC 60947-4
- c. The utilization category of AC-3 shall be applied, for starting of squirrel-cage motor, switching off motors during running. Motors use for application duty such as inching or reversing rotating service, AC-4 utilization category shall be used.
- d. Indicators:

On-off indication of contactors shall be provided on the front of the compartment and clear indication green light for OFF and red light for ON. Indication lights shall be low in power consumption and durable, such as multi-segment LED.

6.12.4 Motor protection relays and meters:

- a. Overload protection relays shall be installed and have single phasing element that will operate at least 10% faster. Relays shall be at least of trip class 20 in accordance to IEC 60947-4-1 and be of the temperature compensated type. They shall be equip with manual resetting features on the front door of the compartment and accidental depression of the reset action shall not trip the circuit.
- b. For motors with long starting time in excess of 5 seconds, a saturable current transformers in combination with standard nominal 5 or 1 A relay shall be provided. Alternatively electronic overload protection with suitable characteristic shall be provided.
- c. Motor earth/ground fault protection for 30kW and above, if it is not incorporated in IED/IMCS shall of core balance type. Relay shall be manually resettable after opening compartment. The sensitivity of the relays shall not exceed 3% of the rated nominal motor current or 6A, whichever is greater.
- d. Motor starter units for motor above 4kW shall be equipped with current transformer for remote ammeter. For motors below 4kW, a direct connection for remote metering from motors cable can be used.
- e. All protection and control setting to be reviewed and approved by Micron engineer for reliable starting and operation requirements without compromising the safety and selectivity.

6.12.5 Integrated Motor Control System

- a) Integrated Motor Control Systems (IMCS), may be used, provided it can be demonstrated that they are cost effective over the total life cycle in comparison with conventional LV motor starters.
- b) System shall comprise Motor Control Units (MCUs), Feeder Control Units (FCUs), Central Control Unit (CCUs), a serial bus connecting MCUs and FCUs to the CCU. A serial communication facility enabling connection of Central Control Unit (CCU) to Supervisory Control and Data Acquisition System (SCADA) and/or operation control system (FMCS)
- c) Scope of supply should include start-up assistance, site installation and training of operational/maintenance staff.
- d) Switchboard Manufacturer shall be responsible for the performance of the IMCS and shall liaise with FMCS and/or SCADA interfaces and testing.
- e) A guarantee of system availability of the IMCS is required a dual redundant CCU and redundant serial link configuration shall be provided. Redundancy arrangement shall have a

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Master-Slave, hot standby system, such that a failure event will have a bumpless transfer not bring the system down.

- f) Manufacturer shall guarantee that spare parts and software support will remain available for the expected lifetime of the system with a minimum of 10 years. Thereafter, any changes to parts and software shall be backward compatible.
- g) Functions of MCU and FCU shall have measurement, control and protection and self-sufficient protection system with a non-volatile memory of all pre-programmed parameters. No internal battery use for backup is permitted. These units shall be able to maintain operation capability after voltage disruption.
- h) Engineering work station (EWS) shall be able to set/change parameters and monitor the operation status of any connected MCU or FCU.

6.12.6 Soft starter Type

- a) Solid state reduce voltage starters design to reduce the inrush current to a motor during starting and limit the amount of starting torque should be used if application and system requires.
- b) The amount of starting current shall be field adjusted to match the site application.
- c) Soft starter shall have features including but limited to soft stop, kick start, stall protection, phase loss, etc.,
- d) Design should contain an integral full rated bypass relay that eliminate heat generation when motor is at speed.

6.12.7 Variable Speed/Frequency Drive type (VSD)

- a) Manufacture shall ensure that VSD in LV MCC does not fail prematurely and/or cause EMC interferences. Heat generated by VSD or by the switchgear loads shall not impact the life expectancy of the MCC.
- b) Depending on the torque applications, design shall select the appropriate controllers of adjustable frequency drives. Example constant torque application for conveyor, mixers, etc., and variable torque application such as pumps, fans, etc.,
- c) Drives for variable torque load shall be rated a minimum of 110% overcurrent for 1 minute.
- d) Drive shall have identical keypad, programable parameters and control terminals.
- e) Control schemes are available for volt/Hz, open loop and close loop vector models.
- f) During the engineering and design stage, contractor shall select features/options to match machine application such as

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- integral line reactor, output reactor, Dv/Dt filter, line fuses, isolation contactors, bypass contactor etc.
- g) Drives shall have communication cards capable of communication on common standard protocol with Scada/FMCS.
 - h) A means of mitigating or shielding control or communication cables from EMI/RFI interference generated by the drive shall be provided.
 - i) The VSD shall operate continuously within the specified voltage and frequency limits, from no-load to full-load conditions without thermal degradation.
 - j) The VSD shall be a Pulse Width Modulation (PWM) type suitable for microprocessor-based digital control. The VSD shall accomplish stepless speed control by adjusting both the output voltage and frequency to the motor.
 - k) Equipment shall be capable of operating in an area where the temperature is 0 degrees C to 40 degrees C (32 degrees F to 104 degrees F), with Relative Humidity 0% to 95% without being derated.
 - l) Field mounted units within buildings shall be afforded an environmental degree of protection of not less than IP31 (NEMA 1) in accordance with IEC 60529. The preference is to not install VSD units outdoors. If they must be installed outdoors, the units shall be afforded an environmental degree of protection of not less than IP55 (NEMA 4X) and will be fitted with a sun cover if appropriate.
 - m) The VSD and associated wiring/controls and settings shall be designed to be resistant to transients and voltage dips due to motor starting and incoming power voltage sags and other system disturbances.
 - n) The VSD and associated wiring shall be designed to be resistant to transients and voltage dips due to motor starting and other system disturbances.
 - o) The VSD assembly shall be supplied with all appropriate listings, certifications, and test reports.
 - p) The VSD shall have sufficient capacity to deliver sequential starting and loading duties.
 - q) If conditions warrant, the unit will be ordered with tropicalization and fungus protection.
 - r) The available fault current at the point of service will be calculated and the VSD and its components shall have suitable withstand rating. The VSD vendor shall provide any addition fusing that may be needed.
 - s) At a minimum, the VSD shall have self-protection and reliability features including overvoltage and undervoltage trip, overload and overcurrent trip, inverter overtemperature, thermal overload protection for the motor, and loss of phase protection.
 - t) The VSD shall have a minimum transient ride through capability of 500 mS without tripping in during a voltage dip

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- of 10% - 100%. The control logic shall have a minimum of a 2 second ride through capability on systems deemed critical.
- u) When motor speed is reduced due to a momentary power failure, the VSD, upon supply restoration shall synchronize with the spinning motor and re-accelerate the drive (catch on the fly).
 - v) The VSD shall be capable of synchronizing with an emergency power system.
 - w) VSD units shall be self-ventilating with powered cooling fans as required. VSD enclosures shall be ventilated for cooling and fitted with easily-replaceable air filters installed over the ventilation louvers. Ventilation openings shall not compromise the rating of the enclosure.
 - x) Applications with long motor lead lengths between the VSD and motor shall be evaluated for the use of dv/dt filters
 - y) VSD shall have input line conditioning to limit current harmonic distortion measured at the input terminals of the VSD while operating at full load to:
 - Motors 35 kW (25 HP) and less - 45% THID
 - Motors between 35 kW (25 HP) and 70 kW (50 HP) - 40% THID
 - Motors between 70 kW (50 HP) and 135 kW (100 HP) - 30% THID
 - Motors over 135 kW (100 HP) - 20% THID
 - z) VSD units shall be functionally tested by the manufacturer at the factory. Testing to include, but not be limited to, insulation test, temperature rise, output frequency and voltage tolerance, harmonic content on the system input and output, efficiency, and voltage dip test. A written test report shall be furnished with each VSD unit.
 - aa) A maintenance bypass shall be considered for drives serving systems deemed critical to allow maintenance to be performed without interruption to the load, Excluded the redundant or N+1 system.

6.12.8 Installation

- a) Refer to section 6.11.8 – similar to LV switchgear and Controlgear assemblies
- b) Arc Shield clearance height shall comply with manufacturer requirement.
- c) Bus torque and all other connections shall follow the manufacturer installation guide.
- d) Installation of conduit and cables shall follow local code and supplier/manufacturer guide.
- e) Remove all movement/transportation securing bolts/pins/brackets/etc., and lifting angle/bars/plates.

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- f) Cables to be secured properly to prevent stress to insulators or cable terminations. Local practice to apply include hardwood brace, cables glanding plates, glass tapes, etc.
- g) Check all starter units for damage and include withdrawal units for free movement. Check functionality of all operating handles/levers and mechanical interlocks.
- h) Qualified person shall follow the manufacturer recommended checklist to ensure all items are checked, including earthing/grounding connections/earth (ground) loop impedance, prior to energisation. A qualified person is defined by NEMA Standards Publication/No. ICS 2.3
- i) All cables shall be enter from the bottom or side of the MCCB/DB, for cases where the installation is outside of the electrical room, unless the MCCB/DB is installed in electrical or control room with properly insulated and covered, to avoid potential risk caused by water, liquid...etc.

6.13 Power Cables

6.13.1 General requirement

- a) In the early phase of the project (Define), cables type, routing and underground/above ground shall be specified. Method, diversity factors and protection against mechanical, fire, temperature/environment, special protection against chemical, etc., needs to be considered.
- b) Design cable layout, cable route, construction method and schedule shall be adopted. Cable system to take into account of proper segregation, earthing/grounding, bonding requirements for electric safety and EMC.
- c) Single core cables should be avoided, if used due to economic, capacity and or HV switchgear connections requirements, then additional precautions needed.
- d) Cables shall be segregated and suitably identified for the applicable category:
- e) Power Cables (HV/MV/LV)
- f) Signaling and instrumentation cables
- g) Telecommunications (CCTV, security, etc.,)
- h) Fire and Gas – detection and protection
- i) Dedicated cables shall be used for CT and VT secondary circuits, interlock/trip circuits, pilot wire or others differential circuits, etc.,
- j) Fibre optic cables is preferred for use for in transmission of signals and data such as Scada/DCS.
- k) All cables should preferably be of copper conductor. If aluminum conductors are to be used, care should be taken to prevent galvanic corrosion.

6.13.2 High Voltage (HV) Power Cables

- a) Single core or three cores HV power cables conforming to standard (AS/NZS 1429, IEC 60840 and IEC 62067) shall be cross-linked polyethylene (XLPE) insulated, copper/Aldrey wire screen over each insulated conductor, single galvanized steel wire armoured or corrugated aluminum/copper and LSHF over-sheathed.
- b) Single core cables shall be designed with derating factor that include sheath current. Cross bonding and earthing/grounding shall be at every 1/3 of the length or multiple of it, if the distance is very long. Other method of bonding/earthing (grounding) can be considered with approval from the owner.
- c) Cross bonding inspection boxes shall be accessible for inspection and maintenance.
- d) Engineering design of HV/EHV cables shall ensure proper load factor applied direct buried or in duct or free air. Considering of cable formation (flat or trefoil), spread/space between, laying depth, Soil temperature and thermal resistivity, soil drying, expose to solar radiation, air temperature, etc.,
- e) Embed optical fibers in cables metal screen can monitor cable temperature to detect cable hot spots and cable capacity.
- f) Partial discharge measurement through capacitive or inductive sensors embed on joints or link boxes.
- g) Cable sheath to be grounded at single end and other end to have sheath voltage limiter of appropriate rating. For extended cable length and joints, the location of the grounding to be decided based on sheath voltage increase.

6.13.3 MV Power Cables

- a) Three core or single core MV power cables conforming to standard (AS/NZS 1429, IEC 60840 and IEC 62067) shall be dry cured, cross-linked polyethylene (XLPE) insulated, copper screen over each insulated conductor, single galvanized steel wire armoured or corrugated aluminum/copper and LSHF over-sheathed.
- b) For greater flexibility, ethylene propylene rubber (EPR) insulation may be specified.
- c) Multicore cables shall have a minimum cross-sectional area of 25 mm² and a maximum cross-sectional area of 300mm²
- d) Depending on the voltage level and cable construction, cable screen may be installed over each insulated conductor.

6.13.4 LV Power Cables

- a) Twin and multicore cables for power, lighting and control shall be cross-linked polyethylene insulated (XLPE), galvanized steel wire armoured and LSHF oversheathed. Steel wire armouring may be replaced by braiding for cable sizes up to and including 10mm². Cables with EPR insulation may be specified when site requires higher flexibility.
- b) Maximum cross-section shall be 185 mm² for motor cables and 240 mm² for distribution/feeders' cables.
- c) Power, lighting and control cables shall have a minimum cross section of 2.5 mm² except for signaling and indication purposes.
- d) Single core cables shall be XLPE or EPR type.

6.13.5 Installation

- a) Check cable for (a) correct size and type; (b) Shipping/transportation damage; (c) End seals; (d) Special instruction in handling
- b) Check reels for (a) damage; (b) protruding nails that might damage cable; (c) Staples.
- c) Use proper pulling equipment/method for different reels and cable types. Cable manufacturer shall be consulted and approve the installation process/procedures.
- d) Care to be taken for cold weather and or high humidity installation of cables, considering the temperature swing affecting the laying process and splicing/termination quality.
- e) Installer shall check required information with vendor/manufacturer on handling and laying/installing of power cables. Information include:
 - Minimum bending radius when pulling and after installation
 - Maximum pull force. Ensure right pulling head is properly secured to the conductor. Tension monitoring device shall be used especially for HV/EHV cables.
 - Total weight of cable drums
- f) HV/EHV cables shall be installed by qualified contractor or manufacturer recommended installer and or jointers.
- g) HV/EHV cables and accessories (joints, terminations, sealing ends, stop joints/bonding, link boxes, etc) shall be supplied by the same vendor/manufacturer.
- h) Protect the cables from storage, transportation and installation against external and mechanical aggression, corrosion and environment (moisture, high temperature, etc)

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- i) Buried cables consideration of cable trench (excavated depth and width), excavated flooring and smooth bedding, distance between cables, earthing/grounding cables, backfilling, warning device/notice. All of these consideration shall be provided in the cable layout and cross section drawings produce by the contractor.
- j) Other installation methods such as in conduits, ducts, laying in galleries, trunking and cable ladders.

6.14 Lighting and Convenience Power Outlets

6.14.1 General Lighting Requirements

- a) Industrial luminaire (LED) in 'white' colour shall be used for illumination unless in low ambient temperatures (sub zero).
- b) Long life lamps in combination with electronic ballasts shall be used, to take advantage of their increased lumen efficiency and economic life.
- c) Lighting fouling factor should be 0.8
- d) Lighting design software shall be used to determine a photometric data of the light distribution. It will also optimize the number of luminaires used and the selection of the appropriate luminaire for the application.
- e) High pressure discharge lamps should be used to light tall buildings or large areas:
 - In view of the restarting time of this type of lighting after a voltage dip, sufficient LED luminaires shall be installed for basic lighting requirements of the area, equivalent to emergency lighting requirements below.
- f) Floodlighting shall be used at the perimeter of process and production plants.
- g) Maintenance free, sealed-for-life, discharge lamps and associated luminaires may be considered with account being taken of their total life-cycle cost subject to approval by the Owner.
- h) Low pressure sodium discharge lamps shall not be used.
- i) Incandescent lighting shall be applied only for decorative lighting.
- j) Siting of lighting fixture should avoid shadows and reflections/glare to monitors/VDUs (video display unit).
- k) Lighting design software shall be use for building/FAB lighting with the consideration of space utilization, partitions, HVAC ducting/units and fire protection system.
- l) Special considerations shall be given to locations with wet, corrosive, or hazardous atmospheres.

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- m) Lighting designs shall be performed with a software program that provides photometric data to help in the selection of products and options that are most appropriate to meet design requirement and application.
- n) Energy efficiency issues shall be reviewed, including avoiding over-illumination, the use of daylighting, energy saving technologies, and control schemes.
- o) Compliance considerations such as security issues, life-safety considerations, energy codes, and building/electrical codes.
- p) Compliance considerations such as security issues, life-safety considerations, energy codes, and building/electrical codes.
- q) If not addressed by local code, a minimum lighting level shall be established for each functional area and incorporated into the site design standards.
- r) Luminaires installed in fire rated structures shall be appropriately rated to maintain the integrity of the area.

6.14.2 Illumination Level

- a) The required illumination levels when measured at working plane (1 meter above the floor level in a horizontal plane). Table below shows the required illumination level.
- b) These are mean values and the uniformity ratio (E_{min}/E_{mean}) is $\frac{1}{4}$ for normal installation.
- c) Values shall be the basis for new design installations unless higher illumination levels are required by local authority.

Table - Required Illuminations Levels	
Location	E _{mean} (Lux)
Cleanroom (Photo area)	500/Yellow
Cleanroom (Except Photo area)	500
Clean Sub-fab (Photo area)	250/Yellow
Clean Sub-fab (Except Photo area)	250
Utility spaces (electrical, mechanical, process, Gas/Chem.)	200
Warehouse	200
Substation	250
Control/monitoring room	500
Office	500
Lab area	500

Mezzanine area / Truss area	150
Pent house	150
Mechanical yard	150
Air returned shaft	100
Corridor, Sub-fab walk way	100
Stair	150
Car Park	50
Roof (equipment area)	50
Outdoor road	10

- d) Higher illumination levels apply where security fence lighting with surveillance camera are used.

6.14.3 Special Lighting

- a) Aviation warning lights, if requires, shall be installed in accordance to ICAO Annex 14, volume1 Chapter 6. Luminaires shall each consist of a double lamp unit with auto change-over if one lamp failed to operate.
- b) Navigation aid light, if applicable, shall comply to the local and international code.

6.14.4 Emergency lighting

- a) Emergency lighting shall provide sufficient level of illumination (150 lux) to allow minimum operation and evacuation areas. It shall be installed at strategic points and ensure uniformity in lumen distribution.
- b) The emergency lighting luminaires shall comprise the following percentage of the total luminaires:
- Control room 50%
 - Cleanroom 25%
 - Substation 20%
 - Utility 20%
 - Office 5%
- c) Emergency lighting for Utility shall be feed from static UPS and the cabling to supply those luminaire shall be of fire resistant (FR) type. Local code and any other requirements in the fire safety regulations shall applied.

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- d) Escape light shall form part of the emergency lighting system and be located so as to illuminate the escape route, walkway and ladders to allow safe movement of personnel. Intensity of the illumination shall be at least 10 lux.
- e) If emergency lighting supply cannot be taken from a secured source, such as UPS, integral battery pack or charger shall be supplied and approved by Owner.

6.14.5 Material and Type

- a) Outdoor light fixture shall be environmentally friendly with high Ultra-Violet (UV) protection, long life, better lumen output and accurate colour temperature.
- b) Reflector design shall be effective to avoid disability glare and multi shadowing.
- c) Heat sinks especially for LED fixtures should be engineered to effectively remove/dissipate heat.
- d) Housing of fixture should preferably be glass fiber reinforced polyester or equivalent.
- e) Reflective/protective bowl should be polycarbonate or strong acrylic material/fiber.
- f) Acrylic lenses use shall not be used near or in areas with airborne contaminants or chemicals that could impact the integrity of the acrylic material.
- g) The luminaire shall be suitable for the environmental conditions at the installation site. It shall be protected from the entrance of rain or snow, and from the maximum wind velocity required by local building codes.
- h) Electronic ballasts shall be rated for a total harmonic distortion of less than 20%.

6.14.6 Control Devices (Switches/Occupancy sensors/Photocell)

- a) Lightings for building and switch room shall have switches control at the internal of the access points.
- b) Emergency lighting switch, if used, shall be clearly identifiable. Luminaires used for security and safety shall normally be ON, switch or means of isolation should not be easily access by the general staff/public.
- c) All outdoor lighting and/or lightings with assisted natural light shall be designed with automatic switching via photo-electric relays.
- d) Controls of photo-electric relays shall have manual/maintenance bypass and with auto "fail-safe" (i.e. switch lights ON when photo-electric relay failed) features.

6.14.7 Protective Devices for lighting

- a) Lighting circuits shall be single phase and neutral or three phase and neutral.
- b) They should be protected with a minimum 16 A fuse or MCBs (miniature circuit breakers), but not be loaded up to 12 A.

6.14.8 Power and Convenience outlets

6.14.8.1 General

- i. For maintenance purposes, three phase and neutral power outlets for movable equipment, and single phase and neutral convenience outlets for the supply of portable tools and hand lamps, shall be provided
- ii. The number and location of outlets should be based on the maintenance activity expected in the area and input from the human factors engineering study.
- iii. The outlets shall be standardised for each rating and type throughout the complex and have an earth/ground connection incorporated.
- iv. Each LV power and convenience outlet circuit shall be protected by phase short circuit protective devices and by current-operated earth/ground leakage protective devices, which are in accordance with IEC 60947-2, i.e., residual current circuit breakers (RCCB).
- v. The RCCB operating current shall be 30 mA for circuits of less than 125 A and 300 mA for circuits equal to or greater than 125 A.
- vi. Components shall be suitable for their particular application as regards to their voltages, rated currents, service life, short circuit withstand, making and breaking capacities

6.14.8.2 Power and Welding outlets

- i. Power outlets shall have a standard supply voltage equal to the LV supply voltage selected for the complex
- ii. Power outlets shall be rated of at least 125A and suitable for outdoor installation.
- iii. Power outlets to be located at high maintenance activities area and space in such a way that, with the aid of extension cables feeding movable secondary supply panels, all points can be served conveniently.
- iv. Power outlets shall be connected so as to have the same phase rotation, ensuring that correct rotation of moveable equipment is obtained from all outlets.

6.14.8.3 Convenience Outlets

- i. Convenience outlets shall have a standard supply voltage equal to the voltage selected for normal lighting.
- ii. The outlets shall be industrial pattern complying to IEC 60309 or equivalent UL1686 standard. Use of local standard material requires the approval of the Owner.
- iii. Plugs shall not be interchangeable with sockets of a different voltage or current rating.
- iv. Socket outlets and plugs shall be at least IP54, degree of protection, both when the plug is removed and when it is fully inserted.
- v. All three phase outlets shall be connected with the same phase sequence.
- vi. For industrial areas, the outlets shall be rated for at least 16A and be suitable for outdoor installation.
- vii. In workshops, the outlets shall be of industrial pattern.

6.14.9 Installation

- a) Lighting installations shall be designed to obviate stroboscopic effects.
- b) Luminaires on tall structures/building shall be design and located so that maintenance and lamp change can be done without the use of scaffolding or complex structure/equipment.
- c) Where a luminaire mounted from elevated area shall be arranged to swivel for maintenance purposes.
- d) Lighting poles shall be hot-dip galvanized in accordance to ISO 1461.
- e) Hinged lighting poles/columns which can be lowered to ease maintenance, may be used if space is available.

6.15 Earthing/Grounding and lightning protection

6.15.1 General Requirement

6.15.1.1 One integrated earthing/grounding system shall be installed reduce and control voltage to an acceptable low level for personal safety, damage to asset and operational protection.

Reasons:

- i. Electrical safety – reduce touch and step voltages in case of earth/ground faults.
- ii. Intrinsic safety – avoid ignition source, prevention of fire and equipment damage prevention.
- iii. Lightning and static electricity protection – prevention of fire, dangerous touch voltage and ignition source
- iv. Electromagnetic Compatibility – reduce uncontrolled voltages and/or frequency at terminals of electronic equipment.

6.15.1.2 Perimeter fence shall be earthed/grounded so as to avoid the danger of touch and transferred voltage, as guided by IEEE 80.

6.15.2 Power System Earthing/Grounding

6.15.2.1 The neutral of earthing/grounding system for electrical distribution systems and uninterruptable power supply system, should be earthed/grounded. Design shall not be for unearthed (ungrounded) system unless they form part of an existing unearthed/grounded system.

6.15.2.2 Substation neutral system earth/ground and all metal supporting structures and equipment shall be earthed/grounded to the substation earthing/grounding system.

6.15.2.3 HV and or MV electrical systems shall be earthed/grounded by means of dedicated earth/ground electrodes connected to the main earthing/grounding Grid.

6.15.2.4 HV and or MV system neutrals shall be earthed/grounded at each source of supply be it transformers or direct connected generator/s.

6.15.2.5 Grid infeed system for voltage 36 kV, the neutral point of transformers should be solidly earthed/grounded unless required by the public utility.

6.15.2.6 Transformer feeders to HV switchboards with a system voltage not exceeding 36kV shall be resistance earthed/grounded and it should limit earth/ground fault

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current to about 100 A. System design should consider multiple earthing/grounding points for each transformer increasing the magnitude of earth/ground fault current and circulating harmonic current to a harmless value.

6.15.2.7 Generators that are directly connected to the main HV switchboard (without generator transformers) should be earthed/grounded via its own earthing/grounding resistor.

6.15.2.8 Each earthing/grounding transformer and resistor shall be rated to withstand the respective earth/ground fault currents for a duration of not less than 10 second. A longer withstand times may be required if earth/ground fault protection coordination requires.

6.15.2.9 LV electrical systems neutral shall be solidly earthed/grounded at each source of supply by means of dedicated earth/ground electrodes that have direct or low impedance connection to the main earthing/grounding Grid.

6.15.3 Electrical equipment earthing/grounding

6.15.3.1 Substation equipment shall be protected against lightning and switching over-voltages by lightning arrestors.

6.15.3.2 AC UPS system shall have their neutrals solidly earthed/grounded, for both single and three phase systems.

6.15.3.3 DC system supplying loads/instruments and switchgear control and protections load shall be earthed/grounded through a high resistance earth/ground fault monitoring unit with a sensitivity of 5mA.

6.15.3.4 Submerged motors or equipment shall be fed by flexible oil resistance cables with earth/ground connector shielding each core.

6.15.4 Control system earthing/grounding

6.15.4.1 Control system reliability and performance depends on effective earthing/grounding and bonding. Hence, adequate and good earthing/grounding network shall be implemented.

6.15.4.2 Earthing/grounding for control system shall ensure it address electro-magnetic compatibility (EMC) to the control instruments/appliances and or the controllers/data centre.

6.15.5 Process system earthing/grounding

6.15.5.1 All conducting support structures and apparatus mounted on non-conducting support structures shall be earthed/grounded at the foot of the support.

6.15.6 Electro-static discharge (ESD) clean room earthing/grounding

6.15.6.1 Building that house the clean room or stand-alone cleanroom shall be designed with lightning protection system, sound earthing/grounding with earth/ground grid and electro-magnetic compatibility.

6.15.6.2 Multipoint earthing/grounding system shall be installed, incorporating all earthing/grounding functions such as safety earth/ground, lightning protection earth/ground, static electricity earth/ground and functional earth/ground for instrument and control systems. The basis of this multipoint earthing/grounding system is the earth/ground grid, with a 10 m x 10 m mesh for all buildings.

6.15.6.3 All cables armouring and screening shall be bonded to the earth/ground grid. Bonding for sensitive buildings and cleanrooms shall be done just outside the building. Selection of material use to eliminate/reduce the possibility of static electricity.

6.15.6.4 Contractor shall be familiar with SEMI E78-0998 "Electrostatic Compatibility – Guide to assess and Control Electrostatic discharge and Electrostatic Attraction for equipment" and apply for clean house design.

6.15.7 Lightning

- 6.15.7.1 Lightning protection systems shall be installed in accordance with local regulations. In the absence of such regulations, the system shall be designed and installed in accordance with IEC 62305.
- 6.15.7.2 Protection against direct lightning strikes shall be provided earth/ground conductors and lightning rods/tapes attached to substations, buildings and other structures.
- 6.15.7.3 For the protection of structures against direct discharges a Lightning Protection System (LPS) should be implemented, comprised of captors to intercept lightning strokes: down conductors to conduct the resulting lightning currents to the earthing/grounding system, and the earth/ground electrode system to spread the lightning currents into the soil/ground.
- 6.15.7.4 Tall expose structures such as buildings, telecom towers, shelters, etc., supporting earth/ground electrodes shall be installed at the base of these structures to lead the lightning currents to earth/ground.
- 6.15.7.5 Metal plating or cladding at the perimeter of buildings, metal gutters, post guard steel pipes and metal downpipes located on the roof shall be bonded to the concrete reinforcement bars and thus be integrated with the external lightning protection installation.
- 6.15.7.6 If cable trunking is located at the highest point and thus exposed to direct lightning strikes, closed ducts with steel covers shall be used.
- 6.15.7.7 Equipment in lightning exposed areas, according to the criteria of BS 6651, shall have an air termination network, via down conductors connected to the earth/ground grid, by means of copper stranded wire of minimum cross section 50 mm².
- 6.15.7.8 For lightning prone areas, a risk assessment of critical parts of the installation where equipment damage due to direct strokes and Lightning Electro Magnetic Pulse (LEMP) is expected and the specification of any additional EMC measures.

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6.15.8 Installation

- 6.15.8.1 Installation of earthing/grounding grid and earth/ground electrodes shall be as design earthing/grounding and lightning layout. Choice of proper earth/ground electrodes/type depend also on the environment and probability of corrosion.
- 6.15.8.2 Materials, appliances and components shall comply to local and international standard.
- 6.15.8.3 Adjacent earth/ground electrodes shall be at least 2.4 meters away or further than the depth of the electrodes driven into the ground.
- 6.15.8.4 Earth/ground electrode shall be driven in earth/ground to at least 3 meters before earth/ground loop measurement is taken to check the effectiveness. If it does not meet the requirement, further electrode shall be driven into the ground to achieve the required value. All earth/ground electrode shall be individually recorded.
- 6.15.8.5 Earth/ground pit shall be installed after completion of installation of earth/ground electrodes connection and have clear gaps for inspection and maintenance check. Each electrode installed shall have connection of a loop-in and out earth/ground cables to form part of the earthing/grounding grid system. Earth/ground pit covers shall be levelled with the finished level of the area.
- 6.15.8.6 PVC sheathed cables of 100 mm² shall be laid between the earth/ground grid and earth/ground bars or other metallic structure/equipment. Compression C-clamp or fused connections shall be used to connect to the earthing/grounding system. All earthing/grounding connections shall have its surfaces thoroughly clean and tightness checked.
- 6.15.8.7 Metallic frame of all equipment shall be connected to the nearest earth/ground bar or earthing/grounding grid.
- 6.15.8.8 Earth/ground continuity for cable ladders/trays and trunking shall be maintained with earth (ground) links/connections to the earthing (grounding) bars/system.
- 6.15.8.9 Earth/ground tapes, clamps, inspection points, tape joints (preferable thermo weld), down conductors, other accessories used for lightning system shall ensure effective lightning protection. All down conductors connecting to the earth/ground pits for lightning protection shall be take the shortest possible route.

6.16 Power Monitoring System/Power Scada (PMCS)

6.16.1 General Requirements

- a) Power Monitoring System or Power Scada shall be ordered directly with manufacturer and approved by Owner. Manufacturer shall be responsible for the correct functioning of the system to specification, including all interfaces with other systems and equipment.
- b) FMCS and IS Machine room for hardware configuration. N+1 redundancy to be maintained at minimum for all critical equipment, SCADA hardware and metering. Power Quality grade metering to be provided at main power intake switchgear for each incoming power source. SCADA system configuration to prevent system outage from any single point of failure (SPOF)
- c) Manufacturer shall provide following data at design and manufacturing stages:
 - Technical specification of the SCADA system describing the design, operation, construction, performance and maintenance aspects.
 - Installed condition and / or environment.
 - System block diagrams showing functional units, hardware and supply boundaries. Propose completed Scada architecture showing all interfaces, components, HMI, EWS (Engineering workstation), Operator station, field devices (IED), RTU/PLC, servers/computers, etc.,
 - Future extension of system
 - Project quality assurance program with proposal for the inspections to be performed by Owner/Contractor
 - Propose tests to be carried out on functional description of each test.
 - Manufacturer and country of origin of main components of the system
 - List of recommended spare parts
 - List of all documents to be delivered with the system
 - List of communication protocols applied
 - Training course available for operation and maintenance
 - Detail schedule of implementation, installation guides and witness points
 - Supply requirement and protection for the system
- d) Scada system shall be able to present all information that is available within the system and agreed with Owner on the screen presentation. The following information shall be available in the Human Machine Interface (HMI):
 - A quick overview of the network components with information; System Configuration;

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- System information of all sub-systems; I/O Programming;
 - Overview of generation/s, transmission (if applicable), distribution and network.
 - Information on control, alarm, object status and measurements;
 - Reports, events record data, trends, recorded transients/disturbance analysis;
 - Information on system, operation and disturbance.
 - Colors shall be used to distinguish network voltages, alarms, operation conditions and color will need to be approved by Owner.
- e) The scope of supply for Scada system shall include all design activities, supply of materials, factory acceptance tests and provision of documentation and instructions.
- f) Integration of the Scada system, Human Machine Interface (HMI) and engineering workstation/facilities and with process system, if applicable.
- g) SCADA Access
Viewer access to be provided to enable the remote monitoring for all electrical engineers, electricians and control room operators for their reference and fast response. Full access to be limited to core group who are trained to handle the SCADA configurations and modifications. ALL access to be controlled by authorized users based on site IT policy.

Electrical supply:

All PLC and remote I/O panels shall be powered with two circuits, one of which shall be Emergency Backed UPS and serve as the primary power for the panel.

All HMI/SCADA servers and control nodes shall be powered with UPS.

All PLC and remote I/O panels shall have redundant power supplies.

When sizing, selecting, and coordinating fusing for the FMCS system, the following factors shall be considered:

- a. Normal operating current - The current rating of a fuse should typically be derated 25% for operation at 25°C to avoid nuisance fuse failure.
- b. Application Voltage - The voltage rating of the fuse shall be equal to or greater than the available circuit voltage.
- c. Ambient Temperature - Derating the fuse shall be considered if the ambient temperature is greater than 25°C.
- d. Overload current and length of time in which the fuse must open

- e. Maximum available fault current - the interrupting rating of a fuse must meet or exceed the maximum fault current for the circuit.

All fusing shall be coordinated to ensure that an overcurrent condition at the instrumentation level will not affect the power distribution to other instruments, PLC modules, or power supplies

Design Parameters

- 1) All PLC and Remote I/O panels shall be built to UL508A or equivalent specifications.
- 2) PLCs shall allow for the configuration and addition of I/O, as well as program downloads in run mode.
- 3) All PLCs and remote I/O shall be provided with short circuit protection.
- 4) All PLCs shall provide the ability to force all I/O.
- 5) All PLCs shall have Ethernet-capable communications.
- 6) All PLC and remote I/O shall provide the ability to remove and replace modules while the system is still powered (hot-swappable).
- 7) All PLC and remote I/O cabinets shall have cooling fans.
- 8) All PLC and Remote I/O power supplies shall be monitored for loss of power.
- 9) All PLC and Remote I/O shall be monitored for loss of communications between components.
- 10) All PLC and Remote I/O shall have diagnostic capabilities that are capable of alarming to control room.
- 11) In general, based on complexity of the facility system being monitored, the SCADA/HMI shall incorporate graphical representation of systems to display status and alarms. Graphics shall be representative of, but are not limited to Piping and Instrument Diagrams (P&ID's), Process Flow Diagrams (PFD's), wiring schematics and/or instrument loop diagrams

Additional I/O should be supplied to include 20% spares and 20% additional rack and terminal block capacity. Enclosure Rating shall comply with IP / NEMA:

IP42/ NEMA 2 indoor for dry area

IP54/ NEMA 3 indoor for wet area

IP54/ NEMA 3S for outdoor with canopy

IP65/NEMA 4X for outdoor

6.16.2 Power Meters and Power Quality Monitoring

- a) Power quality monitoring device shall comply to IEC 61000-4 EMC standard series with continuous acquiring and providing data of the following functions:
- Status reports
 - Alarm reports
 - Disturbance reports (Voltage sags and swell)
 - Harmonics and total harmonic distortion (THD)
 - Frequency variations
 - Transients reports
 - Unbalance
 - Flickers
 - Rapid voltage fluctuation
 - Currents measurements
 - Voltages (Line; phase) measurements
 - Power (VA, active and reactive) measurements
 - Energy measurements
 - Daily logs (such as events and operator actions)
- b) PQM device/s shall have common communicate protocol and able to transfer data and all information to Scada system.

6.16.3 Circuit Monitors

- 6.16.3.1 All circuits that are part of the network to be monitored by Scada will have devices connected via remote terminal units (RTUs) and/or Programmable logic Controllers (PLCs) and/or Port Server/Communication Processors, etc.,
- 6.16.3.2 Information and measurements from circuits include:
- i. Status report; Isolators and/or circuit breakers and/or earthing/grounding position.
 - ii. Alarms from monitored circuits abnormality and device supervision.
 - iii. Measurement of circuits include power flow; voltage; current; etc.,

6.16.4 Transformer temperature monitoring devices

- 6.16.4.1 Transformer temperature should be monitoring both for oil/gas and winding temperatures.
- 6.16.4.2 For liquid or gas filled transformers, temperature measuring device shall be dial types with two adjustable contacts level for operation of cooling fans (for ONAF) and other for trip function. It shall be connected to remote control panel and then to Scada. If numerical relay/device is used, setting of value shall be done on relay/device.

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6.16.4.3 Winding temperature via PT-100 for each phase shall have two setting values for alarm and other for trip. Similarly, it shall be wired to remote control panel and to Scada. If numerical relay/device is used, setting of value shall be done on relay/device.

6.16.5 MCCB electronic trip unit

6.16.5.1 Circuit Breaker 3 or 4 poles shall fully comply with IEC 60947 (1 & 2). Certificate of all the sequences should be made available.

6.16.5.2 The breakers shall be operated by a toggle, which shall clearly indicate the three positions ON, OFF and Tripped.

6.16.5.3 Characteristics of the protective functions have to be selected and with adjustable values and timing. System information (voltage, short circuit withstand, etc.,) shall be also provided to ensure the correct MCCB selected.

6.16.5.4 MCCB electronic trip unit together with circuit breakers status and measurement shall have communication capability to Scada system.

6.16.5.5 MCCB shall be equipped with extended rotary handle with padlocking and door inter-locking facility.

6.16.5.6 The breaker shall have a "push to trip" button or external emergency trip button, to test operation and opening of the poles. Built-in trip button shall avoid accidental depression.

6.16.6 LV ACB trip unit

6.16.6.1 Low voltage Air Circuit Breaker (ACB) shall comply and type tested to IEC 60947-2; ANSI C37.13 or NEMA PUB NO.SG3 standard. All outgoing final circuits shall have utilisation category A and circuit breakers feeding sub-distribution switchboards which shall have utilisation category B.

6.16.6.2 Circuit breakers shall have a rated short-circuit making and breaking capacity in conformity with the prospective short-circuit current as specified in the requisition.

6.16.6.3 Circuit breakers in a change-over supply duty should be withdrawable but shall at least have isolation facilities for maintenance and testing purposes.

6.16.6.4 Air circuit breaker should have intelligent built-in features to provide status, monitoring, control and protection. These features shall have common open type IEC protocol or equivalent, to interface/communicate with SCADA.

6.16.7 Intelligent Electronic Devices (IED); Protection Device Relay

6.16.7.1 Intelligent Electronic Devices (IED) used in power system have many advance functions, like:

- i. Protection function
- ii. Circuit breaker control and programmable logic
- iii. Metering and Power Quality Analysis
- iv. Self-supervision and external circuit monitoring
- v. Fault Diagnosis and Event captured/reporting
- vi. Tools for settings, commissioning and testing
- vii. Programmable LCD Display

IED can function independently and shall communicate with Scada for all those function that the IED can perform.

6.16.7.2 Protection Device Relay may have less functionality as compare to IED due to its main purpose/application. Numerical relay with communication capability shall be selected and interface with Scada system with all its available features displayed.

6.16.8 Facilities Electrical SCADA System Parameters

Following table contains recommended SCADA Parameters and alarm set points for electrical system monitoring purpose. Site specific changes are required based on the system configuration, operating limitations to prevent nuisance alarms. If SCADA system is used for the operation of any equipment, documented operating procedure are to be followed for each step. Any SCADA parameters change, Status monitoring and Alarm changes need to be done only after proper engineering review through Management of Change (MOC) process.

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6.16.9 Facilities Electrical SCADA Alarm priority

Follow FMCS SCADA to select the priority and mode of notification for alarms and out of range operating condition.

Table 1: Recommended SCADA Parameters and Alarm Set Points

Parameters			Alarm			
			Low Low	Low	High	High High
S/N	General	Status	Value			
HV & MV SYSTEM						
1	Voltage % of Nominal	Value + Alarm	92%	95%	105%	108%
2	Current % of OC LT setting /Rating (lowest)	Value + Alarm			80%	90%
3	Power quality-Sag / Swell - Utility intake	Alarm	90%			110%
4	Power quality-Sag / Swell - DVR/UPS/DUPS Output	Alarm	90%			110%
5	KW - (Alarm If required)	Value + Alarm			90%	100%
6	KVA - (Alarm if required)	Value + Alarm			90%	
7	Power factor (Alarm If required)	Value + Alarm	85%		100%	
8	Energy Consumption	Alarm				
9	Load Demand (Alarm If required)	Value + Alarm				
10	Relay Pick up	Alarm				
11	Relay Trip	Alarm				
12	Relay Alarms /General alarm/Disable/failure /Com error	Alarm				
Switch Gear						
1	Mode Selector Switch-Not in normal mode	Value + Alarm	Local / Remote			
2	Breaker Status	Value + Alarm	Breaker Open & Close status			
3	Breaker not in-service position / Not ready -If used for auto mode	Value + Alarm	Breaker status			
4	Trip disable	Alarm				
5	Partial Discharge	Alarm	Value if applicable			
6	SF6 Gas pressure low	Alarm	Low & Low-Low alarm based on OEM			

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Parameters			Alarm			
			Low Low	Low	High	High High
Oil Filled Transformers						
1	Transformer Oil level	Alarm	Low & Low-Low alarm based on OEM			
2	Transformer Winding / Oil temp (% of temp trip setting)	Value + Alarm			75%	95%
3	Transformer tank pressure – High	Alarm	High & High-High alarm based on Alarm OEM			
4	Transformer tank pressure – Low					
5	Online oil quality - If applicable	Alarm				
6	Partial discharge	Alarm			10 pC	
7	Cooling fan operation	Status				
8	Fan controller general alarm/ communication alarm	Alarm				
Dry Type / Cast Transformers						
1	Transformer Winding temp (% of temp trip setting)	Value + Alarm			75%	95%
2	Fan controller general alarm / communication alarm	Alarm				
3	Partial discharge	Alarm			10 pC	
4	Cooling fan operation	Status				
LV Switchgear						
	General					
1	Voltage (for mains) % of Nominal	Value + Alarm	90%	92%	108%	110%
2	Current % of OC LT setting /Rating (lowest)	Value + Alarm			80%	90%
3	Power quality (for main)-Sag / Swell - DVR/UPS/DUPS output	Alarm	95%			105%
4	Harmonics (if more than 5% Vthd)	Value				
5	Power factor (Alarm If required)	Value + Alarm	85%	90%	100%	
6	Relay Pick up	Alarm				
7	Protection setting selection	Alarm	Alarm if not selected for primary protection setting			
8	Relay Trip	Alarm				
9	Relay alarm (general/ disable/ failure /com error)	Alarm				

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Switch Gear						
1	Breaker position (if applicable)	Status + Alarm				
2	Breaker mode (if applicable)	Status + Alarm				
3	Breaker status	Status				
4	Trip disable	Alarm				
Dry Type / Cast Transformers						
1	Transformer Winding temp (% of temp trip setting)	Value + Alarm			80%	95%
2	Cooling fan operation	Status				
Emergency Generator & ATS						
	Generator					
1	Voltage (for mains) % of Nominal (low alarm if running)	Value + Alarm	92%	95%	108%	110%
2	Current % of OC LT setting /or Rating (lowest)	Value + Alarm			80%	90%
3	Power factor (Alarm If required)	Value		80%	100%	
4	Frequency	Value + Alarm		98%	102%	
5	Relay Pick up	Alarm				
6	Relay Trip	Alarm				
7	Relay alarm (General/failure /Com error)	Alarm				
8	Engine General Warning / Alarm	Alarm				
9	Engine Mode (Not in Auto alarm)	Status + Alarm				
10	Engine Status	Status				
11	Engine High Temp	Alarm				
12	Engine Low oil pressure	Alarm				
13	Fuel tank level (Normal 70 to 95%)	Value + Alarm	50%	60%		100%
14	Fuel Leak	Alarm				
15	DC Battery voltage low / Battery Open	Alarm				
16	Battery charger general alarm	Alarm				
17	Room Ventilation interlock status	Alarm				
18	Cooling water system (Pressure/ DP/ Flow/temp)	Alarm				
19	Cooling water system pump status	Status + Alarm				
20	Cooling water system pump / fan mode	Status + Alarm				

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ATS						
1	ATS Status	Status + Alarm				
2	ATS Mode	Status + Alarm				
3	ATS Load current	Value + Alarm			80%	100%
4	ATS General Warning / Alarm	Alarm				
UPS /DUPS/DVR						
	UPS					
1	Source 1 Voltage % of Nominal	Value + Alarm		90%	110%	
2	Source 2 (Bypass source) Voltage % of Nominal	Value + Alarm		92%	108%	
3	Source 1 Current % of OC LT setting /Rating (lowest)	Value + Alarm			75%	95%
4	Source 2 Current % of OC LT setting /Rating (lowest)	Value + Alarm			70%	90%
5	Output Voltage % of Nominal	Value + Alarm		96%	104%	
6	Output Current % of OC LT setting /Rating (lowest)	Value + Alarm			70%	80%
7	DC Bus Voltage % of nominal value	Value + Alarm		95%	105%	
8	DC Breaker status	Status + Alarm				
9	Heat sink temperature	Value + Alarm			70%	95%
10	UPS General warning / Alarm	Alarm				
11	UPS Status	Status + Alarm				
12	Bypass Status	Status + Alarm				
13	Communication alarm	Alarm				
14	Battery room temperature	Value + Alarm				
Battery Monitoring System						
1	Battery Internal Resistance	Value + Alarm				
2	Battery Voltage	Value + Alarm				
3	Battery Current	Value + Alarm				
4	Room Temperature	Value + Alarm				
5	Battery Monitoring system alarms General / Com	Alarm				

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DUPS						
1	Source 1 Voltage % of Nominal	Value + Alarm		90%	110%	
2	Source 2 (Bypass Source) Voltage % of Nominal	Value + Alarm		92%	108%	
3	Source 1 Current % of OC LT setting /rating (lowest)	Value + Alarm			75%	90%
4	Source 2 Current % of OC LT setting /rating (lowest)	Value + Alarm			70%	90%
5	Output Voltage % of Nominal	Value + Alarm		96%	104%	
6	Output Current % of OC LT setting /Rating (lowest)	Value + Alarm			70%	80%
7	DC Bus Voltage % of nominal value -If applicable	Value + Alarm		90%	110%	
8	DC Breaker status -If applicable	Status + Alarm				
9	Winding Temperature	Value + Alarm				
10	Bearing Temperature	Value + Alarm				
11	Vibration level	Value + Alarm				
12	General Warning /Alarm	Alarm				
13	DUPS Status	Status + Alarm				
14	Bypass Status	Status + Alarm				
15	Engine Status	Status + Alarm				
16	Engine Mode	Status + Alarm				
17	Engine & Fuel Alarms to follow Emergency Generator	Status + Alarm				
18	Communication error (if applicable)	Alarm				
DVR						
1	Source 1 Voltage % of Nominal	Value + Alarm		90%	110%	
2	Source 2 Voltage % of Nominal	Value + Alarm		92%	108%	
3	Source 1 Current % of OC LT setting /Rating (lowest)	Value + Alarm			75%	90%
4	Source 2 Current % of OC LT setting /Rating (lowest)				70%	90%
5	Output Voltage % of Nominal	Value + Alarm		94%	106%	
6	General Warning / Alarm	Alarm				
7	DVR Status	Status + Alarm				
8	DVR Bypass Status	Status + Alarm				
9	Communication error (if applicable)	Alarm				

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DC Power System						
1	AC Source Voltage low	Alarm		90%		
2	DC Voltage	Value + Alarm		98%	110%	
3	DC Current (if applicable)	Value + Alarm	<5%	50%		
4	DC circuit Open/Battery Open	Alarm				
5	General Warning / Alarm	Alarm				
Variable Speed Drive VFD/VSD & Soft starters						
1	Source Voltage % of Nominal	Value + Alarm		90%	110%	
2	Source Current % of OC LT setting /Rating (lowest)	Value + Alarm			75%	90%
3	Output Voltage	Value				
4	Output Current % of Motor /Drive Rating /setting (lowest)	Value + Alarm			85%	95%
5	Heat sink Temp (%of trip)	Value + Alarm			85%	95%
6	General Warning & Trip Alarm	Alarm				
7	Operating mode	Status + Alarm				
8	Communication error (if applicable)	Alarm				
MCC / Motor Starters						
1	Starter Mode	Status + Alarm				
2	Starter status	Status				
3	Over load trip	Alarm				
4	Source Voltage % of Nominal	Value + Alarm		90%	110%	
5	Source (Main) Current % of OC LT setting /or Rating (lowest)	Value + Alarm			70%	80%
6	Motor Current % of OL setting /Motor Rating (lowest)	Value + Alarm			90%	98%
7	Motor winding temperature (for large and critical motors)	Value + Alarm			80%	95%
Distribution Panels						
1	Source Voltage % of Nominal	Value + Alarm		90%	110%	
2	Breaker trip	Alarm				
3	Breaker status (For critical application only)	Status + Alarm				
4	Source Current % of OC LT setting /Rating (lowest)	Value + Alarm			70%	80%

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6.16.10 SCADA training and training documents

Manufacturer/contractor's vendor shall provide training to electrical engineers, control room engineers/ operators/ electricians and technicians on the electrical SCADA system. All documents/training materials shall be provided and hand-over to the Owner.

6.16.11 Installation

6.16.11.1 Site verification/preparation for installation of main equipment of Scada.

6.16.11.2 Connection lists, circuit diagrams and information other needed for integration of the system.

6.16.11.3 Terminate functional devices and perform checks.

6.16.11.4 Perform loops, parameters checks, settings and performance.

6.16.11.5 Test reports of functional units and system

6.16.11.6 Submit functional design specification, components/spares handover and all documents of the system.

6.17 General Electrical Work

6.17.1 Cable wiring

6.17.1.1 Cable wiring selected for used in sensitive areas shall not create contaminate/emit particles especially in the cleanroom environment.

6.17.1.2 Cables and accessories, including track cables etc., shall have the following properties unless approve by Owner:

- i. Flame Retardant according to IEC 60332-1;
- ii. Oil resistance
- iii. Halogen Free;
- iv. Weather, Ultra-violet (UV) and ozone resistance.
- v. Chemical resistance to solvent, fluid, acid and alkalis
- vi. Resistance to mechanical stress
- vii. Resistance to tear, impact and abrasion

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6.17.2 Cable Ladder/Tray

6.17.2.1 Depending on the site situation/environment, cable trays and accessories selection will be selected for the correct applications. Part of cable tray system are tray covers, clamping bolts and accessories include coupler plates, tees, bends, reducer, vertical elbow, etc.,

Comply with:

- NEMA VE 1, "Metal Cable Tray Systems"
- NEMA VE-2, "Cable Tray Installation Guidelines".
- NFPA 70.

Aluminum: Straight section and fitting side rails and rungs shall be extruded from Aluminum Association Alloy 6063. All fabricated parts shall be made from Aluminum Association Alloy 5052.

OR

Pre-galvanized Steel: Straight sections, fitting side rails, rungs, and covers shall be made from steel meeting the minimum mechanical properties and mill galvanized in accordance with ASTM A653 SS, Grade 33, coating designation G90.

Cable ladder/tray shall have complied with the NEMA load/span class.

Electrical Power cable ladder shall be Class 20C plus 200-pound concentrated static load applied between side rails at midspan with safety factor.

Install cable ladder/tray according to NEMA VE-2

NEMA VE1 Warning signs with engraved letters to include the load limitation and warning, "Do not use cable ladder/tray as walkway, support, working platform".

All the cable ladder/tray support design to be endorsed by local professional engineer.

Future expansion to be considered in the cable ladder design, spare space and capacity to be provided in the details engineering stage.

6.17.2.2 Commonly used cable trays and accessories are hot-dip galvanized steel. These include cable ladders, perforated type galvanized steel and solid type galvanized steel.

6.17.2.3 Contractors shall prepare Quality Control Plan identifying the various stages of manufacturing, quality checks performed at each stage and customer hold points.

6.17.3 Conduit

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- a) Conduits, fittings and accessories shall comply to BSI 9537 or ANSI C80.5.
- b) Wiring shall be PVC insulated in accordance with IEC 60227 with minimum cross-section of 2.5 mm², considering correct current rating and maximum voltage.
- c) Wiring colors shall follow local regulation. If there is no local requirement, the following shall apply:
 - Blue for neutral
 - Brown for phase
 - Black for switched phase
 - Green/yellow for earth/ground

6.17.4 Busway

a) General

Busway typically used in ~~commercial~~ and industrial location for distribution of power through various switchboards, transformers and panel boards to loads via enclosed busbar connection to conductors in trays or conduits.

Unless specified, the minimum ingress protection for equipment shall be IP54 for installed equipment.

b) Prefabrication

Contractor shall design and select the appropriate type of busway/s for the applications, load harmonic and capacity rating considering future extension. Busway used shall comply to local or international standards such as UL 857 or NEMA BU 1.1 and IEC439/539 and or others.

Standard busway lengths and plug in outlets at both side of the busway should be used unless restricted by site application.

Effective earthing/grounding of enclosure and earthing/grounding bus shall form part of the busway system.

The busway installed in FAB or Sub-FAB or wet area shall have cover with Corrosion-resistant materials.

- c) Busways and POU/PIU shall be assembled and installed by qualified and OEM trained contractor or by manufacturer with OEM approved checklist.
- d) All the Busway and accessories (joints, terminations, sealing ends, POU/PIU, etc.) shall be supplied by the same vendor/manufacturer.
- e) All Busway joints and PIU/POU shall be inspected and tested based on OEM approved checklist with Micron supervision before energizing.
- f) All PIU/POU breaker handle to be protected against accidental operation if there any risk during adjacent activities.g) All

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Busway, and PIU/POU support to be designed for reliable operation, based on OEM recommendations and site-specific seismic conditions. Extra support to be provided for PIU/POU where additional mechanical loading is possible due to cable and termination activities.

- h) Required access and working clearance to be provided around existing and future PIU/POU area and around the busway to support operation, maintenance and repair needs.
- i) Live and exposed electrical work risk to be eliminated or minimized with proper pre facilitation and by design.

6.17.5 Panel Board

6.17.5.1 Selection of panel board features to meet the site requirements/applications. Include enclosure type for indoor or outdoor or special application.

6.17.5.2 Enclosure for indoor panel board should be code-grade galvanised steel, except for column type panel boards which include painted finish. All enclosures shall furnished according to UL/NEMA standard.

6.17.5.3 Live and exposed electrical work risk to be eliminated or minimized with proper pre facilitation and by design.

6.17.5.4 Application consideration for panel boards:

- i. Standard compliance, UL; NEMA; NEC
- ii. Determining factors such as voltage, system, amperage, interrupting rating/type of branch devices and other special requirements.
- iii. Panel short circuit rating
- iv. Selective coordination
- v. Service Entrance equipment
- vi. Panel Boards Type (Column, multi-section, interconnecting multi-section; Through feed lug; Sub-feed lugs; etc)
- vii. Ambient temperature;
- viii. Arc Flash reduction
- ix. Loading of branch circuit for lighting panel
- x. Overcurrent and earth/ground fault protection
- xi. Harmonic contents
- xii. Surge Protection
- xiii. Metering
- xiv. Special Condition (humidity, altitude, vibration, etc)

6.17.6 Safety Switch

6.17.6.1 SEMI standard provide guidelines on safety switches and shall be comply accordingly. Equipment should incorporate emergency off (EMO) circuit to put equipment

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- in a safe state which no harm is posed to workers or equipment.
- 6.17.6.2 Design of safety system include safety switches/emergency off switches shall consider the impact to the process, environment, hazard posed to workers and asset damage.
- 6.17.6.3 When there is a possibility that the emergency off switch could be operated mistakenly, a guard must be installed and switch must use direct opening operation. Button must be red with yellow background and switch itself must include the letters "EMO".
- 6.17.6.4 Safety switch or emergency switch for motors shall be mounted close (visible when operating valve) to motor within 5 metres and with padlockable features.
- 6.17.6.5 Depending on the application (indoor, outdoor, dust, hazardous environment, etc.,) should be compact design, impact and corrosion resistance, durable/heavy duty, ergonomic switch handle, enclosure water resistance, etc.,
- 6.17.7 Shock and Arc Flash Prevention
- 6.17.7.1 Protection against electric shock for installation and equipment standard IEC 61140 apply. Terminals of equipment installed when door is opened, shall be shrouded to a degree of protection of at least IP 2X (Finger safe).
- 6.17.7.2 All practical measures shall be taken to prevent danger to personnel working on a disconnected functional units with adjacent units still operation.
- 6.17.7.3 Means of proving circuit DEAD and earthing/grounding applied by a competent person before work can proceed for disconnection/reconnection of a circuit.
- 6.17.7.4 Assemblies shall comply with IEC 1641 with regard to internal arc testing.
- 6.17.7.5 Arc Flash study and mitigation actions shall be performed at design stage and updated progressively throughout the project. Management of change shall include this document, if there are changes:
- i. to the load;
 - ii. power generation;
 - iii. system configuration;
- 6.17.7.6 Protection grading studies and relay setting schedules should be used as mitigation to arc flash study. Study is not require if High/Medium/Low voltage switchboards with

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internal arc containment and arcing fault is within protection grading time.

- 6.17.7.7 Recommendation of mitigations such as change to protection grading/setting, procedure, caution/warning notice, arc flash PPE (personal protective equipment), arc flash detection, etc., should be approved by the Owner/GFE&C.

6.17.8 Installation

- 6.17.8.1 Busway shall be installed by qualified person with the right tools (laser alignment, measurement, etc) and knowledge (adequate supports) to ensure proper installation and selection of route to avoid offsets.

- 6.17.8.2 Busway passing through walls and floors shall not have joints close to it. Joints for vertical busway should be at least 400 mm (16") away from the floor level and joints for horizontal busway, a minimum of 150 mm (7") away from the walls. For vertical busway section/s extend, it should be at least two metres (6') away from the floor.

- 6.17.8.3 End of busways shall be closed with proper end closer. Minimum clearance to the walls or ceiling shall follow the manufacturer recommendation.

- 6.17.8.4 Busbars connections in the busways shall be tension with set torque wrench of the recommended torque.

6.17.8.5 Safety-

Risk assessment should be conducted to determine the hazards and mitigations to minimize the RISK to ALARP.

Proper personal protective equipment (PPE) should be worn when performing the installation work.

Applied Electrical Safety Rules/procedures, such as safe isolation and earthing/grounding, lock-out-tag-out, permit-to-work, etc., for all work on Electrical equipment.

Take all necessary safety precautions in handling and transporting of equipment for installation such as lifting, siting, transportation route, techniques, barricade, etc., Refer to local safety requirements, owner, manufacturer and contractor safety procedures.

Ensure sound earthing/grounding of electrical equipment and metallic structure/s.

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6.17.9 Temporary Power Provision

6.17.9.1 General

- i. All local statutory regulations shall be mandatory. Additional requirements below shall apply.
- ii. Electrical Installation work for site construction offices, temporary warehouse/s, ancillary buildings, etc., shall adhered to the local regulations and or accepted standard such as IET wiring regulation, NEN 1010, NFPA 70.
- iii. Single line diagram and layout drawing shall be prepared for the temporary power supply system.
- iv. All equipment and cabling shall be clearly marked and uniquely numbered.
- v. HV/MV supplies should preferable be in a ring form, operated with open at one point. HV/MV distribution equipment, transformers and main LV distribution switchboards shall be accessible to authorized persons only.
- vi. LV supplies shall be arranged as a TN-S system, with radial feeders from a distribution board. Example each local transformer or generating sets, to sub-distribution switchboards close to the work areas/high activities area.
- vii. Energy meter at power supply sources should be Class 2 except for generator sets.

6.17.9.2 Diesel Generating set

- i. Generating sets should be sited at a reasonable centralise locations to distribute temporary power to the vicinity.
- ii. Generators shall be star connected with their neutral points solidly earthed/grounded.
- iii. Generating sets shall be complete with circuit breaker and all necessary control equipment. If more than one set is installed suitable paralleling facility should be provided.

6.17.9.3 HV/MV Switchgear and transformer

- i. HV/MV switchboard shall comply with IEC 62271 and have single busbars with incoming switch/es. Number of outgoing feeder circuit breakers should be distributed in an ring or fuse switches for radial distribution.
- ii. Each outgoing feeder circuit breaker shall be protected with IDMT (inverse Definite Minimum Time) overcurrent and earth/ground fault protection.

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- iii. DC tripping and power supplies to switchgear and protection is not required.
- iv. d) Earthing/grounding facilities shall be provided.
- v. Switchgear shall be suitable for outdoor application or mounted in weatherproof package enclosure, with adequate access for operation and maintenance personnel.
- vi. Transformers shall comply with IEC 60076 and preferably delta-star connection with LV neutral points solidly earthed/grounded.
- vii. Transformers and or switchgear packaged units shall be mounted on concrete plinths within a fenced enclosure with padlockable gates.
- viii. The corner posts shall protect the equipment/substation from vehicle impact and should be located at least 5 metres from the edge of any road.

6.17.9.4 LV Switchgear

- i. All distribution boards shall comply with IEC 61439-4. Main LV distribution switchboards supplied by generating set/s shall have an incoming switch for each generator. Outgoing circuits shall be protected by either fuses or MCCBs having a maximum rating or setting of 400 A.
- ii. Sub-distribution switchboards shall have an incomer switch and all outgoing circuit with switching devices and protection short circuit and residual current. Maximum outgoing feeder rating shall be 125 A.
- iii. c) Sensitivity of residual current circuit breakers shall be 300 mA for circuits above 125 A and 30 mA for circuits below 125 A.
- iv. d) Fuses shall be accessible only if the fuse bases are protected to at least IP 2X, or if interlocking exists so that the fuse base is electrically isolated first.

6.17.9.5 Temporary Cable

- i. Depending on the site condition and risk of damage, temporary cables installation method may be above-ground or underground.
- ii. Temporary underground cables for HV/MV or LV services shall be laid at least 0.8 m to avoid damaged during construction. Cables laying route shall be clearly marked.
- iii. Depending on the site construction equipment movement, overhead cables shall be erected at a minimum clear height of 6.0 m for roads/routes designated for vehicular movement and 4 m for crossing where vehicular traffic is prohibited. Cables shall be bound with yellow/black coloured tapes.

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- iv. Above ground cables installed shall be clear of building operation, passageways, ladders, stairs, etc., and are not a hazard. Strips of color fabric/plastic to be attached to cables and/or protective barriers may be required for hazard identification.
- v. Apparatus and accessories other than lighting fixtures shall not be suspended from electric cables.
- vi. All cables that are frequently moved during construction activities shall be flexible type, such as heavy duty neoprene rubber.

6.17.9.6 Earthing/grounding and Components

- i. Earthing/grounding for electrical system and equipment for the installation shall have one common earthing/grounding grid connected to at least two earth/ground electrodes. This may form part of the permanent earthing/grounding system.
- ii. The resistance to earth/ground of this common earthing/grounding grid shall be as low as reasonably practical and not exceeding 1 Ohm. Refer to Section 5.6 on Earthing/grounding System.
- iii. Component selected shall be suitable for their site application with regards to voltages, rated currents, short circuit withstand, making and breaking capacity, service life.
- iv. Components used, having breaking capacity and/or short circuit strength less than the withstand the stresses occur at the installed place shall be protected by means of current limiting devices.
- v. Industrial pattern power and convenience outlets shall comply to IEC 60309. Three phase outlets shall all have the same phase sequence.
- vi. Plugs of different rating currents or voltages shall not be interchangeable. Plugs and socket outlets shall have at least IP54 degree of protection, both with plug removed and when it is fully inserted.

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7.0 Electrical Testing and Commissioning Requirement.

7.1 Overall Requirements

7.1.1 General Requirement

National and/or local regulations exist (incl. Safety, Workers Competency and Environmental impact) in which some of the requirements may be more stringent than this, the Contractor shall determine by careful scrutiny and adapted the more stringent requirements.

7.1.2 Electrical Safety

- a) Safety precautions during work on electrical equipment cannot be over-emphasised. Attention should be given to apply company safety requirements together with local safety codes and practices.
- b) Contractor shall provide an Electrical Safety Rules including but not limited to the following:
 - Personal safety and protection equipment
 - Proper use of equipment/tools/etc.;
 - Lifting, transportation and handling of Electrical equipment, materials and others;
 - Housekeeping and environment;
 - Authorization of Electrical person for testing, work on or close to "LIVE"/energized circuits/equipment. Competency of worker (Qualify Electrical Workers) to avoid danger;
 - LOTO/CoHE procedure
 - Hazard identification and mitigation measures;

This document shall be submitted to the Principal for review and comments prior to the commencement of site construction activities.

7.1.2.1 Competence

Contractor shall provide list of qualified electrical workers to the Principal before implementation of the project. Contractor to demonstrate workers have the necessary/sufficient training and experience to perform roles and responsibility for the design, engineering and construction of the Electrical Installation works.

7.1.3 Environment:

Contractor shall demonstrate sustainable development at all time during the implementation of the project. Which should include:

- Reduce energy
- Limit fuel usage
- Reduce noise

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- Properly disposal of waste
- Utilize reusable technology
- Recycling initiatives
- Promote energy efficiency through the selection and use of tools, equipment, materials and resources (include right people for the right job)
- Environment ecofriendly design; Green Construction Practices

7.1.4 Responsibility, Procedures and Documentation

7.1.4.1 Responsibility:

- i. An electrical installations and equipment erected by Contractors and, in general, any equipment which has been completed and is ready for operation, shall be subject to certain hand-over procedures which shall have been agreed between the Contractor and the Owner well before pre-commissioning stage. Furthermore, the take-over procedure should state clearly the relevant responsibilities of Contractor and Owner.
- ii. The Contractor shall be responsible for organize and execute pre-commissioning and commissioning activities. Unless otherwise agreed by Owner, complex electrical systems, such as HV/MV/LV switchgear, generators, AVRs, Sag mitigation system, DUPSSs, SUPSSs, VSDSs, shall be commissioned with the assistance of the Manufacturer.
- iii. Contractor shall have competent person/s to supervise and or perform electrical work on equipment/system include testing and commissioning.

7.1.4.2 Procedures

- i. The below testing and commissioning procedures may not be pertinent or fully cover the extend of the installation/equipment.
- ii. Contractor is responsible to make assessment and judgement to supplement testing and commissioning (T&C) procedures in those area not covered hereto, including manufacturer's recommendations for site commissioning. Supplementary T&C procedures shall be agreed/approved by Owner.
- iii. Steps required to connect an electrical system, prior to its initial energizing, to an existing network, shall be jointly agreed by the Owner's authorized electrical person in charge of the electrical network.

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- iv. Initial agreement should be reached between Contractor and Owner on the need to perform a full performance tests (e.g. full power output, temperature rise) and If this cannot be realized because of site conditions, such tests shall be performed at a later date.
- v. Owner should have internal procedures for equipment acceptance by Owner's maintenance team from Project/Construction team.
- vi. Contractor shall complete the Inspection Forms provided. General inspection form is applicable for all equipment together with other inspection forms.
- vii. The Contractor shall develop additional inspection forms for equipment not covered inspection/test forms and include manufacturer's inspection and test forms for site commissioning.

7.1.4.3 Documentation

- i. Documents for Type test factory acceptance test (FAT) **installation inspection test, start-up and commissioning test** and these site acceptance tests (SAT) shall be handed over as part of the systemization packages.
- ii. All deviations from the requirements shall be documented on an "exception list" which shall be presented to Owner for agreement and action.
- iii. All the document mentioned in 7.1.3.2 item (f), (g) and 7.1.3.3 (b) shall form part of the official hand-over of system and equipment.

7.2 Testing and Commissioning Methods

7.2.1 Insulation Testing

Insulation tests shall be carried out at the pre-commissioning stage and prior to the energizing of equipment and cables. Two methods shall be used:

- a) Insulation resistance (IR) testing for motors, generators, transformers, MV/LV switchgear, cables and all equipment other than those items susceptible to damage by over-voltage (e.g. electronic components)
- b) High voltage AC or DC testing for HV/MV switchgear and cables

Refer to Appendix A and B for recommended test voltages and insulation resistance values.

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7.2.1.1 Insulation Resistance (IR) tests

- i. Insulation resistance test shall be carried out by applying a d.c. voltage from an insulation tester at voltage levels given in Appendix A.
- ii. Measured resistance value is also dependent on the temperature of the insulation and shall be corrected with the use of the nomogram in Appendix 1.
- iii. As a guide, the insulation resistance value of windings reduces by approximately 50% for every 10oC rise in temperature.

7.2.1.2 High Voltage Test

- i. Before any high voltage test is to be performed, insulation resistance tests shall be carried out.
- ii. High voltage testing can be carried out by applying a.c. (power frequency), very low frequency (VLF) or d.c. voltage at a level above the operating voltage.
- iii. The nature and practicality of these tests (clause b above) are distinctly differences. Testing at power frequency is preferable, the a.c. test equipment must be capable of supplying the capacitive current requirement of the equipment under test. Physically large test equipment is frequently called for which is impractical for field commissioning test.
- iv. D.C. test equipment may be used for on-site commissioning.
- v. High voltage testing of electrical equipment, except LV cables, shall have been carried out at the Manufacturer's works with a.c. at voltage levels in accordance with the relevant International standards.
- vi. Repeated high voltage pressure tests shall not be applied on site for insulation of windings, such as motors, generators or transformers. Applying such tests may weaken the winding insulation.
- vii. If there are special reasons that these tests are considered to be necessary, manufacturer shall be consulted prior to executing the test.
- viii. After completion of the high voltage withstand test, equipment shall be re-tested for insulation resistance as per clause 7.2.1.1. There should be no decrease in insulation resistance due to insulation damage.

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7.2.1.3 Polarisation Index (PI)

- i. The PI is the ratio between the insulation resistance measured after one minute and after 10 minutes of continuous testing at the appropriate voltage ($PI = R_{10 \text{ min}} / R_{1 \text{ min}}$).
- ii. After 10 minutes the capacitive current, the leakage current over the surface of the insulation and the dielectric absorption current will have stabilised.
- iii. The PI obtained gives an indication of the condition of the winding insulation with regard to its dryness and cleanliness; the PI will be lower for a dirty, moist/wet or chemically contaminated winding insulation.
- iv. The PI, being a ratio, is practically independent of winding temperature and test voltage.
- v. Appendix C shows recommended Polarisation Index values.

7.2.1.4 Results and Actions

- i. a) Should the obtained insulation resistance (IR) or polarisation index (PI) values fall below the minimum values given in Appendix B, then the equipment should not be energised or subjected to high voltage testing.
- ii. b) Appropriate action should be taken, such as drying, to normalise the values.
- iii. c) Should a flash-over occur during high voltage testing then the test shall be deemed to be a failure and appropriate measures should be taken (e.g. cleaning, drying, repair) to eliminate the cause, before a re-test shall be performed.

7.2.1.5 Insulating Oil Tests

- i. **High voltage withstand tests and moisture content test shall be carried out.** High voltage withstand tests shall be carried out using a standard insulating oil tester or dielectric oil tester, with 25 mm diameter mushroom-shaped electrodes placed at a gap distance of 2.5 mm from each other.
- ii. The minimum withstand voltage shall be 30 kV. The test sample shall be tested 6 times, using the same oil in the cell, and the arithmetic means shall not be lower than the minimum withstand voltage.

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7.2.1.6 Bearing insulation tests

- i. Controlling shaft current on a rotating machine depending on the method of insulating bearings, the type of construction and methods employed to support the bearings.
- ii. A sleeve or pedestal bearing housings are completely insulated from the body of the machine, the insulation may be checked with the machine coupled and running at normal speed to establish an oil film in the bearing which will prevent electrical contact between shaft and bearing. An ohmmeter applied across the insulation should indicate a resistance of $> 10 \text{ k}\Omega$; figures below this shall be investigated.
- iii. If machines with bearings internally insulated 'shells' or insulated shafts, measurements shall be carried out with the machine uncoupled (and stopped). If, however only one bearing is insulated verification is limited to the inspection of Manufacturer's QC documentation.
- iv. Where double insulation is applied to pedestal bearings, i.e. they are insulated around the bearing 'shell' and at the pedestal base, IR measurement of the 'mid-section' with respect to earth/ground will indicate if either insulation is bridged.

7.2.2 Conductivity Tests

Conductivity tests shall be carried out on switchgear busbar connections and switchgear earth/ground system joints in order to establish their tightness and surface contact condition. It shall be carried out with a low-resistance test set, e.g. a 'Ducter' with a 20 A (d.c.) output.

The measured values for Identical connections shall not differ by more than 20% from each other.

7.2.3 Earth/Ground Resistance Tests

- a) All earth/ground electrodes shall be tested by measuring the individual electrode and group electrode resistance against the mass of earth/ground, using a ground/earth/ground resistance tester.

7.2.4 Wiring/termination checks and Functional Tests

- a) Factory tested assemblies such as switchgear should require only sample checks. Electrical wiring shall be checked against the schematic diagrams and all terminals tightness shall be verified.

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- b) Functional test shall be performed to demonstrate the correct operation of the complete systems.
- c) Special attention should be given to equipment interfaces, where it is likely a functional impairment will happen.
- d) For complicated controls and/or interlocking systems. Contractor shall demonstrate compliance with the design requirements by using special functional check lists and system logic diagrams.

7.3 Testing and Commissioning Procedures

7.3.1 Motors

- a) Contractor should carry out inspection using Form 1, 2a and 2b for induction cage motors.
- b) Motor vibration shall be measured on triple-axis direction and nearest to the bearing housing. Example
 - x axis – side of bearing housing at shaft height
 - y axis – top of bearing housing
 - z axis – axial of bearing housing at shaft height
- c) The measurements shall be executed with an instrument conforming to ISO 2954 with sensor sensitivity that can read a wide range frequency of 10 to 1000 Hz. Instrument should be able to measure vibration for acceleration, velocity, and displacement units of measurement for a wide variety of machines.
- d) The motor vibration velocity shall not exceed 2.8 mm/s RMS, or 4 mm/s PEAK in any direction and with motor running at normal operating temperature.
- e) For bearings fitted with proximity probes, the unfiltered peak-to-peak value of vibration (including shaft 'run-out') at any load between no load and full load, shall not exceed the following values:
 - 50µm for two-pole motors
 - 60µm for four-pole motors
 - 75µm for six-pole or higher motors
- f) The bearing temperature rise for both roller and sleeve bearings, measured by embedded temperature detectors (ETD), shall not exceed 55 K at the end of the heat run.
- g) For sleeve bearings with forced lubrication the lubricating oil temperature rise with the minimum allowable oil flow shall not exceed 28 K at the end of the heat run. However, the oil temperature shall not exceed 82 °C (API 617). For oil ring lubricated bearings, the sump temperature shall not exceed 82 °C (API 610).

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- h) After motor "heat run" of 3.5 to 4 hours the bearing temperature shall not be above the following value:

Sleeve bearing

- Oil temperature 90 °C
- Bearing temperature rise by RTD 50 °C
- Lub oil temperature rise from ambient 30 °C (for force lubrication system)

Rolling Bearings

- Outer ring measurement 90 °C
- Temperature rise from ambient 50 °C

- i) When commissioning motors, precautions shall be taken to avoid excessive vibration caused by the phenomenon known as 'soft foot'; i.e. feet which do not have solid flat contact with the base prior to the tightening of the holding-down bolts.

This may be measured and rectified during installation or detected during running by the loosening of each holding-down bolt in turn while measuring motor vibration.

7.3.2 GENERATORS

7.3.2.1 General requirements

Contractor/Manufacturer shall ensure compliance of any local requirements for testing and commissioning of generators. They should liaise with the authority on the plan and witness tests, if necessary.

Contractor should perform inspection and test using Form 1, 3a, 3b and 3c.

7.3.2.2 Base load Generators

Contractor/Manufacturer should develop and provide a detailed commissioning plan before tests are commenced.

Following tests shall be carried out in order to prove the satisfactory performance of the generator, its governor, automatic voltage regulator, synchronising and protection systems.

Pre-commissioning work (before generator running)

Prior to the running and energisation of the generator, all electrical pre-commissioning

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checks shall have been satisfactory carried out.

Protection relays should be tested in accordance with Section 7.3.4 (Protection) and section 7.3.2.2.2 (d) with some of these tests can only be possible with the generator running.

Pre-synchronising checks (generator running without load)

a) Open Circuit tests

With generator running gradually increase excitation and record by plotting excitation current against generator output voltage. This test is to verify the excitation system with generator characteristics.

b) Phase rotation of Generator

Check phase rotation of generator against the connected bus by means of a LV phase rotation meter connected, as in Appendix 2, to generator VTs. With test 1 showing phase rotation on busbar and test 2 showing for the generator.

c) Synchronising circuit checks

Refer appendix 2, test 2 verify operation of the synchroscope, voltmeter, auto and check synchronising relays follows the "beat frequency", i.e. the difference between bus/grid and generator frequency and voltage.

Test 1 above verify in-phase indication on all of the above-mentioned devices and verify synchronising relay operating parameters and close command.

Last check is using a high voltage "phasing sticks" across the open generator breaker spouts, verify that the synchronising relay "close" command coincides with the in-phase condition on the phasing stick.

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d) Protection tests

- If a suitable 3-phase primary injection test set is unavailable, differential relays operational stability, in-zone and out-of-zone protection, may be verified using generator as source.

- Stability and operational/sensitivity test for Generator Differential relay

Applied shorting links on 3-phases at an appropriate point outside of the differential protection zone. Subsequently, run up generator to speed and slowly start to excite machine. Generator excitation shall be adjusted from zero to value that no more than full load current may flow.

Applied 3-phase shorting links at suitable point within the differential protection zone. Similar to above, run-up generator and adjust excitation from zero to the relay pick-up and trip points. Other checks/tests can be performed at the same time to verify the sensitivity and operation of their protective functions with this primary current injection test. Protection relays/functions such as overcurrent, negative sequence relays, neutral displacement, etc. verified at this time.

- Overspeed Check:

Operation of mechanical and electronic overspeed trip devices shall be verified.

Synchronising checks (generator running with load)

a) Synchronise to the Grid

The electrical network to which the generator is to be connected should be configured to minimise the risk of disturbance to the other part of the electrical infra-structure.

b) Functional checks/tests:

The following shall be verified:

- Manual synchronisation
- Auto synchronising
- Control of load

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- Control of power factor over the full range of generation
 - Transfer of AVR from voltage control to power factor control and vice versa. Change from Grid connection to "island operation.
 - "Bumpless" change-over from Auto AVR (Auto Voltage Regulation) to Manual Voltage Control (MVC)
- c) Reverse Power relay and Field Failure Protection test

Decrease the governor setting and bring the generator into motoring to operate and verified the reverse power relay operation.

The operation of the Field Failure relay shall be verified by decreasing the excitation at minimum load.

d) Dynamic tests

Dynamic tests shall comprise the following:

- i) Active load rejection, reactive load acceptance, active and reactive load sharing tests and islanding operation test. These parameters shall be measured on a recorder:

- Generator voltage/time; current/time and frequency/time.
- Test acceptable provide it shall meet the below criteria:
 - o None of the protection device activated to trip;
 - o None of the parameters exceed 80% of the difference between nominal value and trip set points;
 - o Transient response shall be within the design parameters.

ii) Governor Tests

- Load acceptance and rejection shall be performed by switching IN and OUT, blocks of active loads to measure/record the transient response. Block load value shall be such as to demonstrate the

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equipment's compliance to the design.

- Droop settings in "islanding mode" shall be measured by changing speed between zero to full load.
- Active load sharing shall be verified that the load sharing is equally (or pro rata for different generators ratings) distributed between the generator being commissioned and all other sharing generators.
- Electrical load variations shall be verified to be within the agreed limits, during the change-over of fuels.

iii) Automatic Voltage Regulation (AVR) Tests

- Transient response during reactive load acceptance and load rejection tests. Generator recovery voltage shall be measured during these tests to verified that it is within the limits of time and terminal voltage.
- Reactive load sharing shall be verified with other generators connected in parallel. This test the ability of the generator being commissioned to develop the same power factor/reactive Var equal during changes in reactive load.
- Voltage droop of generator shall be verified that it is proportional to reactive load or in accordance with design. It is acceptable if test is already conducted in the factory.
- For AVRs whose supply is derived from its terminal voltage, current boost test shall be verified from design data that generator terminal voltage is sufficiently maintained to allow operation of protective devices in the event of a short circuit. Similar to clause c3 that if test conducted in the factory, it is acceptable.

Load Trials

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- a) Load trials, a series of load tests, shall be performed between the Contractor and Owner and will usually after satisfactory completion of the above tests. System and sub-systems peripheral to the prime mover and generator shall be completed and tested before this tests are performed.
- b) Final tests, a protracted run under design conditions at the equipment designed maximum power output.
- c) These final tests will verified the generator, prime mover and all ancillary equipment performance under fully rated load and continuously, meets the designed data.
- d) Temperature and vibration reading shall demonstrate to be within the specified limits.
- e) Tests recording, measurements, log sheets, etc., should form part of the tests report deliverables.

7.3.2.3 Emergency and Standby Generators

Testing methods shall be subjected to agreement with the Owner. Relevant tests from above sections 7.3.2.2.1 to 7.3.2.2.3 shall be carried out.

Functional tests shall be carried out to ensure that the unit functions as intended. Generators with Automatic transfer system with mains failure shall be tested with associated switchgear as a complete system.

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7.3.3 SWITCHGEAR

- a) Commissioning test requirements for Gas Insulated Switchgear (GIS) shall be obtained and carried out by or under supervision from the Manufacturer.
- b) The following inspection/tests should be carried out and recorded in the relevant FORMs and incorporated into the complete test pack returns:
 - FORM #1 – Electrical Equipment Inspection (General)
 - FORM #4a – Inspection of HV, MV Switchgear Busbar Assembly
 - FORM #4b – Inspection of HV, MV Switchgear Busbar Conductivity
 - FORM #4c – Inspection of HV, MV Switchgear switching Units
 - FORM #5a – Inspection of LV Switchboard
 - FORM #5b – Inspection of LV Busbar Conductivity
 - FORM #5c – Inspection of LV Switchboard outgoing units
 - FORM #5d – Inspection of LV Switchboard outgoing Units (cont'd)

7.3.3.1 Conductivity tests

Checks to ensure all bolts and nuts in the power current carrying circuits shall have been tightened to the required torque (using a torque wrench), as recommended by the manufacturer, before carrying out conductivity tests. Random checks shall be carried out to verify compliance with the torque values as recommended by manufacturer.

"Dropper" bus of switchboard shall be considered as part of the busbar system in which conductivity tests have to be carried out.

Conductivity tests shall be carried out between adjacent cubicle spouts. Refer to clause 7.2.2

Result of the measured values shall be interpreted on a comparative basis, also considering difference in values with conductors length and size.

7.3.3.2 Insulation Tests

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At pre-commissioning stage, high-voltage tests shall be carried out on high voltage switchgear components (busbars, circuit breakers and contactors).

If high voltage test is to be performed with a DC test set, test voltage shall be limited to 70% of the peak value of the relevant AC voltage levels (IEC 60694 or IEEE C37.100.1).

Low voltage switchgear components such as busbars, contactors and switches, shall be subjected to insulation resistance tests at the pre-commissioning stage.

Manufacturer shall recommend appropriate testing methods for high voltage cast resin insulated switchgear.

For HV GIS, gas purity, moisture and SO₂ test shall be tested and meet the standard in accordance with the system voltage level.

7.3.3.3 Functional Tests

To demonstrate that all features of the switchboard function is as intended, a full check shall be carried out. All checks/tests in the appropriate forms shall be satisfied.

For complex logic/controls and or installations, contractor shall develop specific functional check lists based upon control schematic diagrams in order to demonstrate compliance with the design requirements.

Functional checks include loop checks for remote controls, remote inter-trips/interlocks and data/controls to/from Scada systems.

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7.3.4 PROTECTION

The following relevant inspection/test form/s should be used for testing and commissioning of the protective devices of the system:

- a) Form #6a-1/6a-2 – Voltage Transformers and Under-voltage relays test
- b) Form #7 – Current Transformers an Ammeter test
- c) Form #8 – Overcurrent and Earth/ground Fault Relays test
- d) Form #10 – Differential Protection Relay Test
- e) Form #11 – Low Voltage overcurrent and earth/ground-fault Protection Relay Test

7.3.4.1 General Tests

Relays should be tested and commissioned in accordance with Manufacturer's instructions. Specific forms shall be developed for multi-function microprocessor-based relays or use of manufacturer's instruction test form. Standard test form should be use if such form is unavailable.

All current transformer (CT) ration shall be verified by primary current injection. Once proven by primary injection, CT connections should not be removed without being retested. Any break in CT circuits, after tests, should only be made at test links provided for the purpose. CT primary shall not be energized with the secondary open circuited.

All protection relays shall be tested with primary current injection at normal settings to verify their operating parameters. Where if it impractical because of extreme current requirements, relays may be subjected to secondary current injection tests with approval from Owner.

Generators and generator/transformer combination tests should be carried out using generator as current source. Refer 7.3.2.2 – base load Generator.

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Testing of relays shall also verify the activation and operation of the switching device's tripping mechanism. All alarms and inter-trips functions shall be verified during this test.

If applicable, motor restart relays and systems shall be tested to verify that their settings are as recommended by the motor reacceleration study.

After completion of each test, it shall be verified that the actual relay setting is in accordance with the setting as indicated on the overall protection scheme of the power system.

Finally after completion, the last test date shall be clearly marked on the relay.

7.3.4.2 Electronic/Digital Relays

Electronic/Digital relays shall be removed or short-circuited during IR tests on connected circuits.

Printed circuit boards and or modules of electronic/digital relays are susceptible to electrostatic discharge damage, if it is incorrectly handled. Failure to take precautionary measures could result in premature failure.

Components are usually delivered in anti-static bags/packing. They should not be removed from the packing until required.

Handling of such components (from digital/electronic relays) shall be under controlled conditions and with anti-static devices/tools.

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7.3.5 POWER TRANSFORMERS

7.3.5.1 Commissioning test requirements for Power Transformers shall be obtained and carried out by or under supervision from the Manufacturer. These relevant inspection test forms should be used for testing and commissioning of power transformers:

- i. Form #1 – Electrical Equipment Inspection (General)
- ii. Form #8 – Inspection of Power Transformer

7.3.5.2 Transformers windings shall not be subjected to high voltage testing. For test voltages and acceptable insulation resistance values, refer to Appendix A and B.

7.3.5.3 Full transformer oil test may not be necessary for newly commission transformer that is transported to site as a complete assembly. Unless there is possibility of contamination, a full test (di-electric test, moisture and DGA) shall be carried out to verify its oil quality.

Samples of oil shall be taken from each drum or a single bulk, for test on its di-electric strength before use for refilling/topping-up.

Large transformers that requires site assemble of certain oil filled components (such as conservator, cooling fins and link/termination boxes) shall be refilled with tested oil and properly bleed to remove air trapped in component/s.

7.3.5.4 After initial energization of a power transformer it should be put on no-load or on "soak" for a few hours before being put on load.

7.3.5.5 Before parallel operation of the transformer is attempted, "phasing-out" shall be carried out to verify that the two supplies are in-phase.

7.3.6 On Load Tap Changer (OLTC)

7.3.6.1 OLTC operation to be tested at transformer factory with no load condition with full voltage for all steps and with recorded voltage waveform analysis.

7.3.6.2 OLTC operations to be tested at transformer factory with full load current for all steps with recorded voltage & current waveform analysis.

7.3.6.3 Winding resistance measurement to be conducted with all OLTC steps.

7.3.6.4 At site OLTC to be tested with full voltage and all steps with voltage and current waveform analysis.

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- 7.3.6.5 After complete OLTC testing, OLTC oil sample DGA to be conducted and values to be compared previous DGA data for any changes. Any change to be reported to Micron and to be analyzed for the cause.
- 7.3.6.6 All protection system to be fully tested and operational while energizing new transformers. Sites may consider lower setting (where practical) while first time energizing the transformer for faster fault isolation.
- 7.3.6.7 Room or area fire protection system and oil containment to be fully functional during the energization of transformers.
- 7.3.6.8 Configure the system for minimum possible impact during new transformer switching (like move the manufacturing load to 1 bus or to one side and isolate from the new transformer or other critical equipment)
- 7.3.6.9 For the case of new site or building where two main transformers are available for testing without significant manufacturing interruption risk, Consider creating circulating current by parallel operation of the transformers and increasing the tap on one transformer & decreasing the tap on other transformer and in that way test the OLTC performance, protection relay actual sensing current, differential relay current & balancing.
- 7.3.6.10 Check the protection relay current sensing data and current unbalance data (Including differential protection relay) during parallel operation as part of load transfer switching.
- 7.3.6.11 If there is no practical choice, protection function disabling to be considered based on full risk study, back up protection system and approval from site facilities director.

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7.3.7 CABLES

- 7.3.7.1 These relevant inspection test forms should be used for testing and commissioning of cables:
- i. Form #2 – Inspection of Electric Motors-cage induction type
 - ii. Form #5 – Inspection of LV Switchboard
 - iii. Form #10 – Inspection of High Voltage Cables
- 7.3.7.2 Testing of all power cables shall be carried out whilst isolated from both supply and load circuits.
- 7.3.7.3 Values of test voltage, method and acceptance insulation resistance values should refer to Appendix A and B.
- 7.3.7.4 Underground cables should be tested before the cable trench is back filled.
- 7.3.7.5 7.3.6.5 All unterminated and exposed cable ends shall be sealed against moisture ingress or chemical contamination.

7.3.8 POWER ELECTRONICS

7.3.8.1 General

- a) Power electronic equipment such as SUPS, DUPS, SMS, DC UPS, VSDS, excitation controls, heater controls, etc., shall be commissioned fully in accordance with the Manufacturer's instructions. Commissioning test requirements for Power electronic equipment such as SUPS, DUPS, SMS, DC UPS, VSDS, shall be obtained and carried out by or under supervision from the Manufacturer. Specific test forms shall be developed for the commissioning of power electronic equipment in the absence of Manufacturer's procedures.
- b) Electrical protection relays of the conventional type shall be tested in accordance with 7.3.3, some devices form an integral part of the central processor unit whose parameters have already been tested in the factory. Functional test should be carried out on site.
- c) All protective devices which perform control, trip and alarm functions shall be functionally tested up to and include the operation of the final element/device.
- d) Upon completion of the commissioning, list of all equipment set points and parameters shall be recorded.

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7.3.8.2 Uninterruptible Power Supply (UPS)

- a) These relevant inspection test forms should be used for testing and commissioning of UPS:
 - i) Form #1 – Electrical Equipment Inspection (General)
 - ii) Form #9a & b – Inspection of Static Supply Unit
- b) Commissioning procedure shall follow the Manufacturer's instructions and shall include tests designed to confirm the static and dynamic performance of the equipment, which include:
 - Rectifier float charge voltage value,
 - Rectifier boost/rapid charge voltage level,
 - Battery discharge capacity test,
 - Battery cell voltage and specific gravity measurements,
 - Operation of Inverter and by-pass synchronism
 - Transfer of load from Inverter to By-pass and return
 - Switch-off Rectifier, the battery will take load and UPS output remains within limits,
 - Oscilloscope checks on equipment's output waveform during supply failure and transfer modes (battery accepts load, switching to bypass and return).
- c) Verification of set points for UPS system shall be carried out during site acceptance test. These tests include:
 - Rectifier mains failure
 - Rectifier output voltage, alarms and shutdown
 - Inverter input voltage high/low (alarm and shutdown)

7.3.8.3 Variable Speed Drives System (VSDS)

- i. These relevant inspection test forms should be used for testing and commissioning of VSD:
 - Form #1 – Electrical Equipment Inspection (General)
 - Form #8 – Inspection of Power Transformer
 - Form #2a & b – Inspection of Electric Motor-Cage Induction type
- ii. A full "string" (i.e. transformer, converter, motor and load) test shall be carried out on site, comprising all

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supplied items with particular attention being paid to items not tested in the factory. Such tests/checks including:

- Settings of specific parameters according to site requirement.
- Full load trials, including auto-restart;
- Full load noise tests throughout the speed range;
- Full load vibration tests throughout the speed range.

7.3.9 LIGHTING

7.3.9.1 These relevant inspection test forms should be used for testing and commissioning of lighting:

- Form #1 – Electrical Equipment Inspection (General)
- Form #5c – Inspection of LV Switchboard outgoing units

Form #1 to be completed per equipment type and within that area.

Form #5c shall be completed for all circuits

7.3.9.2 Random checks shall be carried out to confirm the installation work as per the design.

7.3.9.3 Illumination level shall be taken using a luxmeter with reading at a working plane or 1 metre above the floor level in a horizontal plane. Record the measurement and verified with the required illumination level and take the necessary action/s.

7.3.9.4 All emergency and escape lighting system shall be verified for the operation and autonomy time.

7.3.9.5 The illumination levels shall be as specified in section 6.14.2

7.3.10 EARTHING/GROUNDING AND BONDING

7.3.10.1 General

These relevant inspection test forms should be used for testing and commissioning of earthing/grounding and bonding:

- Form #11a – Inspection of Earthing/Grounding System
Form #11b – Inspection of Earthing/Grounding Electrodes
- Partial on earthing/grounding section only:
- Form #4a and 4c-1 – Inspection of HV Switchgear

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- Form #5a and 5c-1 – Inspection of LV Switchboard
- Form #8 – Inspection of Power Transformer

Test methods and acceptable values as mentioned in section X.X.X

7.3.10.2 Electrical System Earthing/Grounding

- i. All earth/ground electrodes shall be measured, with earth/ground cable disconnected, and the resistance value of each electrode shall not exceed 1 Ω .
- ii. Continuity test shall be carried out on all earth/ground connections between the electrode and the supply neutral and the installation earthing/grounding loop.
- iii. Continuity test shall be carried out on all earth/ground connections between Transformers, Switchboards and the Substation earth/ground bar.
- iv. Earth/ground loop impedance shall be measured using an "earth/ground loop tester" for all power and convenience outlets.

7.3.10.3 Clean Earth/Ground

- a) All earth/ground electrodes shall be measured with earth/ground cable disconnected from the electrode, and the resistance value of each electrode shall not exceed 1 Ω .
- b) All earth/ground connections between the electrode and the clean earth/ground bar and the substation earth/ground bar shall be tested for continuity.

7.3.10.4 Electrical Equipment Earthing/Grounding and Non-Electrical Equipment Earthing/Grounding

- a) Sample measurements shall be made, on 10% of total electrical equipment count, on the return earth/ground path of external earth/ground connection to the equipment.

Earth/ground return impedance measurements shall be carried out with electrical equipment in operational condition and all cables connected. Measurements shall be made between the equipment body and the substation earth/ground bar, using a conductivity meter and a calibrated cable. Results shall be evaluated on a comparative basis.

- b) Earth/ground Bonding measurements shall be taken, with a 5% sample of the total, to verify the continuity of the non-electrical equipment

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earthing/grounding cable (with earth/ground cable disconnected from the equipment) and any adjacent earthing/grounding cable (earthing/grounding cable also disconnected).

7.3.10.5 Earth/ground for lightning and static discharge

- i. All earth/ground electrode systems used for lightning or static discharge shall be measured with the main earth/ground grid disconnected, and the resistance value of each electrode shall not exceed 10 Ω .
- ii. Earth/ground electrode connections and the installation earth/ground loop shall be verified for continuity.

8.0 Reference Specifications

- 8.1 SEMI F61 Guide for Ultrapure Water System Used in Semiconductor Processing

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9.0 Document Control

9.1 **Approval:** GLOBAL_FAC_GFTT_APPR

9.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY

9.3 Final Viewer Access

All Micron Team Members at all Sites

9.4 Retention

Adherence to the requirements specified in the
[**Global Facilities – Document Control Procedures**](#)

9.5 Review

Biennially (two years)

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10.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	08/12/20
1	Main Page	Document name change to Global Facilities – Design & Construction Standard - Electrical - Electrical Specification	02/23/21
	Section 3	Added FCT and EHS to the Term/Acronym list	
	Section 5.2	Added the need to perform a preliminary Short Circuit Analysis Prior to purchasing electrical equipment and the subsequent application of the analysis.	
	Section 5.3.24	Added, “..and other power quality improvement equipment..”	
	Section 5.3.25	Added the need for power factor correction and the consideration if capacitors are utilized	
	Section 5.3.26	Mentioned on the use of a secondary selective protection scheme to provide a higher degree of reliability.	
	Section 5.4.3	Added a reference to APPENDIX H: Facility Electrical Power Supply Matrix.	
	Section 5.5	Added, “..all electrical equipment that needs live electrical work or exposed electrical work..”	
	Section 5.7c	Added: NFPA 70E – Standard for Electrical Safety in the Workplace	
	Section 6.1ii	Update the ambient temperature and humidity requirements. Was - Ambient temperature – 10 Deg Celsius (50oF) to 40 Deg Celsius (104oF) for electrical equipment design and required the electrical room temperature is 24+/- 2 Deg Celsius (71.6~78.8 oF) unless specified in the project specification. Is - Ambient temperature – 10 Deg Celsius (50oF) to 40 Deg Celsius (104oF) for electrical equipment design and required the electrical room temperature is 24+/- 2 Deg Celsius	

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		(71.6~78.8 oF) unless specified in the project specification" from "Ambient temperature – 10 Deg Celsius (50oF) to 40 Deg Celsius (104oF) with average room temperature is 25 Deg Celsius (77oF) unless specified in the project specification."	
		Was - Humidity 40% to 70% with average room humidity is 65% unless specifically mentioned in the project specification.	
		Is - Humidity 0% to 100% with average room humidity is 70% unless specifically mentioned in the project specification.	
	Section 6.1 viii	Added a subsection on special condition area installation: I.S. Machine room.	
	Section 6.2.7 c	Added the need for DC power supply to be monitored and be alarmed on the SCADA system.	
	Section 6.4.5c	Added the need for the DUPS system to automatically switch to bypass and without interruption in an event of an internal failure in the system	
	Section 6.4.6.b	Added Fuel tank level and control.	
	Section 6.5.2c	Added the need for the SUPS system to automatically switch to bypass and without interruption in an event of an internal failure in the system	
	Section 6.5.2d	Mentions the need to check on malfunction possibilities, detection and switching of the equipment to safe mode for UPS and other power quality improvement devices especially those use the DSP	
	Section 6.7.5.2 vii	Added a subsection to the section on Site installation to mention the need for the QA/QC to be approved by Manufacturer's authority before energized.	
	Section 6.7.4.4 iii	Removed the following:	

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Revision #	Section Changed	Details of Changes	Date
		iii. Three phase power supply from LV distribution board for local tap changer motor, heater and control. iv. Supply to the remote tap changer cubicles for transformer shall include: - o Supply from their respective voltage transformer (VT) to the AVR for voltage regulation/line compensation. - o LV supply source for remote controls/operation and monitoring of transformer's cooling fans and tap changer. - o DC power supply to remote tap changer cubicle (OLTC). Include power supply for the AVR. Added a subsection to the section on Low Voltage Switchgear and Control Gear installation to mention the entry point for all of the cable to be from the bottom or side of switchboard Added details to the section on Variable Speed/Frequency Drive type (VSD) Added a subsection to the section on LV MCC Installation to mention the entry point for all of the cable to be from the bottom or side of MCCB/DB Added section on Busway: Busway installation personnel, supplier requirements for Busway and accessories and inspection/testing requirements of Busway joints and PIU/POU Added a section on "On Load Tap Changer (OLTC)" Added Appendix H: Facility Electrical Power Supply Matrix	

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Revision #	Section Changed	Details of Changes	Date
2	5.3	HV power intake feed routing redundancy requirements and main substation scheme updated.	12/08/2023
	5.4.2	Tool Power UPS needs examples removed.	
	5.5	Use of ETAP for all new system studies other than expansion projects updated.	
	6.1	Ambient condition for non-specified electrical rooms added.	
	6.5.3	UPS Battery requirements to include Li Ion batteries. Wet cells use to be limited for electrical protection, engine starting and control circuits.	
	6.8.3.3	Reactor installation clearance, protection against flashover included. Measures to reduce the circulating current also added.	
	6.9.4	Update on emergency generator fuel storage requirements included.	
	6.11.11	Update on LV switchgear & Switchboards on maintenance breaker/slot, busbar insulation requirements, ACB lifting provisions and fuse requirements updated.	
	6.12.4	Protection setting and logic review and approval requirement added.	
	6.13.2	High voltage cable sheath grounding and SVL needs included.	
	6.17.4	Update on busway & PIU assembly, installation, supports and testing needs. Update on busway PIU breaker handle protection needs. Update on busway & PIU maintenance space requirement. Update to reduce the live and exposed work requirements.	12/08/2023
	6.17.5	Live exposed electrical work risk reduction section added.	

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11.0 APPENDIX

Appendix A: Recommended Test Voltage for Commissioning**A1.1 CABLES**

Insulation resistance tests and high voltage tests shall be performed between each phase and earth with the remaining phases connected to earth and loads disconnected.

A1.1.1 Insulation resistance tests:

System Voltage	Test Voltage
LV 50 V to 1 kV	500 V (DC)
HV up to 4.6 kV	2500 V (DC)
HV above 4.6 kV	5000 V (DC)

A1.1.2 High Voltage Tests:

Cable voltage designated in according with following principles:

Example:

U_0/U (U_m) : 3.8/6.6 (7.2)

U_0 : 3.8 kV phase to ground voltage

U : 6.6 kV rated phase to phase voltage

U_m : 7.2 kV maximum permissible voltage of the system

Test Duration: 3 minutes

Applied Voltages as follows:

Cable voltage designation kV (AC) U_0/U (U_m)	Applied Test voltage kV (DC)
Less than (<) 1.0	Insulation Resistance Test only
1.9/3.3 (3.8)	10
3.8/6.6 (7.2)	15
6/10 (12)	25
8.7/15 (17.5)	35
12/20 (24)	50
18/30 (36)	70

NOTES: 1. Cables specifically manufactured for unearthed systems,

i.e. $U_0 = U$, shall be tested at the value given above for U_0 . E.g. cable type 3/3 (3.6) should be tested at 11 kV.

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A1.2 MOTORS, GENERATORS, TRANSFORMERS (coil wound equipment).

A1.2.1 Insulation resistance tests:

System Voltage	Test Voltage
LV Below 1kV	500 V (DC)
HV up to 4.6 kV	2500 V (DC)
HV above 4.6 kV	5000 V (DC)

A1.3 SWITCHGEAR

A1.3.1 Insulation Resistance Test:

Insulation tests on busbar systems shall be carried out between each phase and earth with the remaining phases connected to earth. Voltage transformers and load disconnected.

Insulation resistance tests on circuit breakers and contactors may be carried out together with or separately from the busbar and shall be carried out with the breaker/contactors closed, with loads disconnected.

Insulation resistance test voltage:

System	Test Voltage
LV Systems	500 V (DC)
MV System up to 4.6 kV	2500 V (DC)
HV/MV System above 4.6 kV	5000 V (DC)

A1.3.2 High Voltage test voltages:

High Voltage tests on busbar systems shall be carried out between each phase and earth with the remaining phases connected to earth (voltage transformers and load disconnected).

High Voltage tests on circuit breakers and contactors may be carried out together with or separately from the busbar and shall be carried out with the breaker/contactors closed, with loads disconnected.

Duration: 1 minute

Max System voltage kV					
Test Voltage kV (DC)					

Ensure any surge diverters or voltage transformers are disconnected.

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Appendix B Recommended Insulation Values for Equipment

A1.1 CABLES Insulation resistance: HV and LV minimum length 100 m (NOTE 1)	Minimum for acceptance at commissioning
	$\frac{10 \times \text{kV}}{\text{km}} \text{ M}\Omega$
A1.2.1a MOTORS and GENERATORS A1.2.1.1 Polarisation Index: (NOTES 2 and 3) LV and HV machines Class B and F A1.2.2 Insulation resistance (at 25 °C) LV and HV machines: (NOTES 4 and 5)	2.0 (2) 10(kV+1) MΩ
A1.2.1b POWER TRANSFORMERS (max. 36 kV) A1.2.1b OIL IMMERSED Insulation resistance: A1.2.1b DRY TYPE Insulation resistance: HV side: LV side: A1.3 SWITCHGEAR Insulation resistance: HV bus LV bus LV wiring (NOTE 6)	75 MΩ 100 MΩ 10 MΩ 200 MΩ 20 MΩ 5 MΩ
A1.4 EQUIPMENT and COMPONENTS A1.4.1 FIXED INSTALLATIONS Insulation resistance : A1.4.2 MOVABLE EQUIPMENT. A1.4.3 Hand Tools. Insulation resistance Class I: Class II: Class III: A1.4.4 Distribution Equipment (cables, distribution boards, transformers) Insulation resistance :	5 kΩ/volt 2 MΩ 7 MΩ 2 MΩ 5 MΩ

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- NOTES:
1. IR to be measured with load disconnected.
 2. PI values below those given can be accepted if IR is $> 100(kV+1) \text{ M}\Omega$.
 3. PI measurements on insulation class 'F' machines with IR in the $G\Omega$ range may be difficult to obtain due to meter scale compression.
 4. Minimum insulation resistance values are given for 25 °C equipment temperature; apply corrections for differing temperatures.
 5. For machines $< 10 \text{ MVA}$ energisation is possible if IR or PI is above the minimum given.
 6. Figures are to be used unless local regulations are more stringent in which case the latter shall prevail.

APPENDIX C: Recommended Polarisation Index Values for Equipment

Polarisation Index	Insulation Condition
1.1	Dirty - Moist Insulation
>1.5	Adequate for Maintenance
>2	Good - New
3 and above	Clean-Dry Insulation

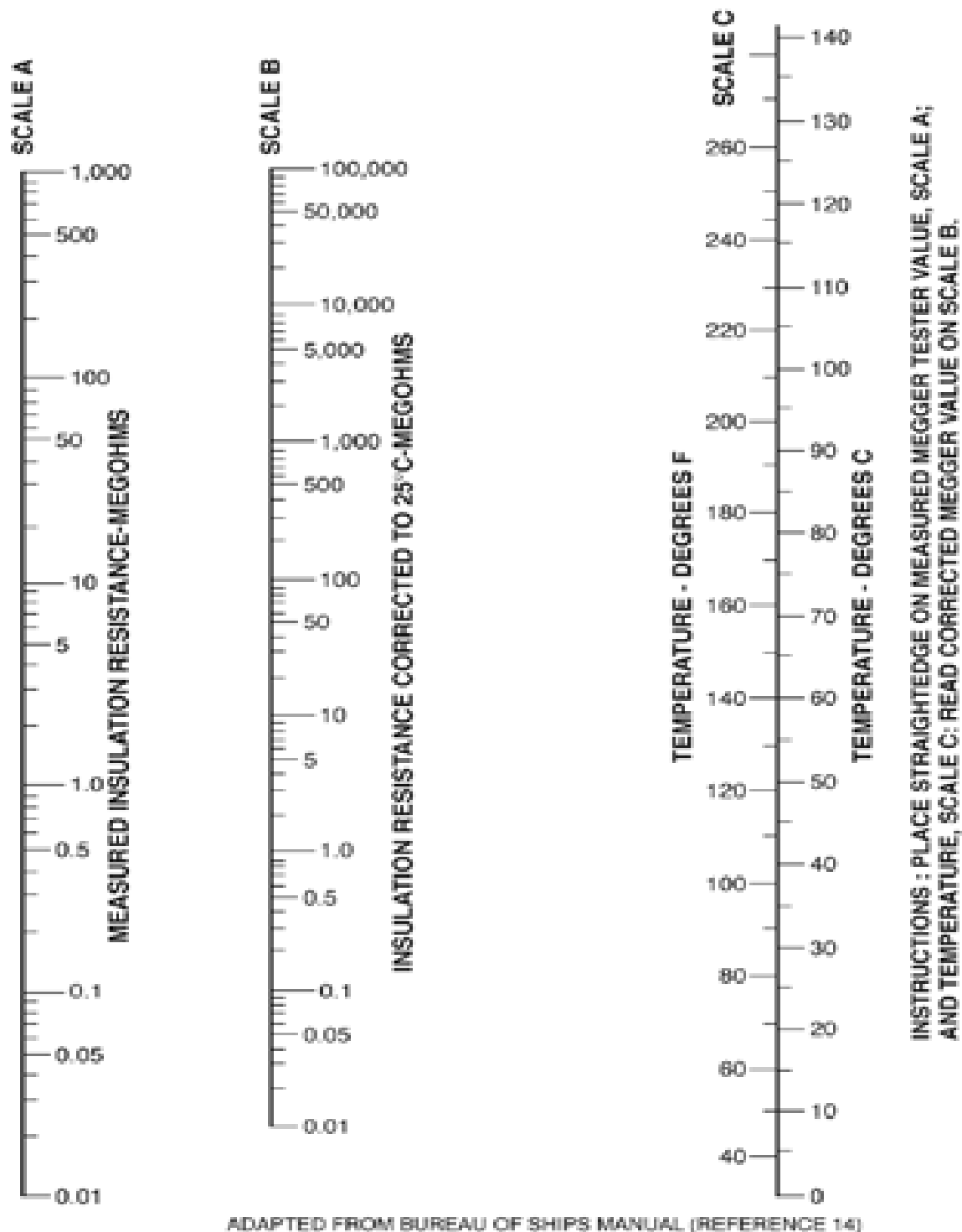
Polarisation Index (PI) value below those given can be accepted if Insulation Resistance (IR) is $> 100(kV+1) \text{ M}\Omega$.

Polarisation Index (PI) measurements for class "F" machines with insulation resistance in the range of G range may be difficult to obtain due to scale compression of the measuring instrument.

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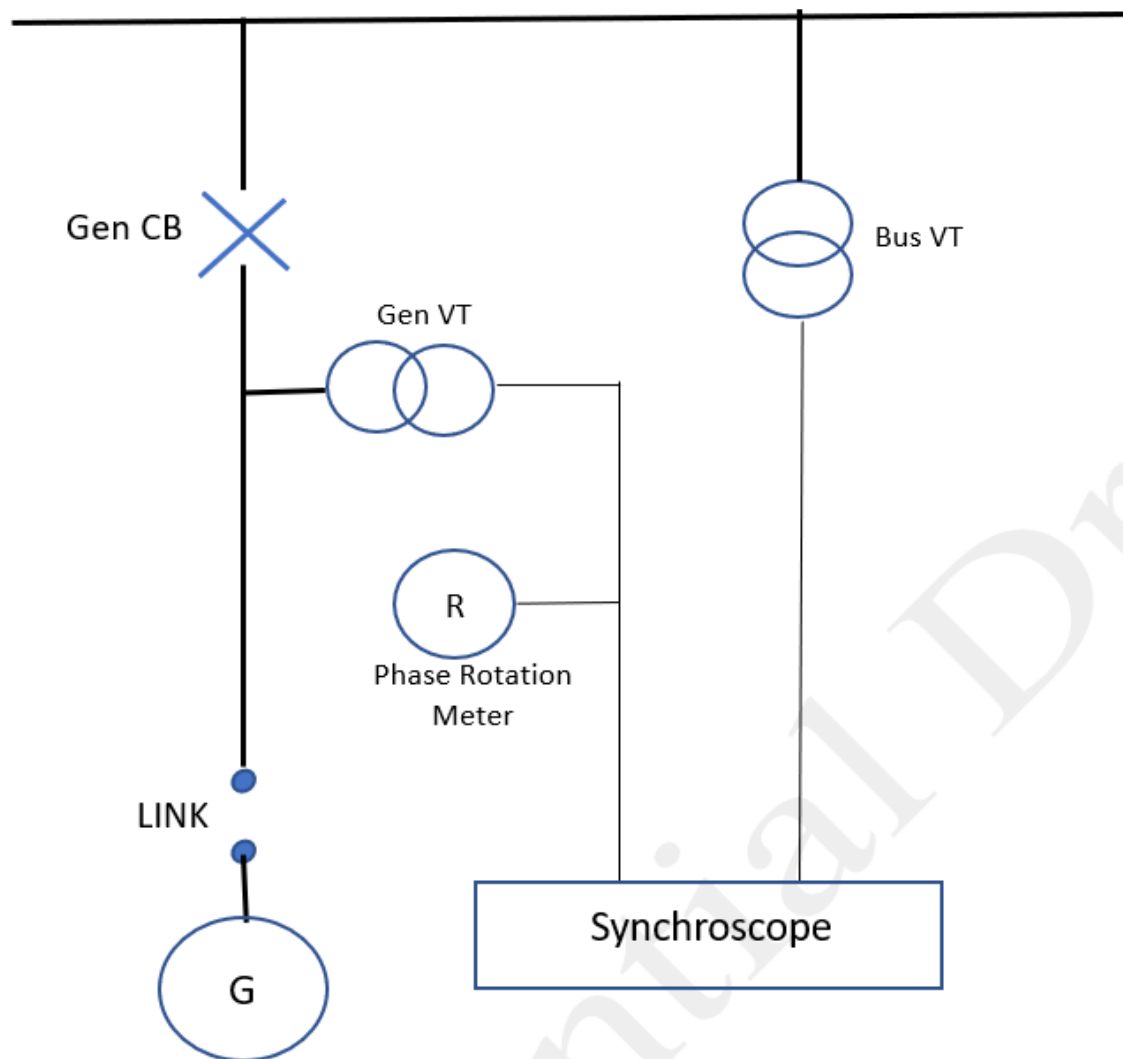
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Appendix D: NOMOGRAM FOR TEMPERATURE CORRECTION

Example: Measured resistance: 200 MΩ

Temperature at measurement: 25 °C

Corrected Resistance: 170 MΩ

Appendix E: TESTING OF GENERATOR SYNCHRONISING SYSTEM

Test 1 – Test condition with generator Circuit Breaker closed, Generator Links removed and phase rotation meter (R) connected.

Verify indication of synchroscope, sync check and voltmeter are in-phase.

Test 2 – Test conditions with generator energised (Generator circuit breaker Opened and Generator link in placed) and phase rotation meter (R) connected.

This test is to verify that synchroscope, voltmeter, auto and check sync relays following the “beat frequency”

Test 3 – From phase rotation meter (R) for test 1 and 2, verification of identical phase rotation is determined.

APPENDIX F: EARTH/GROUND ELECTRODE RESISTANCE

An alternating current is passed between the earth electrode "E" and an auxiliary earth electrode T1 placed at a distance from "E" that the resistance areas of the two electrodes do not overlap.

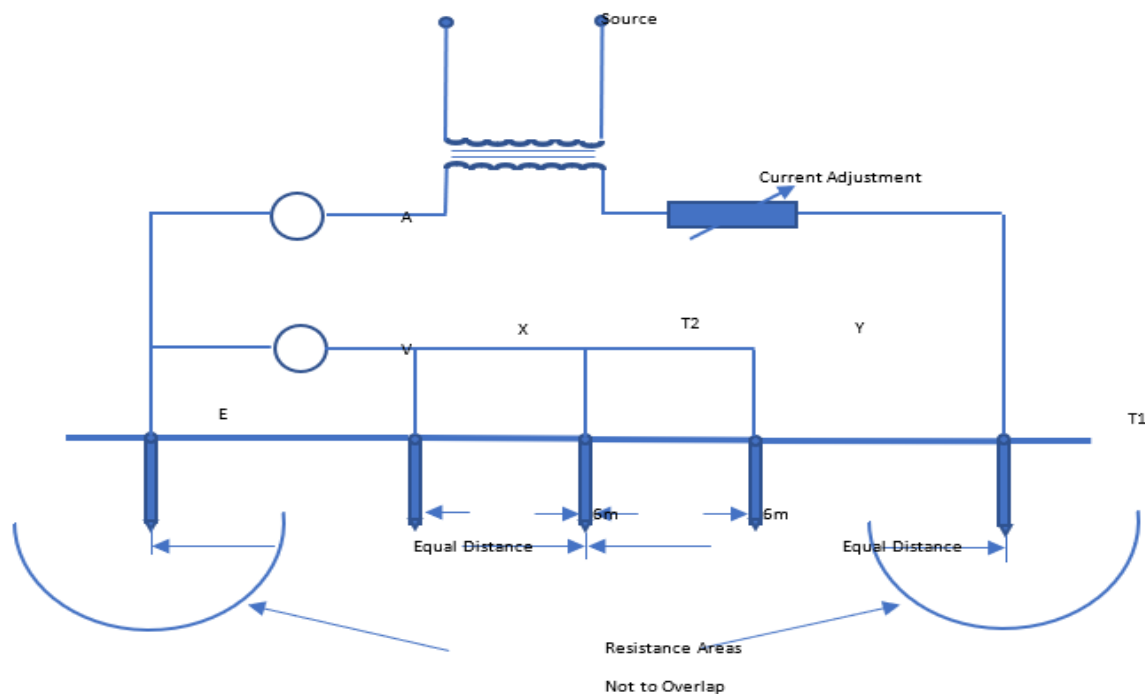
A second auxiliary earth electrode T2, which may be a metal spike driven into the ground, is then inserted half-way between "E" and T1 and the voltage drop between "E" and T2 is measured.

The resistance of the earth electrodes is then the voltage between "E" and T2, divided by the current flowing between "E" and T1, provided that there is no overlap of the resistance areas.

To check that the resistance of the earth electrode is a true value, two further readings are taken with the second auxiliary electrode T2 moved 6 metres further from and 6 metres nearer to "E" respectively. If the three results are substantially in agreement, the mean of the three readings is taken as the resistance of the earth electrode "E". If there is no such agreement the tests are repeated with the distance between "E" and T1 increased.

The test is made either with current at power frequency, in which case the resistance of the voltmeter used must be high (of the order of 200 Ω per volt), or with alternating current from an earth tester comprising a hand-driven generator, a rectifier (where necessary), and a direct-reading ohmmeter.

If the tests are made at power frequency the source of the current used for the test is isolated from the mains supply (e.g. by a double-wound transformer), and in any event the earth electrode "E" under test is disconnected from all sources of supply other than that used for testing.



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Measurement of earth electrode resistance

- E - earth electrode under test, disconnected from all other sources of supply.
- T₁ - auxiliary earth electrode.
- T₂ - second auxiliary earth electrode.
- X - alternative position of T₂ for check measurement.
- Y - further alternative position of T₂ for check measurement.

APPENDIX G: Electrical Equipment Inspection & Testing Forms

The following are forms to be used as documentation for Facilities electrical equipment inspection and testing during project construction, testing and commissioning.

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FORM # 1 ELECTRICAL EQUIPMENT INSPECTION – GENERAL

Equipment Type:	_____	Location:	_____
System voltage:	_____	Equipment Tag/No.:	_____
Rating:	_____	Serial No.:	_____

1. Check nameplate information with equipment record. _____
Any deviations should be recorded under the remarks below.

2. Establish and record the following:

- a. Equipment/circuit identification _____
- b. Degree of enclosure protection (e.g. IPxx) _____
- c. Actual maximum load/ammeter reading. _____
- d. Electrical protection setting/rating _____
- e. Temperature Class/Type of Ex-protection/Gas or dust group _____
- f. Area classification (Zone/Class Division) _____

3. Check the following items and report any deviations/defects under remarks:

- a. Correct labelling (e.g. circuit/location/tag/etc) _____
- b. No deliberate modifications _____
- c. Seals and gaskets satisfactory _____
- d. Bolts and glands/barriers seal complete and tight _____
- e. No damage to enclosures and/or glasses _____
- f. No accumulation of dust and dirt _____
- g. No oil/compound leaks _____
- h. No damage to above-ground part of connected cables _____
- i. Adequate protection of equipment and cables against corrosion, weather, vibration and other adverse factors _____
- j. Motor fans and couplings not rubbing on cowls/guards _____
- k. Earthing in good condition _____
- l. Oil levels in sight glasses correct level _____
- m. Padlock facilities satisfactory _____
- . _____
- n. Silica gel/Dry agent in breathers satisfactory _____
- o. Pressurisation system correct & tested _____

4. Equipment suitable for use in Hazardous location Y/N/NA*

5. Remarks Tester/Inspector: _____
Date: _____

6. Equipment Witness/Accepted Y/N* Engineer: _____
Date: _____

*NOTE: One Form per type of bulk materials, such as light fixtures, boxes etc.

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FORM #2a INSPECTION OF ELECTRIC MOTOR - CAGE-INDUCTION TYPE (INCLUDING CONTROL UNIT)

Location: _____ Equipment Tag/No.: _____

Driven equipment: _____

Manufacturer: _____ Serial No.: _____

System Voltage: _____ V Rating: _____ kW

1. Check Motor record sheet against installed equipment completed. _____

2. Inspection of the switchgear unit supplying this motor is completed:
(Form #5 or #8 – HV or LV Outgoing) Attached Sheet No.: _____

3. Insulation Resistance (IR) tests:

Before cable connection:

Test Voltage: _____ V

IR of motor cables (Phase + other phases to earth)

L1 – E _____ MΩ L2 – E _____ MΩ L3 – E _____ MΩ

Polarisation Index (IP)/IR Test of motor (phases to earth)

After 1 min. _____ MΩ After 10 min. _____ MΩ

Polarising Index ($R_{10 \text{ min}}/R_{1 \text{ min}}$): _____**After cable connection:**

- a) Check cable termination tightness/connection _____
- b) Insulation resistance of motor and cable (phases to earth) _____ MΩ
- c) Insulation resistance of control unit + cable to earth _____

4. Earthing

- a) Check condition of earth connections at Motor: _____
- b) Check condition of earth connections at Control unit: _____
- c) Earth continuity measurement on motor frame to closes earthing point _____ Ω

5. Special items:

- a) Inspect system auxiliary motors _____
- b) Check motor magnetic centre _____
- c) Check installation work of anti-condensation heater _____
- d) Use a 500V insulation resistance tester to test motor heater _____ MΩ
- e) Use a 500V insulation resistance tester to test motor temperature detectors _____ MΩ
- f) Test bearing insulation, if applicable _____ MΩ

NOTE: Polarisation Index may be waived for motors below 30kW or 40 HP rating.

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FORM #2b INSPECTION OF ELECTRIC MOTOR - CAGE-INDUCTION TYPE (INCLUDING CONTROL UNIT) cont'd

6. Electrical Equipment Inspection – General, completed

(Form #1)

Attached Sheet No.:

7. Test/record the following items in uncoupled condition:

a) Record rotation viewed from NDE: CW/CCW* (Confirm correct)

b) Measure vibration (refer Specification 7.3.1 b):

DE Side value (x) mm/s Top value (y)mm/s Axial value (z)mm/s

NDE Side value (x)mm/s Top value (y)mm/s Axial value (z)mm/s

Max, Vibration reading: DE/NDE* mm/s RMS

c) Record motor starting current: A

d) Motor no load Current: A

e) Motor bearing temperature:

Temp./Time	min	0	20	40	60
Temp. DE	°C				
Temp. NDE	°C				
Ambient Temp.	°C				

8. Test/record the following items in full load condition:

a) Measure vibration (refer Specification 7.3.1 b):

DE Side value (x) mm/s Top value (y)mm/s Axial value (z)mm/s

NDE Side value (x)mm/s Top value (y)mm/s Axial value (z)mm/s

Max, Vibration reading: DE/NDE* mm/s RMS

b) Motor bearing temperature:

(motors < 100 kW, for 60 min)

Temp./Time	min	0	30	60	90	120	150	180	210
Temp. DE	°C								
Temp. NDE	°C								
Ambient Temp.	°C								

9. Record in coupled condition:

Motor starting/run-up time (if over 1 s) s

10. Remarks

Tester/Inspector:

Date:

11. Equipment Witness/Accepted Y/N*

Engineer:

Date:

NOTE:

For motors below 30kW or 40 HP rating the following tests may be waived:

1) Uncoupled vibration measurements (Clause 7b, c & e);

2) Full load test and motor starting time (Clause 8 & 9)

* Delete as necessary.

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FORM #3a INSPECTION OF SYNCHRONOUS MACHINES (INCLUDING EXCITER)

Location: _____ Equipment Tag/No.: _____

Driven equipment: _____

Manufacturer: _____ Serial No.: _____

System Voltage: _____ V Rating: _____ kW

1. Inspection of the switchgear unit complete: **(Form No. 4 or No. 11)** _____
- Electrical Equipment Inspection – General, completed: **(Form No. 1)** _____

2. Insulation resistance tests:

Insulation resistance test of stator (phase to earth):

after 1 min: _____ MΩ Test voltage: _____ V

after 10 min: _____ MΩ Polarisation index: _____

Insulation resistance test of stator + cable: _____ MΩ

Insulation resistance test of rotor circuit (diodes short circuited): _____ MΩ

Insulation resistance test of exciter field: _____ MΩ

Insulation resistance test of exciter field + cable: _____ MΩ

3. Earthing, check the condition of earthing connections at:

machine: _____ exciter: _____ neutral: _____

4. Special items:

- check anti-condensation heater installation working _____
- test insulation resistance of heater at 500 V _____ MΩ
- test insulation resistance of temperature detector _____ MΩ
- check magnetic centre _____
- test bearing insulation (if fitted) _____ MΩ
- inspect system auxiliary motors _____

5. Vibration, measure at full load condition:

refer sheet No.: _____

Maximum readings DE _____ mm/s RMS

NDE _____ mm/s RMS

NOTES: - HV cables tested using Form No. 14

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FORM #3b INSPECTION OF SYNCHRONOUS MACHINES (INCLUDING EXCITER)
GENERATOR LOAD TRIALS

DATE:	TIME, HOURS				
LOAD:					
- Voltage					
- Current					
- PF					
- MW					
- Mvar					
EXCITATION:					
- Current					
- Voltage					
TEMPERATURE:					
Stator Frame/Body DE					
Stator Frame/Body NDE					
Windings RTDs:					
U1 Phase reading					
V1 Phase reading					
W1 Phase reading					
U2/UV winding reading					
V2/UW winding reading					
W2/VW winding reading					
Air:					
Ambient					
Inlet					
Outlet					
Water/Other medium:					
Inlet					
Outlet					
Bearings:					
DE					
NDE					

Remarks

Tester/Inspector:

.....

Date:

.....

.....

.....

.....

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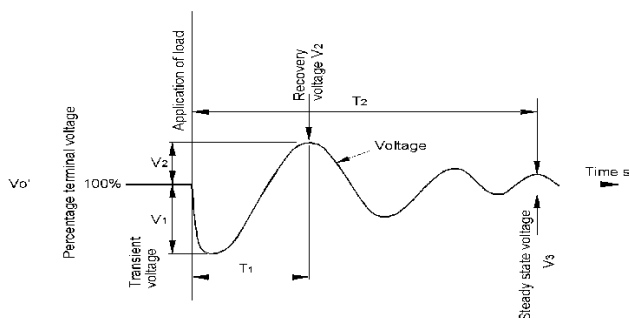
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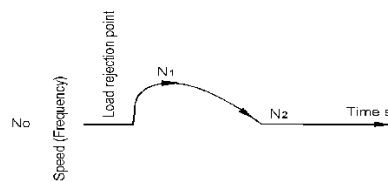
FORM #3c SYNCHRONOUS A.C. GENERATOR DYNAMIC TESTS LOAD (ACCEPTANCE/REJECTION* TEST RESULTS)

* Delete as applicable

Date & Time				
Generator Initial Load (kW)				
Load Acceptance / Rejection * (kW/kVA*)				
Generator Speed:				
Initial Frequency (N0)				
Transient Frequency (N1)				
Steady State Frequency (N2)				
Time to N1 Peak/Min* Frequency				
Time to Steady State Frequency				
Generator Voltage:				
Initial Voltage (V0)				
Transient Voltage (V1)				
Recovery Voltage (V2)				
Steady State Voltage (V3)				
Recovery Time (T1) to V2				
Time to Steady State (T2) to V3				
Governor Position:				
Initial %				
Transient %				
Steady State %				



Typical voltage response characteristic
(Load Acceptance)



Typical speed response characteristic
(Load rejection)

Remarks

Tester/Inspector:

Date:

Equipment Accepted Y/N*

Engineer:

Date:

Current ECN: 101154991

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FORM #4a INSPECTION OF BUSBAR ASSEMBLY - HV SWITCHGEAR

Location: _____ Manufacturer: _____
Switchboard Tag: _____ Section: _____
System Voltage: _____ Rating: _____

1. Insulation Resistance test – Busbar System

Test Voltage : _____ V Tester Used : _____
Phase L1 _____ L2 _____ L3 _____ MΩ

2. HV pressure test – Busbar System

Test Voltage DC : _____ KV Tester Used : _____
Leakage Current : _____
Phase L1 _____ L2 _____ L3 _____ mA

3. Conductivity Test – Busbar System

Spout to spout
Use Form #4b

4. Earthing System:

Check earth connections and earthing bar _____
Measure the continuity of earthing bar to main earth _____ mΩ

5. Voltage Transformer Test (test to include Undervoltage relay and voltmeter/s)_

Sheet No. : _____

Use Form #6a & b

6. Electrical Equipment Inspection – General, completed (Form #1)

Sheet No. : _____

7. Remarks :

Tester/Inspector : _____

Date : _____

8. Equipment Acceptance : Y/N *

Engineer : _____

Date : _____

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FORM #4b INSPECTION OF HV SWITCHGEAR BUSBAR CONTINUITY
(INCLUDING INCOMING AND SECTIONALISER)

Location: _____

Manufacturer: _____

Switchboard Tag: _____

Section: _____

System Voltage: _____

Rating: _____

1. Conductivity Test Instrument (Ducter) : _____
2. Test Current (Injection) : _____ A

Notes: a) *Measurements shall be carried out between adjacent outgoing panels or adjacent outgoing panel and incomers in order to check the entire busbar length including "droppers and/or risers" bars.*

b) *Measured values of identical connections shall not differ by more than 20% from each other.*

Circuit designation		Busbar conductivity, mΩ				
From Pnl #	To Pnl #	L1	L2	L3	N	E

3. Remarks : _____ Tester/Inspector : _____
- Date : _____
4. Equipment Acceptance : Y/N * Engineer : _____
- Date : _____

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FORM #4c_1**INSPECTION OF SWITCHING UNITS (HV SWITCHGEAR)**

Location: _____

Switchboard No.: _____

Manufacturer: _____

Switching Device Type: _____

Panel No.: _____

Rating: _____

1. Check the items below and report any deviations/defects under remarks:

- | | | |
|----|---|----------|
| a. | Check correct labelling | _____ |
| b. | Safety shutters operation | _____ |
| c. | Padlocking facilities | _____ |
| d. | Insulation bushings and barriers | _____ |
| e. | Alignment of main and auxiliary contacts | _____ |
| f. | Cable box and terminations | _____ |
| g. | Mechanical position indicators, indication lights | _____ |
| h. | Mechanical closing and tripping | _____ |
| i. | Interlocking facilities, mechanical/electrical | _____ |
| j. | Electrical closing – local/remote | _____ |
| k. | Electrical tripping – local/remote | _____ |
| l. | Alarm indications – local/remote | _____ |
| m. | Anti-condensation heater working | _____ |
| | - Insulation resistance test at 500 V | _____ MΩ |
| n. | Secondary and auxiliary wiring | _____ |
| | - Insulation resistance test at 500 V | _____ MΩ |
| 2. | Earthing: | |
| | Check earthing connections and earthing bar | _____ |
| 3. | Record fuse/MCB type/rating | |

Continue next page - Form #4c_2 Cont'd

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FORM #4c_2**INSPECTION OF SWITCHING UNITS (HV SWITCHGEAR) – Cont'd**

4. Record operation counter reading : _____
-
5. Insulation resistance – circuit breaker/contactors main contacts:
 Test Voltage (DC) _____ V Tester used: _____
 Phase L1: _____ Phase L2: _____ Phase L3: _____ MΩ
-
6. High Voltage Test
 Test Voltage (DC) _____ kV Tester used: _____
 Leakage Current L1: _____ L2: _____ L3: _____ mA
-
7. Insulation resistance – (After HV test):
 Test Voltage (DC) _____ V Tester used: _____
 Phase L1: _____ Phase L2: _____ Phase L3: _____ MΩ
-
8. Conductivity Test: Tester used: _____
 Phase L1: _____ Phase L2: _____ Phase L3: _____ mΩ
-
9. Vacuum circuit breakers/contactors: Check for loss of vacuum: _____
 For SF6 circuit breakers/contactors: SF6 pressure: _____
 High voltage test: _____ kV leakage current: _____ mA
 Across open contacts (refer to Manufacturer's instructions)
 High voltage test: _____ kV leakage current: _____ mA
-
10. Current transformer test completed: _____ Sheet No.: _____
-
11. Protection relay test completed: _____ Sheet No.: _____
-
12. Restart relay tested, if any: _____ Result (delay): _____ sec.
-
13. Remarks: Tester/Inspector: _____
 Date: _____
-
14. Equipment Acceptance : Y/N * Engineer : _____
 Date : _____
-

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FORM #5a INSPECTION OF LV SWITCHBOARD
(INCLUDING INCOMERS AND SECTIONALISER UNITS)

Location: _____

Switchboard No.: _____

Manufacturer: _____

Section: _____

System voltage: _____

Rating: _____

1. Electrical Equipment Inspection – General, (Form #1) completed

Sheet No. : _____

2. Insulation resistance test – busbar system at 500 V

Tester used: _____

Phase L1-E _____

L2-E _____

L3-E _____ MΩ

Anti-condensation heater (if any) working. Y/N *

Insulation resistance of heater at 500V _____ MΩ

3. Conductivity test for busbar system. Use Form #5b

4. Earthing system

Check connections and condition of earthing bar _____

Measure continuity of switchboard to main earth grid _____ mΩ

5. Current transformer test, including ammeters, completed.
Refer to Form 7

Sheet No.: _____

6. After energisation:

Check voltmeter reading: _____

Check phase sequence/rotation: _____

Check phasing at sectionaliser:
(for parallel operation Switchboard) _____

7. Remarks:

Tester/Inspector: _____

Date: _____

8. Equipment Acceptance : Y/N *

Engineer : _____

Date : _____

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**FORM #5b INSPECTION OF LV SWITCHBOARD BUSBAR CONTINUITY
(INCLUDING INCOMING AND SECTIONALISER)**

Location: _____

Switchboard No.: _____

Manufacturer: _____

Section: _____

System Voltage: _____

Rating: _____

1. Conductivity Test Instrument (Ducter) : _____

2. Test Current (Injection) : _____ A

Notes:

a) Measurements shall be carried out between adjacent outgoing panels or adjacent outgoing panel and incomers in order to check the entire busbar length including “droppers and/or risers” bars.

b) Measured values of identical connections shall not differ by more than 20% from each other.

Circuit designation		Busbar conductivity, mΩ				
From Pnl #	To Pnl #	L1	L2	L3	N	E

3. Remarks :

Tester/Inspector : _____

Date : _____

4. Equipment Acceptance : Y/N *

Engineer : _____

Date : _____

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FORM #5c-1 INSPECTION OF LV SWITCHBOARD OUTGOING UNIT

Location:	_____	Switchboard No.:	_____
Manufacturer:	_____	Function:	_____
Switching Device Type:	_____	Panel No.:	_____
System Voltage:	_____	Rating:	_____

1. Check the items below and report any deviations/defects under remarks:

- | | |
|--|----------|
| a. Check correct labelling | _____ |
| b. Mechanical damage | _____ |
| c. Door interlock | _____ |
| d. Mechanical position indication, indication lights | _____ |
| e. Alignment of contacts, withdrawal type only | _____ |
| f. Electrical closing – local/remote | _____ |
| g. Electrical tripping – local/remote | _____ |
| h. Alarm indications – local/remote | _____ |
| i. Wiring connection and marking | _____ |
| - Insulation resistance test at 500 V | _____ MΩ |

2. Check Earthing connections and condition of earth bar _____

3. Insulation resistance test -power/switching device at 500 V

Phase	L1-L2/L3/E _____	L2-L1/L3/E _____	L3-L1/L2/E _____	MΩ
-------	------------------	------------------	------------------	----

4. Protection relay test, Form #11

Sheet No.: _____

5. Restart relay tested, if applicable _____

Result: _____ sec.

6. Check and record fuse/MCCB:

Class/Characteristic: _____ Rating: _____ A Condition: _____

For MCCB:

Current setting range: _____ A Operating Short Circuit Rating: _____ kA

7. Remarks :

Tester/Inspector : _____

Date : _____

8. Equipment Acceptance : Y/N *

Engineer : _____

Date : _____

*Note: Complete page 2 for lighting circuits and convenience outlets circuits (use one form per board)

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FORM #5c-2**INSPECTION OF LV SWITCHBOARD OUTGOING UNIT (Cont'd)**

1. Lighting Circuits Insulation Resistance Test

- Test conducted with fittings disconnected.

Date :		Date :	
Circuit Number	IR Value (MΩ)	Circuit Number	IR Value (MΩ)

2. Convenience Power Outlets

Date	Circuit Number	IR Value (MW)	Polarity as marked	Earth loop impedance	Residual Current Device

3. Remarks :

Tester/Inspector : _____

Date : _____

4. Equipment Acceptance : Y/N *

Engineer : _____

Date : _____

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FORM #6a-1**TEST FOR VOLTAGE TRANSFORMER AND UV RELAY**

Location: _____ Manufacturer: _____

Switchboard Tag: _____ Section/Panel No.: _____

Voltage transformer: _____

Type: _____ Serial No.: _____ Ratio: _____

Vector Group: _____ Rating (VA): _____ Class: _____

1. Record the protective device/fuse rating _____ A

2. Check the following and record deviations/defects under remarks below:

a) Mechanical damage	_____	b) Alignment of contacts	_____
c) Connections	_____	d) Earthing points	_____
e) Locking facility	_____	f) Core marking	_____
g) Gas/liquid pressure/level	_____		

3. Insulation Resistance (IR) Tests:

Secondary wiring at 500 V: _____ MΩ

Secondary winding with Primary winding earthed, at 500 V:

Phase to Phase:

L1-L2 : _____ MΩ L2-L3 : _____ MΩ L3-L1 : _____ MΩ

Phase to Earth:

L1-E: _____ MΩ L2-E: _____ MΩ L3-E: _____ MΩ

IR test on Primary Winding with secondary winding earthed

Phase to Phase:

L1-L2 : _____ MΩ L2-L3 : _____ MΩ L3-L1 : _____ MΩ

Phase to Earth:

L1-E: _____ MΩ L2-E: _____ MΩ L3-E: _____ MΩ

4. Test Voltage Transformer polarity using battery flick test

L1 – E : _____ L2 – E : _____ L3 – E : _____

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FORM #6a-2 TEST FOR VOLTAGE TRANSFORMER AND UV RELAY (Cont'd)
(After Energisation Test)

5a. Measure Voltage Transformer secondary voltage:

Phase to phase L1-L2 _____ V, L2-L3 _____ V, L3-L1 _____ V

Phase to Earth L1-E _____ V, L2-E _____ V, L3-E _____ V

5b. Check phase rotation : _____

5c. Voltmeter Reading secondary voltage _____ V

Meter Reading _____ kV

6. Undervoltage Relay Test

Manufacturer: _____ Type: _____

	Drop of voltage	Pick-up voltage
Setting Range:	_____	_____ V
Selected Setting:	_____	_____ V
Measured Value:	_____	_____ V

7. Remarks :

Tester/Inspector : _____

Date : _____

8. Equipment Acceptance : Y/N *

Engineer : _____

Date : _____

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FORM #7 TEST FOR CURRENT TRANSFORMER AND AMMETER

Location:	_____	Manufacturer:	_____
Switchboard Tag:	_____	Section/Panel No.:	_____
Current transformer:	_____	Function:	_____
Type:	Bar/Wound*	Serial No.:	_____
Ratio:	_____	Class:	_____
Rating (VA):	_____		

1. Check item for:

- (a) Mechanical damage _____
- (b) Connections tightness _____
- (c) Star-point connection and earth link _____

2. Insulation resistance test on secondary winding at 500 V

Phase to phase L1-E _____ L2-E _____ L3-E _____

3. Resistance test of secondary circuit, including transformer winding and all connected secondary loads.

L1 _____ L2 _____ L3 _____ Ω

4. Current transformer (CT) polarity test by battery flick test

L1 _____ L2 _____ L3 _____

5. Ratio test of CT (can be combined with relay testing)

Phase :	L1	L2	L3
Primary Current (A)			
Secondary Current (A)			
*Ammeter Reading (A)			

* Include current transducers, if any

6. Remarks :

Tester/Inspector :

Date :

7. Equipment Acceptance :

Y/N *

Engineer :

Date :

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FORM #8 TEST FOR OVERCURRENT/EARTH FAULT PROTECTION RELAY

Location: _____
 Switchboard Tag: _____

Manufacturer: _____
 Panel No.: _____

Relay: Overcurrent inverse time

Manufacturer: _____

Serial No.: _____

Type: _____

CT Ratio: _____

Current Setting Range: _____ A

Time Setting Range: _____ sec/%*

Set Current Ip: _____ A

Set Time Tp: _____ sec/%*

Characteristic: _____

1. Secondary Injection Test:

Test at Current Setting (Ip) and Time setting Tp at 100%

Secondary Injection Current				Trip Time (seconds)					
Multiple Set Current (Ip)	Current (A)			Curve			Actual		
	R	Y	B	R	Y	B	R	Y	B
Ip				N/A			No Trip		
1.3 x									
2 x									
4 x									

* Inject and measure minimum relay operating current.

Record values for all phases & earth fault elements, injected.

2. Primary Injection Test:

Test at Current Setting (Ip) and Time setting Tp at 100%

	Primary Injection Current		Trip Time (seconds)	
Multiple Ip	Pri. Current	Current (A)	Curve	Actual
1.3 x				
2 x				

a) Check correct trip action: _____

b) Check correct flag/indication operation: _____

3. Relay setting after test:

Instantaneous element _____ A
 Current Setting _____ A
 Time Setting _____ s/%*
 Adhesive label provided _____ Y/N*

4. Remarks :

Tester/Inspector : _____

Date : _____

5. Equipment Acceptance : Y/N *

Engineer : _____

Date : _____

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FORM #9 TEST FOR MOTOR PROTECTION RELAY (NUMERICAL)

Location: _____

Switchboard Tag: _____

Ratio Core Balance/Ground CT, if any: _____ A

Panel No.: _____

CT Ratio: _____

Relay: Motor Protection Relay (MPR)

Manufacturer: _____

Type: _____

Serial No.: _____

1. Site verification of Motor Detail against protection schedule:

Full Load Current (FLC): _____ A

Locked Rotor Current (ILRT): _____ A

Locked Rotor Time (Cold/Hot): _____ / _____ sec.

Starting Current: _____ A

Starting Time: _____

RTD Sensor Type, if any: _____

Cooling Time, if any (kc): _____

Number of starts (Cold/Hot): _____

Safe stall time (t6x): _____ sec.

2. Motor Protection Relay settings:

Overload current setting: _____ A

Starting current: _____ A

Set safe stall time (t6x): _____ sec.

Cooling Time Constant: _____ s/min*

Short circuit protection (Isc): _____ A

Characteristic: DT/IDMT/RI

Time Delay: _____ sec.

Earth-fault protection current: _____ A

Operating Time: _____ sec.

Under current setting, if applicable: _____ A

Time Delay: _____ sec.

3. MPR injection and functional test:

a) Secondary Injection test

Nominal injection current I_s = setting current x rated current: _____ A

Injection Current		Operating Temperature	Trip Time	
x I_s	A		Curve	Actual
4x		Cold		
1x		Warm	Running load indication: A	
2x		Warm		
4x		Warm		

Instantaneous elements: Short circuit/earth fault

Setting: _____ x I_n = _____ A

Pick-up current: _____ A

b) Primary Injection Test: Injection current 100%

FLC _____ A; Indication: _____ A

Functional verification/test:

Event Recorder: _____

Recording function: _____

Communication to remote/Scada: _____

Alarms/indication: _____

Trip function/operation: _____

Remote controls/indication/measurements: _____

Relay Supervision: _____

Latched: _____

* Check setting normalise as per protection schedule and Adhesive label provided (Y/N*).

4. Remarks :

Tester/Inspector : _____

Date : _____

5. Equipment Acceptance : Y/N *

Engineer : _____

Date : _____

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FORM #10 DIFFERENTIAL PROTECTION RELAY TEST

Location: _____

Switchboard Tag: _____

Relay: _____

Upstream/Downstream* _____

Manufacturer: _____ Type: _____

Current setting range: _____

Time setting range: _____

Panel No.: _____

CT ratio: side 1: _____

Interposing CT: _____ if any

CT ratio: side 2: _____

Interposing CT: _____ if any

Selected setting: _____ %

Selected setting: _____ s

1. Secondary injection test:

Sensitivity test: at max. setting _____ A

at min. setting _____ A

at selected setting _____ A

Stability test: at 100% nominal current _____ A

at 500% nominal current _____ A

2. Primary current injection tests:

Sensitivity test (supply from test set) at selected setting _____ A

Stability test: supply from system _____ A

At actual load or 100% nominal current for generators: Y/N*

3. Verified functions:

Check correct reset action _____

Check correct flag/indication operation _____

Check correct trip action (include operation CB/VCB) _____

Check communication to Scada & relay supervision _____

4. Relay setting after test verification:

Check correct setting _____ %/A

Time setting _____ s

Adhesive label provided Y/N *

5. Remarks :

Tester/Inspector : _____

Date : _____

6. Equipment Acceptance : Y/N *

Engineer : _____

Date : _____

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FORM #11 TEST FOR LV OVERCURRENT AND EARTH-FAULT PROTECTION RELAY

Location: _____

Switchboard Tag: _____

Panel No.: _____

Relay: Motor Protection Relay

Manufacturer: _____

Type: _____

Contactor rating: _____ A

Trip setting range: _____ A

Motor Full load current (FLC): _____ A

1) Primary injection test at selected trip setting _____ A (all phases connected in series).

Test current in % setting	300	300	120	105
Current (A)				
Relay condition	Cold	Warm	Warm	Warm
Relay time, seconds				No trip after _____ min.
Curve time, seconds				

Earth-fault relay setting: _____ A

Measured pick-up current: _____ A

2. Relay setting after test:

Thermal relay _____ A

Earth-fault relay _____ A

Setting sealed _____

3. Functional verification/test (for numerical/digital type):

Event Recorder: _____

Recording function: _____

Communication to remote/Scada: _____

Alarms/indication: _____

Relay Supervision: _____

Trip function/operation: _____

Latched: _____

Remote controls/indication/measurements: _____

* Check setting normalise as per protection schedule and Adhesive label provided (Y/N*).

4. Remarks :

Tester/Inspector : _____

Date : _____

5. Equipment Acceptance : Y/N *

Engineer : _____

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FORM #12-1 INSPECTION/TEST OF POWER TRANSFORMER

Location: _____
 Transformer Tag: _____
 Enclosure: _____
 Rating: _____ kVA/MVA*
 Vector group: _____

Manufacturer: _____
 Serial No.: _____
 Voltage: _____ / _____ V
 Current: _____ / _____ A

1. Inspection of the switchgear transformer feeder unit completed: _____
 Protection relay setting: _____ Sheet No.: _____

2. Insulation resistance tests:

	Cable incl/excl*	Test Voltage	IR (MΩ)	Meter used
Pri. winding to earth				
Sec. winding to earth				
Pri to Sec winding				

Polarisation Index test for Dry type transformer (Phase to earth):

Insulation resistance test of stator :

after 1 min: _____ MΩ Test voltage: _____ V
 after 10 min: _____ MΩ Polarisation index: _____

3. Continuity test – tap changer: after moving tap changer, switch set at tap number: _____
 Continuity test – Primary (HV) winding:
 L1-L2 _____ mΩ L2-L3 _____ mΩ L3-L1 _____ mΩ

4. Earthing/Grounding:
 Check condition of neutral earthing _____
 Measure neutral earthing continuity _____ mΩ
 Measure equipment earthing continuity _____ mΩ

5. For breather type transformers:
 Check the position of the conservator valve/s _____
 Check desiccant condition/colour _____
 Result of oil test for main tank _____
 Result of oil test for conservator _____
 Test gas/oil relay (Buchholz) alarm and trip _____
 Pressure relief tripping device function/condition _____
 Check thermometer for oil and winding _____
 Reset max. indicator/s for thermometer gauge/s _____
 Check oil level indicator/s _____

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FORM #12-2 INSPECTION/TEST OF POWER TRANSFORMER (cont'd)

6. Transformer Tap Changer:

Check manual tap operation _____

Check automatic tap changer operation _____

Check remote tap operation _____

Check parallel operation; master & slave, individual
(if applicable) _____

Verify voltage regulation operation/function, if applicable _____

Verify alarm, controls and trip functions include Scada comm. _____

7. Cooling fans/pumps Control & Operation, if any

Check manual cooling fans operation _____

Check automatic cooling fan operation activation
via oil/winding temperature _____

Check remote fan operation, if applicable _____

Verify alarm, controls and trip function _____

8. Check single core cable bonding, if applicable _____

Verify transformer surge arresters, if applicable _____

9. Transformer turn ratio test at designated tap position _____

Test transformer neutral grounding impedance device _____

Polarity Test _____

Phase Relation Test _____

10. Electrical Equipment Inspection – General, completed (Form #1)

Sheet No.: _____

11. Remarks :

Tester/Inspector : _____

Date : _____

12. Equipment Acceptance : Y/N *

Engineer : _____

Date : _____

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FORM #13-1 INSPECTION/TEST OF STATIC SUPPLY UNIT

Location: _____

Equipment Tag: _____

Manufacturer: _____

Rectifier/Inverter: _____

Serial No.: _____

Battery Type: _____

No. of cells/volts: _____

1. AC/DC Distribution Board – Check

Good condition – no mechanical damage _____

Correct labelling _____

Correct earthing _____

Test insulation resistance at 500 V:

Bar 1 to earth: _____ MW

Bar 2 to earth: _____ MW

Bar 1 to Bar 2: _____ MW

Test earth-fault indication and detection _____

Record :

Fuse/Circuit breaker size: _____ Type: _____ Actual load: _____ A

2. Check the following:

Cleanliness _____

Correct earthing _____

No abnormal noise _____

Parallel Capacitors same temperature _____

No leaking capacitors _____

Cable connections _____

Operating temperature _____

3. Check correct operation of:

Cooling fans _____

Common alarm _____

Test/Reset button/key _____

Indication lights/meters _____

Interlock system _____

Scada communication _____

4. Check Rectifier operating set points:

Float charge voltage: _____ V

Boost Charge Voltage: _____ V

Check indication of alarm or trip function for:

AC Failure: _____ V _____ s

Low DC voltage: _____ V

Earth fault: _____

Rectifier failure: _____ V

High DC voltage: _____ V

Fuse failure/MCB: _____

Rectifier output oscilloscope check:

Check at normal load _____

5. Electrical Equipment Inspection – General, completed (Form #1)

Sheet No.: _____

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FORM #13-2 INSPECTION/TEST OF STATIC SUPPLY UNIT (cont'd)

6. Batteries

6a. Check that all connections are tight and no corrosion stain

- Coating (Vaseline) is provided
- Check the liquid level of each cell
- Check the cells for leakage

6b. Battery cells tests:

- Unit No.: Test Date:
- Temperature: °C Test Meter No:

Cell No.	Volts	S.G.	Cell No.	Volts	S.G.	Cell No.	Volts	S.G.

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FORM #13-3 INSPECTION/TEST OF STATIC SUPPLY UNIT (cont'd)
(INVERTER)

7. Inverter:

Inverter output voltage and frequency

_____ V _____ Hz

Inverter in synchronism

Auto change-over:

a) from inverter to by-pass _____

b) from by-pass to inverter _____

Inverter output check with oscilloscope at normal load _____

Check alarm indication, function, trip and measurement:

a) Low DC input _____ V b) High DC input _____ V

c) Low AC output _____ V d) High AC output _____ V

e) Inverter fail _____ f) Load transfer to bypass _____

g) Fuse/MCB Failure _____ h) Fan failure, if appl. _____

i) Scada communication _____

8. Remarks :

Tester/Inspector : _____

Date : _____

9. Equipment Acceptance : Y/N *

Engineer : _____

Date : _____

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FORM #13-4**INSPECTION/TEST OF STATIC SUPPLY UNIT (cont'd)****BATTERY DISCHARGE CAPACITY TEST**

Equipment Tag: _____

Test Date: _____

Normal load: _____ A

Meter No.: _____

Time	DC Volts	DC Amps	AC Volts	AC Amps	Hz
0 min					
5 min					
10 min					

1. Discharge current to load shall be within +/- 10% of the normal full load

2. Discharge should be performed by simulating rectifier power failure. Discharge cells shall be recorded until the design end cell voltage and further discharge till the point of inverter trip or the specified minimum voltage.

3. Detection of defective cell should be performed by measuring individual cell voltages during battery discharge.

4. Upon completion of the battery discharge test, system should be switched to boost charge to restored the accumulator energy.

5. Remarks :

Tester/Inspector : _____

Date : _____

6. Equipment Acceptance : Y/N *

Engineer : _____

Date : _____

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FORM #14 INSPECTION/TEST OF HIGH VOLTAGE CABLE

Location: _____

Cable Tag: _____

System Voltage: _____ V

Cores: _____

Size: _____ mm² Cable rated voltage: _____ V

1. Check and record the length of cable: _____ metres/kM *

* Cable to be isolated and unearth at both ends during following tests.

2. Insulation resistance test

Test voltage: _____ V Meter No.: _____

Phase to earth and remaining phases:

Phase L1: _____ MΩ L2: _____ MΩ L3: _____ MΩ

3. High Voltage Tests

Test voltage: _____ kV Time: _____ s Meter No.: _____

Measured leakage current:

Phase L1: _____ mA L2: _____ mAΩ L3: _____ mA

* Refer to Appendix A & B

4. Insulation resistance test, after High voltage test:

Test voltage: _____ V Meter No.: _____

Phase to earth and remaining phases:

Phase L1: _____ MΩ L2: _____ MΩ L3: _____ MΩ

5. Check correct cable termination: _____

6. Remarks :

Tester/Inspector : _____

Date : _____

7. Equipment Acceptance : Y/N *

Engineer : _____

Date : _____

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APPENDIX H: Facility Electrical Power Supply Matrix

- N: Normal Power (Main-Tie-Main)
- E: Emergency Power (E-Gen)
 - # 1. Preferred 100% with DC-Bank/DVR/SMS to be used and 50% with E-Gen back up. (Not applicable to Back-end Site)
- U: UPS (Back up by E-Gen)

System	Equipment	Front-End			Back-End			Note
		<u>N</u>	<u>E</u>	<u>U</u>	<u>N</u>	<u>E</u>	<u>U</u>	
Mechanical - FAB								
Acid Exhaust	Scrubber	50%	50%		50%	50%		
Acid Exhaust	Fans	50%	50%		50%	50%		#1
Acid Exhaust	PLC, Controllers			100%			100%	
Ammonia Exhaust	Scrubber	50%	50%		50%	50%		
Ammonia Exhaust	Fans	50%	50%		50%	50%		#1
Ammonia Exhaust	PLC, Controllers			100%			100%	
Arsenic House Vacuum	Pumps	100%			100%			
Caustic Chemical Room Exhaust	Fans		100%			100%		
Caustic Chemical Room	MUA	50%	50%		50%	50%		#1
Clean Room	Fan Filter Units (FFU)	50%	50%		50%	50%		
Clean Room	MUA Fans	50%	50%		50%	50%		#1
Clean Room	MUA Air Washer Pumps	50%	50%		50%	50%		
CLF3 Exhaust	Fans	50%	50%		50%	50%		
CLF3 Room	MUA	50%	50%		50%	50%		
Computer Rooms	CRAC		100%			100%		
Computer Rooms	DXU		100%			100%		
Cold Room	Cooler Units		100%			100%		
Corrosive/Inert Gas Room	MUA	50%	50%		50%	50%		
Corrosive/Inert Gas Room Exhaust	Fans		100%			100%		
Emergency Exhaust	Fans		100%			100%		
GHVAC	AHU	100%			100%			
GHVAC	FCU	100%			100%			
Flammable Gas Room	MUA	50%	50%		50%	50%		

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System	Equipment	Front-End			Back-End			Note
		<u>N</u>	<u>E</u>	<u>U</u>	<u>N</u>	<u>E</u>	<u>U</u>	
Flammable Gas Room Exhaust	Fan		100%			100%		
General Exhaust	Fans	50%	50%		50%	50%		#1
General Exhaust	PLC, Controllers			100%			100%	
Wet Exhaust (MMP)	Fans	50%	50%		50%	50%		
House Vacuum	Pumps	100%			100%			
Utility Subfab (i.e. Level 1)	MUA	50%	50%		50%	50%		
Utility Subfab (i.e. Level 1)	AHU/RCU	100%			100%			
Plantrooms	RCU	100%			100%			
Probe	MUA	50%	50%		50%	50%		
Probe	RCU	100%			100%			
Process Vacuum	Pumps	50%	50%		100%			#1
PCW	Pumps	50%	50%		100%			#1
PCW	PLC, Controllers			100%			100%	
Slurry Chemical Room	MUA	50%	50%		50%	50%		
Slurry Chemical Room Exhaust	Fans		100%			100%		
Solvent Chemical Room	MUA	50%	50%		50%	50%		
Solvent Chemical Room Exhaust	Fans		100%			100%		
Solvent Exhaust	Fans	50%	50%		50%	50%		#1
Solvent Exhaust	Abatement Units	50%	50%		50%	50%		
Solvent Exhaust	PLC, Controllers			100%			100%	
Staircase Lobbies	Fan		100%			100%		
Staircase Ventilation	Fan	100%			100%			
Toilet Ventilation	Fan	100%			100%			
VOC Abatement Room Exhaust	Fans	80%	20%		80%	20%		#1
Mechanical - CUB								
CDA	Compressors	100%			100%			
CDA	Oil Pumps	100%			100%			
CDA	Dryers		100%			100%		

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		<u>N</u>	<u>E</u>	<u>U</u>	<u>N</u>	<u>E</u>	<u>U</u>	
CDA	Compressor Controller			100%			100%	
CDA	Dryer Controller			100%			100%	
CDA	PLC			100%			100%	
CDA Compressor Intake	MAU	100%			100%			
CDA Room Exhaust	Fans	50%		50%	50%		50%	
Chiller Plant Room	RCU	100%			100%			
Cooling Towers	Fans	100%			100%			
Cooling Towers	Dosing Unit	100%			100%			
Cooling Towers	Reuse Water Makeup Pumps	50%	50%		50%	50%		
Cooling Towers	Pumps	100%			100%			
Cooling Towers	Sidestream Filter Pumps	100%			100%			
Cooling Towers	CDA/HPCDA Cooling Pumps	50%		50%	50%		50%	
Genset DUPS Room Exhaust	Fans	100%			100%			
Guard Room	FCU	100%			100%			
Guard Room	DXU	100%			100%			
GHVAC	FCU	100%			100%			
GHVAC	MUA		100%			100%		
GHVAC	RCU	100%			100%			
Heat Pump Room Exhaust	Fans	100%			100%			
HPCDA	Compressors	100%			100%			
HPCDA	Oil Pumps	50%	50%		50%	50%		
HPCDA	Dryers	50%	50%		50%	50%		
HPCDA	Compressor Controller			100%			100%	
HPCDA	Dryer Controller			100%			100%	
HPCDA	PLC			100%			100%	
Heat Recovery Water (HRW)	Pumps	50%	50%		50%	50%		
Heating Water (HW)	Pumps		100%			100%		
Heating Water (HW)	Heat Pumps	50%	50%		50%	50%		

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System	Equipment	Front-End			Back-End			Note
		<u>N</u>	<u>E</u>	<u>U</u>	<u>N</u>	<u>E</u>	<u>U</u>	
Heating Water (HW)	Control Panels			100%			100%	
LTCHW	Chiller Compressors	100%			100%			
LTCHW	Chiller Control Panels			100%			100%	
LTCHW	Refrigerant Leak Detection		100%			100%		
LTCHW	Primary Pumps	100%			100%			
LTCHW	Secondary Pumps	100%			100%			
MTCHW	Chiller Compressors	100%			100%			
MTCHW	Chiller Control Panels			100%			100%	
MTCHW	Refrigerant Leak Detection		100%			100%		
MTCHW	Primary Pumps	100%			100%			
MTCHW	Secondary Pumps	50%	50%		50%	50%		
Glycol CHW (MMP / A&T Backend)	Chiller Compressors	N/A			100%			
Glycol CHW (MMP / A&T Backend)	Chiller Control Panels	N/A					100%	
Glycol CHW (MMP / A&T Backend)	Refrigerant Leak Detection	N/A				100%		
Glycol CHW (MMP / A&T Backend)	Primary Pumps	N/A			100%			
Glycol CHW (MMP / A&T Backend)	Sidestream Filter Pump	N/A			100%			
Glycol CHW (MMP / A&T Backend)	Accumulator/Makeup Pump	N/A			100%			
Glycol CHW (MMP / A&T Backend)	Secondary Pumps	N/A			100%			
New Water	Pumps	100%			100%			
Non Potable Water	Pumps	100%			100%			
Potable Water	Pumps	100%			100%			
Rainwater Reuse	Pumps	50%	50%		50%	50%		
Solvent Chemical Room	MUA	50%	50%		50%	50%		
Solvent Chemical Room Exhaust	Fans	50%	50%		50%	50%		

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		<u>N</u>	<u>E</u>	<u>U</u>	<u>N</u>	<u>E</u>	<u>U</u>	
GHVAC	AHU	100%			100%			
UPW Room	FCU	50%	50%		50%	50%		
WWTP Room Exhaust	Fans	100%			100%			
XCDA	Purifier			100%			100%	
Fire Protection	Jockey Pump (Wet Pipe System)		100%			100%		
Fire Protection	Fire Pump (Wet Pipe System)		100%			100%		
Fire Protection	Jockey Pump (Hose Reel System)		100%			100%		
Fire Protection	Fire Pump (Hose Reel System)		100%			100%		
UPW Plant								
	Raw Water Pump	100%			100%			
	ACF Backwash Water Pump	100%			100%			
	RO Water Transfer Pump	100%			100%			
	MAU Make Up Supply Pump	100%			100%			
	Regen. Water Supply Pump	100%			100%			
	HCL Transfer Pump	100%			100%			
	Regen. Waste Transfer Pumps	100%			100%			
	Control Panels			100%			100%	
	PLC			100%			100%	
	Vacuum Pump	100%			100%			
	1st Pass Feed Water Pump	100%			100%			
	2nd Pass Feed Water Pump	100%			100%			
	Primary TOC UV (St)	100%			100%			
	EDI Train	100%			100%			
	UPW Feed Pump	50%		50%	50%	50%		

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		<u>N</u>	<u>E</u>	<u>U</u>	<u>N</u>	<u>E</u>	<u>U</u>	
	Polishing Loop Booster Pump	50%		50%	50%	50%		
	Polish GTM Vacuum Pump	50%		50%	50%		50%	
	Primary TOC UV (Polishing Ox)	50%		50%	50%	50%		
	DIRC Recirculation Pump	100%			100%			
	DIRC Lift Pumps	100%			100%			
WWT Plant								
	Transfer Pump	100%			100%			
	Mixer	100%			100%			
	Controls			100%			100%	
	PLC			100%			100%	
	IW Lift Station	50%	50%		50%	50%		
	Sump Pumps		100%			100%		
Bulk Chemicals								
	All							
	ACID STORAGE	50%	50%		50%	50%		
	ACID DISTRIBUTION	100%		100%	100%		100%	
	CAUSTIC STORAGE	50%	50%		50%	50%		
	CAUSTIC DISTRIBUTION	100%		100%	100%		100%	
	SOLVENT STORAGE	50%	50%		50%	50%		
	SOLVENT DISTRIBUTION	100%		100%	100%		100%	
	OXIDISER STORAGE	50%	50%		50%	50%		
	OXIDISER DISTRIBUTION	100%		100%	100%		100%	
	SLURRY STORAGE	50%	50%		50%	50%		
	SLURRY DISTRIBUTION	100%		100%	100%		100%	
Bulk + Specialty Gases								
(With two source/supply)	Feeder to gas plant (main equipment)	100%			100%			
(UPS for gas plant to be provided by supplier)	UPS supply to gas plant			100%			100%	

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		<u>N</u>	<u>E</u>	<u>U</u>	<u>N</u>	<u>E</u>	<u>U</u>	
(Applicable to storage tanks not located in the gas plant.)	Storage tank (pump power socket outlet)		100%			100%		
(Supplier's responsibility)	Gas Plant area lightning							
(Supplier's responsibility)	Gas Plant equipment control Systems (PLC) located in the gas plant							
Micron's responsibility (i.e., purifiers, CQC analyzers)	Gas Plant equipment control Systems (CQC) located in Micron site	100%		100%	100%		100%	
	Spec Gas Equipment Control Systems (GC, BSGS, MVB, DVB, VMB, LDC, heater controllers, etc.)	100%		100%	100%		100%	
	Gas Equipment Heaters (heat trace, cyl heating, JT heater, etc.)		100%			100%		
	Abatement/Scrubber	100%	100%		100%	100%		
	LDC Degasser Pump		100%			100%		
	LDC Vacuum Pump	100%			100%			
Miscellaneous Equipment								
	Passenger Lift		100%			100%		
	Goods Lift		100%			100%		
	Dumb Waiter		100%			100%		
I&C/ELV Equipment								
Dual power with at least one from SUPS (critical equipment)	FMCS (SCADA, PLC/RIO)	100%		100%	100%		100%	
Dual power with at least one from SUPS (critical equipment)	TGMS (Toxic flammable Gas Monitoring System) - SCADA, PLC/RIO, Gas Detectors.	100%		100%	100%		100%	
	LSS (Fire Alarm, HSSD and PA)			100%			100%	

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		<u>N</u>	<u>E</u>	<u>U</u>	<u>N</u>	<u>E</u>	<u>U</u>	
	Security (Card Access and CCTV)			100%			100%	
	Walkie Talkie			100%			100%	
	IT equipment (in MDF, IDF, and Data Center)			100%			100%	